

QUANTUM COMPUTING FOR PARTIAL DIFFERENTIAL
EQUATIONS AND INVERSE PROBLEMS

by
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A dissertation submitted to the Faculty and the Board of Trustees of the Colorado School of Mines in partial fulfillment of the requirements for the degree of Doctor of Philosophy (Geophysics).

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ABSTRACT

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TABLE OF CONTENTS

ABSTRACT	iii
LIST OF FIGURES	viii
LIST OF TABLES	xi
LIST OF SYMBOLS	xiii
LIST OF ABBREVIATIONS	xiv
ACKNOWLEDGMENTS	xv
DEDICATION	xvi
CHAPTER 1 INTRODUCTION	1
CHAPTER 2 BACKGROUND	2
CHAPTER 3 CONVOLUTION ABSORBING BOUNDARIES FOR EXPLICIT-CIRCUIT QUANTUM SIMULATION OF THE WAVE EQUATION	3
3.1 Abstract	3
3.2 Introduction	3
3.2.1 Related work and positioning	5
3.2.2 Contributions	6
3.3 Background	6
3.3.1 The wave equation as Hamiltonian simulation	6
3.3.2 The convolutional perfectly matched layer	6
3.3.3 Schrödingerisation	7
3.4 Absorbing generators	7
3.4.1 One dimension: exact collapse of the convolution	7
3.4.2 Two dimensions: memory form and classical validation	8
3.4.3 The recovery-threshold obstruction	10
3.4.4 The Lyapunov symmetrizer	12
3.5 Explicit circuits	14

3.5.1	Projector-extended Bell-basis term evolution	14
3.5.2	The Schrödingerised circuit	16
3.5.3	End-to-end demonstrations	17
3.6	Complexity	17
3.7	Discussion and outlook	20
3.8	Proof of Lemma 3.1	20
CHAPTER 4 SEISMIC TRAVELTIME INVERSION WITH QUANTUM ANNEALING		24
4.1	Abstract	24
4.2	Introduction	24
4.3	Results	25
4.4	Discussion	29
4.5	Conclusion	29
4.6	Methods	30
4.6.1	Quantum Annealing	30
4.6.2	Seismic Data Acquisition	31
4.6.3	Transforming Ray Equations to QUBO	32
4.6.4	Tikhonov Regularization Linear Least Squares Inversion	34
4.6.5	Cost Analysis	35
4.7	Acknowledgments	36
4.8	Author Contributions	36
4.9	Competing Interests	36
4.10	Data Availability	36
CHAPTER 5 ACCELERATING PHYSICS-INFORMED NEURAL NETWORKS FOR FULL WAVEFORM INVERSION USING A HYBRID QUANTUM-CLASSICAL FINITE-BASIS ARCHITECTURE		37
5.1	Abstract	37
5.2	Introduction	37
5.3	Methodology	40

5.3.1	Acoustic wave equation	40
5.3.2	Finite-basis physics-informed neural network	41
5.3.3	Parameterized quantum circuits	42
5.3.4	JAX-based statevector simulator	44
5.3.5	Hybrid quantum-classical architecture	45
5.3.6	Training setup	47
5.3.7	Network architectures and experimental setup	48
5.4	Results	49
5.4.1	Anomaly benchmark	50
5.4.2	Checkerboard benchmark	51
5.5	Discussion	52
5.6	Conclusion	55
5.7	Data and Code Availability	55
5.8	Appendix: Fourier Structure of the Angle-Encoded PQC	56
5.9	Acknowledgment	56
CHAPTER 6 CONCLUSIONS		59
REFERENCES		60
SELECTED BIBLIOGRAPHY		71
APPENDIX A THE CONVOLUTIONAL PML IN THE HAMILTONIAN-SIMULATION VARIABLES		72
A.1	From the CFS stretch to the memory ODE	72
A.2	One dimension: memory generator and collapse	73
A.3	Proof of the exact collapse	73
A.4	Two dimensions: the eight-block generator	74
A.5	The staggered sampling rule	75
A.6	Profile formulas	76
APPENDIX B SCHRÖDINGERISATION: DISCRETE FORMULAS AND RECOVERY		77
B.1	Hermitian split and warped transform	77

B.2	Discrete setup and per-mode evolution	77
B.3	Initial data and the C^1 profile	78
B.4	Recovery: threshold, wraparound, plateau	78
B.5	Measurement interpretation	79
APPENDIX C VALIDATION PROTOCOL AND CONVERGENCE DATA		80
C.1	Gold-standard reflection protocol	80
C.2	Reflection versus layer width and design reflection	80
C.3	Recovery order in the p register	82
C.4	Trotter orders at gate level	83
C.5	Stagger ablation	83
C.6	Test suite	84
APPENDIX D THE LYAPUNOV SYMMETRIZER: CONSTRUCTION, EXACT PROPERTIES, AND COMPRESSIBILITY DATA		85
D.1	Construction and classical cost	85
D.2	An exact identity for the transformed Hermitian part	85
D.3	The full conditioning table	86
D.4	Saturation of $\kappa_2(S)$ with grid size	88
D.5	Compressibility of S	88
D.6	Choosing the shift ϵ	89

LIST OF FIGURES

- Figure 3.1 The four-step structure: second-order acoustic physics, first-order quantum encoding, convolution absorbing layer applied in that first-order form, and Schrödingerisation restoring unitarity. The cost of step three—the indefiniteness quantified by Lemma 3.1—and its remedy (the Lyapunov symmetrizer) are the subject of Section 3.4.4. . . . 4
- Figure 3.2 Classical and Schrödingerisation convergence. (a) Gold-standard reflection (truncated domain versus $4\times$ reference) versus layer width n_{pml} at design reflection $R_0 = 10^{-3}$. The curve is non-monotonic: the transition error falls with layer width while the wall return rises toward the design R_0 from below, because node sampling of the polynomial profile over-damps thin layers (dashed curve: the predicted discrete wall-return $R_{\text{eff}} = R_0^{\Sigma\sigma h/\int\sigma}$, which the measurement tracks within 5–20% for $n_{\text{pml}} \geq 8$). Red squares: the discretely calibrated profile pair (Appendix C), for which the realised reflection is flat at the design value. (b) Reflection versus R_0 at $n_{\text{pml}} = 12$: the measured reflection tracks the Collino–Tsogka design value within a factor ~ 2.3 over two decades. (c) Recovery error versus p -register size n_p for the kinked $e^{-|p|}$ warping profile $[O(\Delta p)]$ and the C^1 cubic profile $[O(\Delta p^2)]$ of Ref. , 1D CPML benchmark. (d) Trotter splitting error versus step count at orders 1 and 2 (fitted slopes -1.02 and -2.01), measured against the exact Schrödingerised evolution to isolate the splitting error. . . . 9
- Figure 3.3 The recovery threshold in action (2D CPML, 8×8 grid, $\sigma_{\text{max}} = 0.5$, $T = 10$). Relative recovery error versus the slice position p^* : below the threshold $p^* = \lambda^+ T$ (dashed line, $\lambda^+ = 0.683$) the recovered state is $O(1)$ wrong; just above it the error drops by four orders of magnitude, then grows slowly as e^{p^*} amplifies the p -grid discretisation floor. . . . 11
- Figure 3.4 The Lyapunov symmetrizer. (a) Relative recovery error versus evolution time T for the 2D CPML system (8×8 grid, $\sigma_{\text{max}} = 0.5$): untransformed recovery at the default small p^* (squares) is $O(1)$ garbage; untransformed recovery above the threshold $p^* \geq \lambda^+ T$ (diamonds) degrades with T as e^{p^*} amplifies the p -grid floor and leaves the admissible p -domain entirely by $T = 40$; the Lyapunov-symmetrized recovery at the same small p^* (circles, $\epsilon = 10^{-3}$, shifted transform) stays at the 10^{-6} level, essentially flat in T . (b) Post-selection cost on the 16×16 grid: the direct $e^{2\lambda^+ T}$ penalty (solid) against the T -independent symmetrizer penalty $\kappa_2^2(S)$ at $\epsilon = 10^{-2}$ (dashed) and 10^{-3} (dotted); the crossovers are the T^* values of Table 3.2. 15
- Figure 3.5 Explicit circuits. (a) Bell-basis evolution $e^{-i\delta t(cP+\text{h.c.})}$ of the projector-containing term $P = \sigma_{01} \otimes \sigma_{00} \otimes I \otimes I \otimes \sigma_{01} \otimes \sigma_{10}$ of the separated 2D wave Hamiltonian (string `mzIImp`, qubit q_k carrying string character $5 - k$; $\lambda = \arg c$, $\gamma = |c|$; $\lambda = 0$ for the real wave-equation coefficients): CNOTs rotate the ladder qubits to the Bell basis, and the single multi-controlled R_z carries the block projector σ_{00} (q_4) as an *open-circle* 0-control—the entire compilation rule of Section 3.5.1. The gate sequence is unitarily identical to the library-compiled circuit (verified to machine precision). (b) The Schrödingerised pipeline (Section 3.5.2) on bundled registers: product-state preparation $|g\rangle \otimes |\psi_0\rangle$, QFT † on the p register, Strang-split steps alternating the Bell wave circuit $e^{-i\Delta t H}$ with the diagonal damping phases $e^{+i\Delta t \eta \Sigma}$ (half steps at the ends), closing QFT, and recovery by measuring the p register, post-selecting the slice at p^* , and rescaling classically by e^{p^*} 16

Figure 3.6	Gate-level absorbing 2D wave evolution (sponge layer, 8×8 grid, 17 qubits = 6 grid + 2 block + 9 p -register). Top row: snapshots of the recovered field from the Trotterised quantum circuit at $t = 2.5, 5, 7.5, 10$. Bottom row: the exact non-unitary reference $e^{At}\psi_0$. The expanding pulse is absorbed in the boundary layer with no visible reflection; snapshot errors are at the 10^{-3} level (Section 3.5.3).	18
Figure 3.7	Gate-level absorbing 2D wave evolution at scale: 32×32 grid, 21 qubits (12 system + 9 p -register), up to 240 second-order Trotter steps (983 native operations per step). Top row: recovered field from the quantum circuit at $t = 0, 8, 16, 24$ (dashed lines mark the absorbing-layer strips). Bottom row: exact reference $e^{At}\psi_0$. The outgoing ring is absorbed across the layer; the total energy decays from 0.97 to 1.25×10^{-3} , and the recovered field matches the reference to 7.7×10^{-3} – 1.7×10^{-2}	19
Figure 3.8	Gate cost per Trotter step after transpilation to the $\{CX, u\}$ basis, versus the number of system qubits n : 1D steps (squares) and 2D Strang steps (circles). The dotted line is the power-law fit $\propto n^{2.63}$ ($r^2 = 0.99997$) to the 2D series; the term count itself is exactly linear in n (Section 3.6).	21
Figure 4.1	The starting model v_{ini} and the inverted velocity model v_{inv} over the first 9 iterations with exact, noise-free traveltimes data.	26
Figure 4.2	Velocity models and error: (a) final noiseless inverted model after 10 iterations $v_{inv,ij}$, (b) true model $v_{true,ij}$, and (c) the component-wise relative errors e_{ij} between the final inverted and true velocity model.	27
Figure 4.3	Traveltimes inversion results with different noisy data using linear least squares (a, b, c) and quantum annealing method (d, e, f).	28
Figure 4.4	The carbon storage velocity model and ray coverage patterns. Red dots are sources, blue dots are receivers, and white lines represent the ray paths. (a) The synthetic velocity model with a wedge-shaped low-velocity carbon storage formation. (b) Ray coverage from sources and receivers placed in a uniform grid within two boreholes. (c) Ray coverage from sources and receivers placed in a non-uniform pattern, enhancing coverage and constraints for the quantum annealing process.	32
Figure 5.1	Multilayer PQC $\mathcal{Q}$. Each input feature z_i is angle-encoded via an $R_y(\beta_i z_i)$ rotation (blue encoding layer), where β_i is a trainable scaling parameter. Variational layers of single-qubit rotations R_x, R_y, R_z followed by a linear CNOT ladder are repeated L times (orange block). Each qubit is measured in the Pauli- Z basis (green meters), and the outputs are averaged to yield a scalar in $[-1, 1]$	44
Figure 5.2	Comparison of parameter gradients $\partial f / \partial \theta_q$ computed by our JAX statevector implementation and PennyLane’s <code>default.qubit</code> backend across eight circuit configurations (2–8 qubits, 1–6 layers). All values fall on the identity line $y = x$ with maximum absolute deviation $\Delta < 2 \times 10^{-7}$, confirming numerical equivalence of the two implementations.	46
Figure 5.3	Schematic of the finite-basis classical-quantum hybrid architecture, combining classical neural networks and quantum circuits with physics-informed constraints. Blue boxes denote classical networks $\mathcal{N}$, orange boxes quantum circuits $\mathcal{Q}$, green circles the predicted fields ϕ and α , and purple nodes the physics- and data-based residuals contributing to $\mathcal{L}_{total}$. The inset depicts the observed (solid purple) versus predicted (dashed orange) seismograms at the receivers, whose misfit defines the data residual $\mathcal{R}_{data}$	47

Figure 5.4	Training loss decomposition for the classical FBPINN baseline on the anomaly benchmark: total loss, PDE residual, displacement-snapshot initial conditions (two components), seismogram misfit, and free-surface boundary condition, plotted versus training iteration. Inset: early-iteration behaviour showing rapid initial loss reduction. . . .	49
Figure 5.5	Anomaly benchmark inversion and convergence. (a) True velocity model used for the spectral-element forward simulation, with source (red star) and receiver array (green triangles). (b) Initial homogeneous velocity model. (c) Inverted velocity model from the quantum hybrid FBPINN at 65,000 training iterations. (d) L1 velocity error versus training iteration for the classical FBPINN baseline (blue), the quantum hybrid (red markers labeled 65k), and 14 of the 15 classical hyperparameter variants (thin colored lines). Traces begin at different initial L1 values because each velocity (α) network initializes to a different (nominally homogeneous) field; this initial spread reflects initialization rather than training.	57
Figure 5.6	Checkerboard benchmark inversion. (a) True velocity model: a 3×3 checkerboard of alternating ± 0.4 km/s perturbations on a 3.0 km/s background, with source (red star) and receiver array (green triangles). (b) Inverted velocity model from the quantum hybrid FBPINN, using the identical network configuration from the anomaly benchmark with only the collocation point density increased.	58

LIST OF TABLES

Table 3.1	CPML versus sponge at matched resources: 32×32 grid, equal layer width n_{pml}, default parameters [$m = 2$, $R_0 = 10^{-3}$, Collino–Tsogka amplitudes $\sigma_{\text{max}} = 2.59/1.30/0.86$ at $n_{\text{pml}} = 4/8/12$]. Reflection is the gold-standard interior error against a single shared $4 \times$ enlarged (128×128) hard-wall reference, maximised over $T \in \{8, 16, 24, 32, 40\}$ ($\ z_0\ = 1$). $\lambda^+ = \lambda_{\text{max}}^+(H_1)$ sets the CPML post-selection penalty $e^{2\lambda^+ T}$ (quoted at $T = 40$) and tracks $\sqrt{\sigma_{\text{max}}}$ to within 5%; the sponge has $\lambda^+ = 0$, post-selecting at the T- and n_{pml}-independent $e^{2p^*} = 1.2$ for the default recovery slice $p^* = 0.11$.	11
Table 3.2	Lyapunov-symmetrizer conditioning across all measured configurations [grids (n_x, n_y) qubits per dimension, i.e. 8×8, 16×8, 16×16 points; n is the full state dimension including the eight blocks]. $\lambda^+ = \lambda_{\text{max}}^+(H_1)$ is the indefiniteness of the untransformed generator; $\kappa_2(S) = \sqrt{\text{cond}_2(W)}$ the conditioning of the symmetrizer; $\lambda_{\text{max}}(\tilde{H}_1^{(\epsilon)})$ the (strictly negative) largest Hermitian-part eigenvalue of the ϵ-shifted transform $SA_\epsilon S^{-1}$; and $T^* = \ln \kappa^2/(2\lambda^+)$ the crossover time beyond which the symmetrizer beats the direct above-threshold recovery. All Lyapunov residuals are $\leq 1.4 \times 10^{-14}$ (relative).	13
Table 3.3	End-to-end recovery tradeoff: direct versus symmetrized recovery, errors $\ z_{\text{rec}} - e^{AT} z_0\ _2 / \ e^{AT} z_0\ _2$. “Small p^*” is the library-default slice three grid points above zero. For the untransformed system above threshold, $p^* = \lambda^+ T + 1$ and the post-selection penalty is e^{2p^*}; entries marked “–” exceed the p-domain ($\lambda^+ T + 1 > p_{\text{max}}$). The symmetrized column uses the ϵ-shifted transform at $\epsilon = 10^{-3}$ recovered at the same small p^*, with the T-independent penalty $\kappa_2^2(S)$ of Table 3.2. The 16×16 run uses a reduced p register ($n_p = 8$), which sets its higher error floor.	13
Table 5.1	Network configurations and full hyperparameter sweep for the anomaly benchmark. The classical baseline (first row) and quantum hybrid (last row) are the two compared configurations; the intervening rows are the 15 classical hyperparameter variants. $\mathcal{Q}(n_q^{\times n_L})$ denotes a parameterized quantum circuit with n_q qubits and n_L layers. Values marked “–” are identical to the classical baseline. Subdomains column shows $\phi(\cdot)$ for wavefield decomposition.	52
Table A.1	Staggered sampling of the damping profiles in 2D (forward-difference convention; the backward convention flips the half-cell shifts). Each profile is sampled on the grid of the field its equation updates.	76
Table C.1	Gold-standard true reflection (1D CPML, $N = 128$ versus the $N = 512$ hard-wall reference, $m = 2$): plateau error $\max_{T \in \{100, 120, 140\}} E(T)$ and relative plateau spread. Upper block: layer-width sweep at $R_0 = 10^{-3}$; lower block: design-reflection sweep at $n_{\text{pml}} = 12$.	81
Table C.2	Schrödingerisation recovery error versus p-register size n_p for the kinked $e^{- p }$ profile and the C^1 cubic profile of Ref. [1D CPML benchmark, $N = 32$, $n_{\text{pml}} = 8$, $\sigma_{\text{max}} = 1$, $T = 30$, $p \in [-18, 18]$, $\Delta p = 36/2^{n_p}$]. Errors are $\ z_{\text{rec}} - e^{AT} z_0\ _2$ with $\ z_0\ _2 = 1$, recovered at the library-default slice $p^* = 3\Delta p$ (valid at any $p^* > 0$ since $H_1 \leq 0$ here). The fitted orders are least-squares slopes of $\log_2 E$ versus n_p.	82

Table C.3	Gate-level Trotter error versus step count [1D CPML, $N = 8$, $n_{\text{pml}} = 2$, $\sigma_{\text{max}} = 1$, $T = 4$, $n_p = 5$, $p_{\text{max}} = 8$; transpiled circuits on a statevector simulator; errors $\ z_{\text{circ}} - z_{\text{ref}}\ _2$ with $\ z_0\ _2 = 1$. “vs exact S.” uses the exact Schrödingerised evolution (exact per-mode propagation at the same p discretisation) as reference, isolating the splitting error; “vs $e^{AT}z_0$ ” uses the full matrix-exponential reference and therefore contains the p -discretisation floor, measured independently as 1.29×10^{-2} . Order 2 symmetrises both Trotter levels (Strang outer split, palindromic term order inner).	83
Table C.4	Stagger ablation under the gold-standard protocol (Section C.1; 1D CPML, $n_{\text{pml}} = 12$, $R_0 = 10^{-3}$): plateau reflection versus the stagger offset s of the σ_w sampling, $x_j = (j + s)h$	83
Table D.1	Full Lyapunov-symmetrizer data set; Table 3.2 of the main text is the excerpt obtained by dropping the $\lambda_{\text{min}}(W)$ and unshifted-transform columns. Grids are listed in points (n is the full state dimension including the eight blocks). $\lambda^+ = \lambda_{\text{max}}^+(H_1)$ is the indefiniteness of the untransformed generator; $\kappa_2(S) = \sqrt{\text{cond}_2(W)}$; $\lambda_{\text{max}}(\tilde{H}_1)$ is the largest Hermitian-part eigenvalue of the transform SAS^{-1} of the original generator and $\lambda_{\text{max}}(\tilde{H}_1^{(\epsilon)})$ that of the ϵ -shifted transform $SA_\epsilon S^{-1}$; $T^* = \ln \kappa^2 / (2\lambda^+)$ is the crossover time. Every row satisfies the identities of Proposition 1: $\lambda_{\text{max}}(\tilde{H}_1)$ equals $\epsilon - 1/(2\lambda_{\text{max}}(W))$ to a relative deviation $\leq 3.8 \times 10^{-12}$, and $\lambda_{\text{max}}(\tilde{H}_1^{(\epsilon)}) = \lambda_{\text{max}}(\tilde{H}_1) - \epsilon$. Relative Lyapunov residuals are $\leq 1.4 \times 10^{-14}$	87
Table D.2	$\kappa_2(S)$ and $\lambda_{\text{min}}(W)$ versus grid size at $\sigma_{\text{max}} = 0.5$ (grids in points; n is the state dimension). The $n = 8192$ point was gated off by the projected runtime and memory (Section D.1).	88
Table D.3	All 25 candidate approximate symmetrizers on the $n = 1024$ (16×8) confirmation grid at $\epsilon = 10^{-2}$, $\sigma_{\text{max}} = 0.5$. Density is the fraction of nonzero entries of the sparse object; for the $\alpha I + \text{rank-}r$ family it is $(r+1)/n$ (the number of LCU terms relative to n), and for the banded- S^{-1} family it is the density of the banded S^{-1} (state preparation by sparse linear solve). $\lambda_b = \lambda_{\text{max}}(\text{Herm}(S_b A_\epsilon S_b^{-1}))$ is the residual indefiniteness. Status: “pass” = $\lambda_b \leq 2 \times 10^{-3}$, valid in ϵ mode at the default slice; “rescue” = run end to end with the enlarged shift $\epsilon + \lambda_b$; “elig.” = rescue-eligible ($\lambda_b \leq 0.5$) but not run end to end; “fail” = S_b indefinite or $\lambda_b > 0.5$. The top-50% threshold and the support-of- A^2 patterns reproduce the exact S to machine precision (relative Frobenius deviation $\leq 6 \times 10^{-16}$); the end-to-end errors of every usable candidate are obtained at conditioning $\text{cond}(S_b) = 5.06 \times 10^2$ – 5.11×10^2 , equal to the exact value 5.06×10^2	90

LIST OF SYMBOLS

Average radius of cylindrical shell	c
activation energy of oxidation reaction of a-C in excited state	E_{act}^*
hello	...

General Nomenclature

absorption cross section	α_{σ}
------------------------------------	-------------------

LIST OF ABBREVIATIONS

Bio Force Gun, Model 9000	BFG9000
Colorado School of Mines	CSM
Field flow fractionation-inductively coupled plasma-mass spectrometry	FFF-ICP-MS
Mammoth Armed Reclamation Vehicle	MARV
Stone of Jordan	SoJ

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I would like to thank and acknowledge <advisor> <family> <funding sources> <committee>.

For those that shall follow after.

CHAPTER 1

INTRODUCTION

This chapter is a placeholder for the introduction.

CHAPTER 2

BACKGROUND

This chapter is a placeholder for the background.

CHAPTER 3

CONVOLUTION ABSORBING BOUNDARIES FOR EXPLICIT-CIRCUIT QUANTUM SIMULATION OF THE WAVE EQUATION

A paper submitted to *Quantum*.

Hoang Anh Nguyen^{1,2}

3.1 Abstract

Explicit quantum circuits for the wave equation via Hamiltonian simulation [Sato *et al.*, Phys. Rev. Research **6**, 033246 (2024)] are restricted to closed (Dirichlet/Neumann/periodic) domains, in which outgoing waves reflect back into the computational region. We add absorbing boundaries of the convolution type—the complex-frequency-shifted perfectly matched layer (CPML) implemented through exponential-kernel memory variables—and make the resulting non-unitary dynamics quantum-implementable by Schrödingerisation. We obtain three results. (i) Explicit gate-level circuits for the absorbing evolution, obtained by extending the Bell-basis term-evolution circuit of Sato *et al.* to projector-valued operator strings and Trotterising to second order, demonstrated end to end on 17–21 qubits with 10^{-3} – 10^{-2} accuracy against exact references. (ii) A structural obstruction: the memory-form CPML generator has an irreducibly indefinite Hermitian part, $\lambda_{\max}(H_1) = \Theta(\sqrt{\sigma_{\max}c/h})$ under any diagonal rescaling of the memory fields, which by the recovery threshold of Jin, Liu, and Ma costs $e^{2\lambda_{\max}T}$ in post-selection. (iii) A remedy: a classically precomputed Lyapunov symmetrizer renders the transformed generator dissipative, replacing the T -dependent penalty by a T -independent conditioning factor, measured at $\kappa_2(S) = 4.0 \times 10^2$ – 8.6×10^2 at shift $\epsilon = 10^{-2}$ on grids with up to $n = 4096$ unknowns, saturating under mesh refinement. The measured crossover times $T^* = \ln \kappa^2/(2\lambda^+) \approx 7$ – 16 are short: beyond them the symmetrized recovery wins by 4–26 orders of magnitude in post-selection cost and, at the longest horizons tested, is the only working recovery.

3.2 Introduction

Quantum algorithms for partial differential equations (PDEs) divide into oracle-based constructions, which assume block-encoded or sparse-access operators [1, 2], and the explicit-circuit line initiated for hyperbolic problems by Sato *et al.* [3], in which every gate of the time-stepping circuit is written down and the cost is dominated by a number of elementary operations polynomial in the register size. Explicit-circuit

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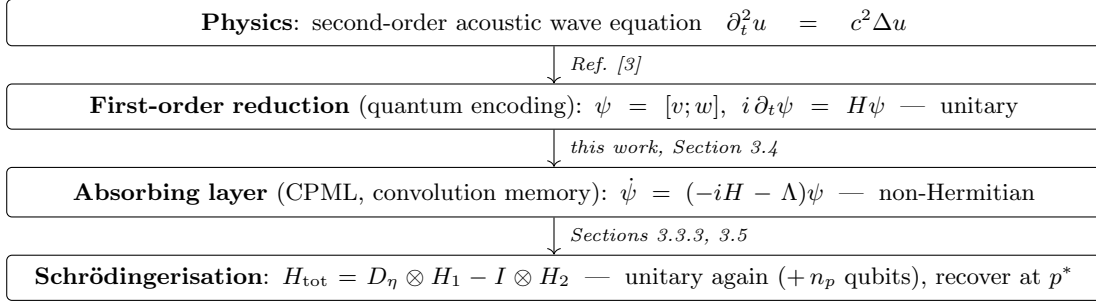


Figure 3.1 The four-step structure: second-order acoustic physics, first-order quantum encoding, convolution absorbing layer applied in that first-order form, and Schrödingerisation restoring unitarity. The cost of step three—the indefiniteness quantified by Lemma 3.1—and its remedy (the Lyapunov symmetrizer) are the subject of Section 3.4.4.

constructions for wave-type equations published to date operate on *closed* domains: Dirichlet, Neumann, or periodic boundaries, for which the semi-discrete generator is skew-Hermitian and the evolution is genuinely unitary [3, 4]; the extension to nonconservative systems [5] admits a bulk damping term $\zeta(x) \partial_t u$ via a linear combination of Hamiltonian simulations, but its boundaries remain closed walls and its demonstrations are confined to the undamped acoustic and uniform heat equations. Scattering and radiation problems, however, are posed on open domains; truncating the domain with hard walls reflects every outgoing wave back into the region of interest. Classical solvers handle this with absorbing layers, the de facto standard being the perfectly matched layer (PML) and its convolutional, complex-frequency-shifted variant (CPML) [6–9]. Absorption is dissipation: the generator acquires a non-trivial Hermitian part, and the dynamics ceases to be unitary.

Schrödingerisation [10] is a general route for running such non-unitary linear dynamics on a quantum computer: the warped phase-space transform turns $\dot{z} = Az$ into a genuine Hamiltonian evolution on one extra register, at the price of a post-selected recovery slice. Figure 3.1 summarises the resulting four-step structure of this work.

The programme now spans artificial boundary conditions for the time-dependent Schrödinger equation [11], inhomogeneous terms and the recovery-threshold theory [12], ill-posed problems [13], Maxwell’s equations [14, 15], elastic waves [16], gate-level circuits for parabolic problems [17, 18], query-optimal formulations [19], and implicit-explicit integrators for multiscale dynamics including the bulk-damped telegraph equation [20]. What it does not yet contain is an absorbing boundary for a classical wave-type equation, at either matrix or circuit level.

3.2.1 Related work and positioning

The positioning below was verified by a full-text literature review through June 2026.³

The core niche is open. No published or preprinted work through June 2026 combines any absorbing boundary (PML, CPML, complex absorbing potential, sponge, or Dirichlet-to-Neumann map) for *classical wave-type equations* with Schrödingerisation or any other unitarisation, at matrix or circuit level. Jin and Zhang [16] (elastic waves) state verbatim that non-periodic boundaries “remain an open issue” left “to the future.” Maxwell circuits exist only for perfect-electric-conductor boundaries [15] and for impedance boundaries at matrix level [14]; the 2026 circuit papers [21, 22] (Robin/Neumann/Dirichlet) contain no absorbing-boundary content. Two 2025 works corroborate this independently. Bösch *et al.* [4], simulating lossless wave equations by Hamiltonian simulation, state that owing to their “fundamentally nonenergy conserving nature” absorbing boundaries and PMLs are “not straightforward to implement... within the HS framework,” and resort to approximate *random* boundaries that scramble, rather than absorb, outgoing energy. Sato *et al.* [5] admit spatially varying bulk damping $\zeta(x) \geq 0$ in an LCHS formulation, but the resulting Hermitian part is semidefinite by construction—the diagonal sponge/CAP case that Lemma 3.1 explicitly excludes—no layer-localised damping or absorbing-layer construction appears, and no damped-wave circuit is demonstrated.

PML plus Schrödingerisation per se is prior art for the time-dependent Schrödinger equation [11], at query-complexity level and without circuits. Our contribution is therefore framed as *wave-type equations plus explicit circuits*.

The recovery threshold is established theory [12, Thm. 3.1], and an indefinite boundary-augmented Hermitian part was already handled through that threshold for the heat equation [18, Remark 1]. Our delta is the *structural* result: Lemma 3.1 proves that the indefiniteness of CPML memory forms cannot be removed by any diagonal rescaling and quantifies its rate, $\Theta(\sqrt{\sigma_{\max}c/h})$.

Diagonal similarity preconditioning is prior art. Liang and Liu [21] use a diagonal transform to enforce a positive-semidefinite Hermitian part before applying a linear-combination-of-Hamiltonian-simulation algorithm to parabolic problems. This is the foil that sharpens our symmetrizer contribution: Lemma 3.1 proves that diagonal transforms are *insufficient* for CPML generators; the full (non-diagonal) Lyapunov symmetrizer of Section 3.4.4 is both necessary and sufficient.

Bell-basis circuits for non-unitary PDEs are prior art [17, 18]. Our circuit delta is narrower and is framed accordingly: projector-valued operator strings for separated multi-field (block) Hamiltonians, the

³Adversarially cross-checked sweep over published and preprinted work, 2023–2026; all primary claims below were confirmed against the full texts of the cited papers.

two-level symmetrised splitting, and the absorbing-boundary application at 17–21 qubits (Section 3.5).

3.2.2 Contributions

Ordered by verified novelty:

1. the first absorbing boundaries for wave-type equations in any unitarisation framework, closing the open problem stated in Ref. [16];
2. the threshold lemma (Lemma 3.1) and the Lyapunov symmetrizer that circumvents it, with the diagonal-insufficiency contrast to Ref. [21];
3. end-to-end validated explicit circuits, extending the Bell-basis constructions of Refs. [3, 17] to projector-valued block structure, with gold-standard classical validation.

3.3 Background

3.3.1 The wave equation as Hamiltonian simulation

Following Ref. [3], the wave equation $\partial_t^2 u = c^2 \Delta u$ is first-order-reduced through $\psi = [v; w]$ with $v = \partial_t u$ and $w = ic \partial_x u$ (discretely, $w = icDu$), so that $i \partial_t \psi = H \psi$ with

$$H = \begin{bmatrix} 0 & D \\ D^\dagger & 0 \end{bmatrix}, \quad (3.1)$$

where D is the forward-difference operator and $D^\dagger = -D_b$ its negative backward counterpart. Mixed Dirichlet/Neumann boundary conditions emerge from the operator products (the conventions of Ref. [3], verified numerically in our implementation). In two dimensions we use the separated-block form $\psi = [v; w_x; w_y]$ —one block per field, in the style of Eq. (32) of Ref. [3] with $d = 3$ —instead of the complex-combined two-block form, because per-direction damping requires separated gradient components.

3.3.2 The convolutional perfectly matched layer

The CPML stretches each coordinate by the complex-frequency-shifted (CFS) factor

$$s(\omega) = \kappa + \frac{\sigma}{\alpha + i\omega}, \quad (3.2)$$

which in the time domain replaces every spatial derivative by a causal convolution with an exponential kernel [6],

$$\partial_x f \longrightarrow \frac{1}{\kappa} \partial_x f + \zeta * \partial_x f, \quad \zeta(t) = -\frac{\sigma}{\kappa^2} e^{-(\sigma/\kappa + \alpha)t} \theta(t). \quad (3.3)$$

Each convolution is carried by one memory field obeying the local ODE

$$\partial_t \phi = -\left(\frac{\sigma}{\kappa} + \alpha\right) \phi - \frac{\sigma}{\kappa^2} \partial_x f. \quad (3.4)$$

The damping profile is polynomially graded inside a layer of n_{pml} points, $\sigma(d) = \sigma_{\text{max}}(d/L)^m$ with $\sigma_{\text{max}} = -(m+1)c \ln R_0/(2L)$ [8]; we use the standard recipe $m = 2$, $R_0 = 10^{-3}$ with the parameter conventions of Komatitsch and Martin [7]. The profile is sampled at the staggered locations where each field effectively lives (v at integer nodes, w at half-cells). This stagger is load-bearing: an ablation shows that unstaggered sampling degrades the measured reflection by factors of 120–235.

3.3.3 Schrödingerisation

Write the semi-discrete absorbing dynamics as $\dot{z} = Az$ and split $A = H_1 + iH_2$ with $H_1 = (A + A^\dagger)/2$ and $H_2 = (A - A^\dagger)/(2i)$, both Hermitian. Only H_1 obstructs unitarity: $\frac{d}{dt}\|z\|^2 = 2z^\dagger H_1 z$, so e^{At} is unitary precisely when $H_1 = 0$, whereas iH_2 is already skew-Hermitian. The warped transform $w(t, p) = e^{-p}z(t)$ trades that obstruction for one extra dimension: because $\partial_p w = -w$, the offending term can be rewritten as $H_1 w = -H_1 \partial_p w$, giving

$$\partial_t w = -H_1 \partial_p w + iH_2 w. \quad (3.5)$$

The rewrite is applied to the H_1 term alone— iH_2 needs no repair—and that is precisely what it buys: ∂_p is skew-Hermitian, hence so is $-H_1 \otimes \partial_p$, and the enlarged generator is skew-Hermitian in full. A Fourier series in p , under which ∂_p acts as multiplication by $i\eta_k$ with η_k the k th grid frequency, decouples Eq. (3.5) into unitary evolutions per mode [10],

$$\hat{w}_k(T) = e^{-iT(\eta_k H_1 - H_2)} \hat{w}_k(0), \quad (3.6)$$

i.e. a Hamiltonian simulation of $H_{\text{tot}} = D_\eta \otimes H_1 - I \otimes H_2$ on the system register plus one p register of n_p qubits. Since e^{-p} is not square-integrable on $p < 0$, the profile is extended to the left half-line by a normalisable g ; the initial data $w(0, p) = g(p)z(0)$ is then a product state across the two registers. The solution is recovered from a slice,

$$z(T) = e^{p^*} w(T, p^*), \quad p^* \geq \lambda_{\text{max}}^+(H_1) T, \quad (3.7)$$

where the threshold on p^* is Theorem 3.1 of Ref. [12]. The kinked profile $g(p) = e^{-|p|}$ is only $O(\Delta p)$ accurate; the C^1 cubic profile of [12, Remark 4.9] restores $O(\Delta p^2)$ while keeping $g(p) = e^{-p}$ for $p \geq 0$, leaving Eq. (3.7) unchanged.

3.4 Absorbing generators

3.4.1 One dimension: exact collapse of the convolution

With $\kappa = 1$, $\alpha = 0$ the convolution collapses *exactly* (for data supported outside the layer) to block-diagonal damping,

$$\dot{\psi} = A\psi, \quad A = -iH - \text{diag}(\sigma_v, \sigma_w), \quad (3.8)$$

so $H_1 = -\text{diag}(\sigma) \preceq 0$ and the recovery threshold (3.7) is unconditional ($\lambda^+ = 0$, any small $p^* > 0$ is valid). The proof is the memory ansatz $\phi = -\sigma v/\gamma$, which satisfies the memory ODE (3.4) identically for data supported outside the layer; we verify the collapse numerically to 6×10^{-13} . The general CFS memory form (two extra block qubits) is provided for $\kappa > 1$ or $\alpha > 0$.

3.4.2 Two dimensions: memory form and classical validation

In 2D the damping acts per direction, so the state uses the separated-block form padded to eight blocks (three extra block qubits),

$$[v; w_x; w_y; \phi_{vx}; \phi_{vy}; \phi_x; \phi_y; 0], \quad (3.9)$$

with the memory injections masked to the layer strips. Corners need no special treatment—both directions’ memory fields are simultaneously active, the structural advantage of unsplit CPML over split formulations [6]. Long-time stability is a serious concern for PMLs in general [9, 23, 24]; here we measure $\max \text{Re eig}(A) = 0$ to round-off across all tested configurations.

Gold-standard validation. Comparing the truncated domain against a $4\times$ enlarged reference, the true reflection is 3.4×10^{-4} (1D, $n_{\text{pml}} = 12$) and 2.8×10^{-4} (2D 32×32 , $n_{\text{pml}} = 8$), at the floor reported by Komatitsch and Martin [7].

Convergence [1D gold standard, $N = 128$ versus an $N = 512$ reference, plateau over $T \in \{100, 120, 140\}$ with plateau spread $< 10^{-9}$, i.e. fully converged; Figure 3.2(a,b)]:

- versus n_{pml} at $R_0 = 10^{-3}$: 6.7×10^{-4} (4), 2.6×10^{-4} (6), 2.6×10^{-4} (8), 3.4×10^{-4} (12), 4.2×10^{-4} (16)—*non-monotonic*, with a minimum at $n_{\text{pml}} \approx 6$ –8. The two contributions are the discrete transition error, which decreases with layer width, and the wall-return reflection, which *increases toward the design value from below*: node sampling of the polynomial profile overestimates $\int \sigma dx$ by a factor $1 + O(1/n_{\text{pml}})$, so the effective round-trip reflection is $R_{\text{eff}} = R_0^{\Sigma \sigma h / \int \sigma}$ ($R_0^{1.38} = 7.1 \times 10^{-5}$ at $n_{\text{pml}} = 4$ rising to $R_0^{1.09} = 5.2 \times 10^{-4}$ at 16); for $n_{\text{pml}} \geq 8$ the measured reflection equals this prediction within 5–20%, i.e. the layer performs at design, converging to R_0 from below [Figure 3.2(a)]. Jointly rescaling the staggered profile pair so the discrete sum matches the design integral (`calibration='discrete'`, Appendix C) makes the realised reflection flat at 0.81–1.08 R_0 over the whole sweep—a one-line calibration fix that applies to classical CPML codes as well;
- versus R_0 at $n_{\text{pml}} = 12$: 4.5×10^{-3} (10^{-2}), 3.4×10^{-4} (10^{-3}), 4.3×10^{-5} (10^{-4})—tracking the design reflection R_0 within a factor ~ 2.3 over two decades. The Collino–Tsogka amplitude formula [8] is predictive in this discretisation.

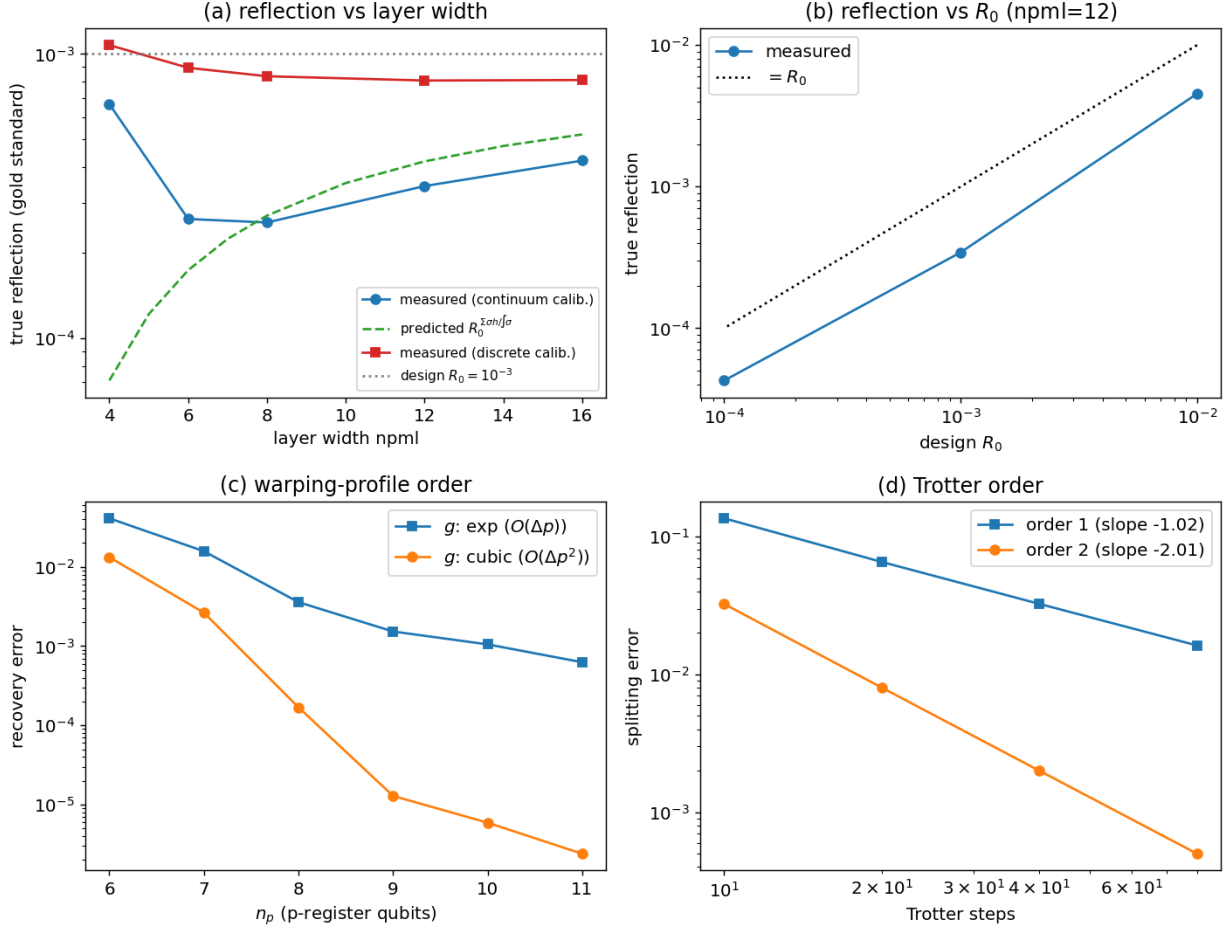


Figure 3.2 Classical and Schrödingerisation convergence. (a) Gold-standard reflection (truncated domain versus $4\times$ reference) versus layer width n_{pml} at design reflection $R_0 = 10^{-3}$. The curve is non-monotonic: the transition error falls with layer width while the wall return rises toward the design R_0 from below, because node sampling of the polynomial profile over-damps thin layers (dashed curve: the predicted discrete wall-return $R_{\text{eff}} = R_0^{\Sigma\sigma h/\int\sigma}$, which the measurement tracks within 5–20% for $n_{\text{pml}} \geq 8$). Red squares: the discretely calibrated profile pair (Appendix C), for which the realised reflection is flat at the design value. (b) Reflection versus R_0 at $n_{\text{pml}} = 12$: the measured reflection tracks the Collino–Tsogka design value within a factor ~ 2.3 over two decades. (c) Recovery error versus p -register size n_p for the kinked $e^{-|p|}$ warping profile $[O(\Delta p)]$ and the C^1 cubic profile $[O(\Delta p^2)]$ of Ref. [12], 1D CPML benchmark. (d) Trotter splitting error versus step count at orders 1 and 2 (fitted slopes -1.02 and -2.01), measured against the exact Schrödingerised evolution to isolate the splitting error.

3.4.3 The recovery-threshold obstruction

The memory form pays a price. The injection column (γ on one side) and the memory-source row ($\sigma\partial_x/\gamma$ on the other) of each memory field are structurally asymmetric, and this asymmetry cannot be scaled away. Throughout this section $\beta_j = \sigma_j/\kappa_j + \alpha_j \geq 0$ denotes the CFS decay rate at node j ; we set the wave speed $c = 1$ (so the stencil entries obey $|D_{ji}| = 1/h$), the general rate restoring a factor of c .

Lemma 3.1 (diagonal irreducibility) *Let A be the memory-form CPML generator of Section 3.4.2 and let $G = \text{diag}(\gamma_j) > 0$ be any diagonal rescaling of the memory fields. Let s_j denote the magnitude of the memory-source coupling ($s_j \sim \sigma_j/(\gamma_j h)$ for the difference stencil). Then the Hermitian part $H_1(G)$ of the rescaled generator satisfies*

$$\lambda_{\max}(H_1(G)) \geq \max_j \max\{f(\gamma_j, \beta_j), g(s_j, \beta_j)\} > 0, \quad (3.10)$$

with $f(\gamma, \beta) = (\sqrt{\beta^2 + \gamma^2} - \beta)/2$, arising from the 2×2 principal submatrix $\begin{bmatrix} 0 & \gamma/2 \\ \gamma/2 & -\beta \end{bmatrix}$ on a (v_j, ϕ_j) pair by Cauchy interlacing, and $g(s, \beta) = f(s, \beta)$ from a (w_i, ϕ_j) pair (the one-directional coupling entry s enters H_1 as $s/2$, so both pairs are governed by the same f). Optimising γ against the two bounds gives

$$\lambda_{\max}(H_1) = \Theta\left(\sqrt{\sigma_{\max} c/h}\right) \quad \text{as } h \rightarrow 0, \quad (3.11)$$

independent of the scaling.

The full proof is given in Appendix 3.8. Its consistency check is instructive: the collapsed 1D form of Section 3.4.1 has *no* injection pair (the memory is eliminated and the damping is diagonal), so the lemma does not apply—and indeed $H_1 = -\text{diag}(\sigma) \preceq 0$ there.

Consequence. By the threshold of Eq. (3.7) [12, Thm. 3.1], recovery requires $p^* \geq \lambda^+ T$ with $\lambda^+ \equiv \lambda_{\max}^+(H_1)$, and the post-selection probability of the recovery window scales as $e^{-2p^*} = e^{-2\lambda^+ T}$. Numerically: at $\sigma_{\max} = 0.5$ on an 8×8 grid, $\lambda^+ = 0.683$; recovery below the threshold is $O(1)$ wrong, while above it the error drops to 2×10^{-3} relative (10^{-4} with $n_p = 11$); see Figure 3.3. The sponge/CAP alternative (diagonal damping $\sigma_x + \sigma_y$, four blocks) has $H_1 \preceq 0$ unconditionally at the price of matched-layer quality.

CPML versus sponge at matched resources. A matched-resource sweep quantifies this tradeoff: both absorbers on the same 32×32 grid with equal layer widths $n_{\text{pml}} \in \{4, 8, 12\}$ and default parameters, each measured against one shared $4\times$ enlarged (128×128) hard-wall reference by the gold-standard protocol of Section 3.4.2, the reflection being the maximum interior error over $T \in \{8, 16, 24, 32, 40\}$ (Table 3.1). At equal layer budget the CPML reaches its discrete wall-return level of 2.7×10^{-4} by $n_{\text{pml}} = 8$ [consistent with the 2.8×10^{-4} gold-standard value of Section 3.4.2], while the sponge stays at the percent level, improving only from 6.2×10^{-2} to 3.6×10^{-2} across the sweep—a quality gap of a factor of 170 at

Table 3.1 CPML versus sponge at matched resources: 32×32 grid, equal layer width n_{pml} , default parameters [$m = 2$, $R_0 = 10^{-3}$, Collino–Tsogka amplitudes $\sigma_{\text{max}} = 2.59/1.30/0.86$ at $n_{\text{pml}} = 4/8/12$]. Reflection is the gold-standard interior error against a single shared $4\times$ enlarged (128×128) hard-wall reference, maximised over $T \in \{8, 16, 24, 32, 40\}$ ($\|z_0\| = 1$). $\lambda^+ = \lambda_{\text{max}}^+(H_1)$ sets the CPML post-selection penalty $e^{2\lambda^+T}$ (quoted at $T = 40$) and tracks $\sqrt{\sigma_{\text{max}}}$ to within 5%; the sponge has $\lambda^+ = 0$, post-selecting at the T - and n_{pml} -independent $e^{2p^*} = 1.2$ for the default recovery slice $p^* = 0.11$.

n_{pml}	Reflection (max over T)		CPML recovery cost	
	CPML	sponge	$\lambda^+(H_1)$	$e^{2\lambda^+T}$, $T=40$
4	4.2×10^{-3}	6.2×10^{-2}	1.54	2.2×10^{53}
8	2.7×10^{-4}	4.7×10^{-2}	1.13	1.6×10^{39}
12	2.8×10^{-4}	3.6×10^{-2}	0.93	1.5×10^{32}

$n_{\text{pml}} = 8$ (130 at 12). The quantum-side cost inverts the ranking. The sponge has $\lambda^+ = 0$ exactly, so its recovery post-selects at the T -independent $e^{2p^*} \approx 1.2$ (default slice $p^* \approx 0.11$); the CPML pays $e^{2\lambda^+T}$, with $\lambda^+ = 0.93$ – 1.54 tracking the $\Theta(\sqrt{\sigma_{\text{max}}c/\hbar})$ rate of Lemma 3.1 to within 5%—already 7×10^7 at $T = 8$ and 1.6×10^{39} at $T = 40$ for $n_{\text{pml}} = 8$. Neither absorber dominates at matched resources: the sponge trades two orders of magnitude in reflection for an essentially free recovery, and the CPML trades an exponentially growing post-selection penalty for matched-layer accuracy—precisely the penalty that the symmetrizer of Section 3.4.4 replaces by a T -independent conditioning factor.

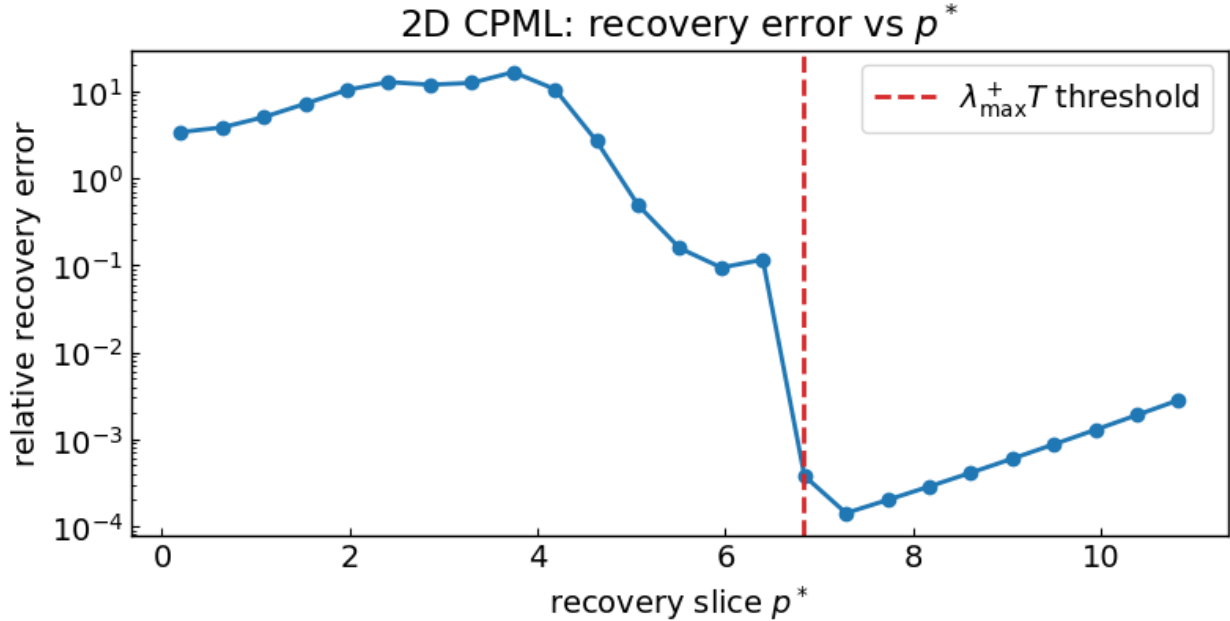


Figure 3.3 The recovery threshold in action (2D CPML, 8×8 grid, $\sigma_{\text{max}} = 0.5$, $T = 10$). Relative recovery error versus the slice position p^* : below the threshold $p^* = \lambda^+T$ (dashed line, $\lambda^+ = 0.683$) the recovered state is $O(1)$ wrong; just above it the error drops by four orders of magnitude, then grows slowly as e^{p^*} amplifies the p -grid discretisation floor.

3.4.4 The Lyapunov symmetrizer

A matrix $W > 0$ solving the Lyapunov inequality $WA + A^\dagger W \preceq 0$ gives, with $S = W^{1/2}$, a transformed generator $\tilde{A} = SAS^{-1}$ whose Hermitian part satisfies

$$\tilde{H}_1 = \frac{1}{2} S^{-1} (WA + A^\dagger W) S^{-1} \preceq 0 \quad (3.12)$$

by congruence. Recovery of $\tilde{z} = Sz$ is then valid at any $p^* > 0$, and $z = S^{-1}\tilde{z}$ is recovered classically (or folded into the readout). Since A is only marginally stable (interior modes are neutral), we construct W from the ϵ -shifted generator:

$$A_\epsilon = A - \epsilon I, \quad A_\epsilon^\dagger W + WA_\epsilon = -I, \quad S = W^{1/2}, \quad (3.13)$$

the shift being undone classically by the factor $e^{\epsilon T}$. An exact identity follows (Proposition 1 in Appendix D: congruence gives $\text{Herm}(SA_\epsilon S^{-1}) = -W^{-1}/2$ exactly; verified to better than 10^{-11} relative in all 18 measured configurations): the transform of the *original* A has

$$\lambda_{\max}(\tilde{H}_1) = \epsilon - \frac{1}{2\lambda_{\max}(W)} \approx \epsilon, \quad (3.14)$$

while the transform of A_ϵ has $\tilde{H}_1 \preceq 0$ strictly. We therefore use the shifted transform; the undo factor $e^{\epsilon T} \leq e^{0.4}$ over the horizons tested is trivial.

Measured results [grids 8×8 , 16×8 , 16×16 ($n = 512$ – 2048), $\sigma_{\max} \in \{0.5, 1.0\}$, $\epsilon \in \{10^{-2}, 10^{-3}, 10^{-4}\}$; Lyapunov residuals $\leq 1.4 \times 10^{-14}$; Table 3.2 and Table 3.3, Figure 3.4]:

- *End-to-end recovery at the default small p^** ($\epsilon = 10^{-3}$): transformed errors 1.2×10^{-6} – 3.5×10^{-5} on the configurations with $n_p \in \{10, 11\}$, essentially flat in T over $T \in \{10, 20, 40\}$; the 16×16 run with a reduced p register ($n_p = 8$) gives 1.1×10^{-4} – 3.8×10^{-3} , consistent with its $O(\Delta p^2)$ floor. The untransformed system at the same p^* returns $O(1)$ – $O(10)$ garbage growing with T .
- *The direct above-threshold method is worse than its nominal $e^{2\lambda^+ T}$ cost*: at $T = 20$ it degrades (errors 0.24–0.45 at $\sigma_{\max} = 0.5$) and at $T = 40$ it explodes (error up to 7×10^{15} at $p^* = 38$) because e^{p^*} amplifies the p -grid floor. At long T the symmetrizer is not merely cheaper—it is the only working recovery.
- *Conditioning*: $\kappa_2(S) = 4.0 \times 10^2$ – 8.6×10^2 at $\epsilon = 10^{-2}$ (the recommended setting), flat in $\sigma_{\max}$, scaling as $\sim \epsilon^{-0.6 \text{ to } -1.0}$ —and *saturating* in n : extending the measurement to $n = 4096$ gives $\kappa_2(S) = 564.5$ at $\epsilon = 10^{-2}$ versus 566.9 at $n = 2048$ (local slope -0.006 ; over the full range $n = 512$ – 4096 the fitted power-law exponent drops from 0.25 to 0.16 with degrading fit quality, i.e. a power law is the wrong model because κ is flattening). λ^+ itself saturates ($0.683 \rightarrow 0.694$ over $8\times$ in n) and $\lambda_{\min}(W)$ stays at 0.66–0.68, so the κ growth is driven entirely by $\lambda_{\max}(W) \sim 1/(2\epsilon)$: the

Table 3.2 Lyapunov-symmetrizer conditioning across all measured configurations [grids (n_x, n_y) qubits per dimension, i.e. 8×8 , 16×8 , 16×16 points; n is the full state dimension including the eight blocks]. $\lambda^+ = \lambda_{\max}^+(H_1)$ is the indefiniteness of the untransformed generator; $\kappa_2(S) = \sqrt{\text{cond}_2(W)}$ the conditioning of the symmetrizer; $\lambda_{\max}(\tilde{H}_1^{(\epsilon)})$ the (strictly negative) largest Hermitian-part eigenvalue of the ϵ -shifted transform $SA_\epsilon S^{-1}$; and $T^* = \ln \kappa^2/(2\lambda^+)$ the crossover time beyond which the symmetrizer beats the direct above-threshold recovery. All Lyapunov residuals are $\leq 1.4 \times 10^{-14}$ (relative).

Grid	n	$\sigma_{\max}$	ϵ	$\lambda^+(H_1)$	$\kappa_2(S)$	$\lambda_{\max}(\tilde{H}_1^{(\epsilon)})$	T^*
8×8	512	0.5	10^{-2}	0.683	4.0×10^2	-4.6×10^{-6}	8.8
8×8	512	0.5	10^{-3}	0.683	3.0×10^3	-8.1×10^{-8}	11.7
8×8	512	0.5	10^{-4}	0.683	1.1×10^4	-6.0×10^{-9}	13.6
8×8	512	1.0	10^{-2}	0.927	5.1×10^2	-5.2×10^{-6}	6.7
8×8	512	1.0	10^{-3}	0.927	2.9×10^3	-1.6×10^{-7}	8.6
8×8	512	1.0	10^{-4}	0.927	9.8×10^3	-1.4×10^{-8}	9.9
16×8	1024	0.5	10^{-2}	0.687	5.1×10^2	-3.0×10^{-6}	9.1
16×8	1024	0.5	10^{-3}	0.687	7.0×10^3	-1.5×10^{-8}	12.9
16×8	1024	0.5	10^{-4}	0.687	3.1×10^4	-8.0×10^{-10}	15.0
16×8	1024	1.0	10^{-2}	0.956	7.5×10^2	-2.4×10^{-6}	6.9
16×8	1024	1.0	10^{-3}	0.956	7.0×10^3	-2.8×10^{-8}	9.3
16×8	1024	1.0	10^{-4}	0.956	2.7×10^4	-1.8×10^{-9}	10.7
16×16	2048	0.5	10^{-2}	0.694	5.7×10^2	-2.4×10^{-6}	9.1
16×16	2048	0.5	10^{-3}	0.694	9.9×10^3	-7.7×10^{-9}	13.3
16×16	2048	0.5	10^{-4}	0.694	6.6×10^4	-1.7×10^{-10}	16.0
16×16	2048	1.0	10^{-2}	0.972	8.6×10^2	-1.9×10^{-6}	6.9
16×16	2048	1.0	10^{-3}	0.972	1.2×10^4	-8.7×10^{-9}	9.7
16×16	2048	1.0	10^{-4}	0.972	6.3×10^4	-3.4×10^{-10}	11.4

Table 3.3 End-to-end recovery tradeoff: direct versus symmetrized recovery, errors $\|z_{\text{rec}} - e^{AT} z_0\|_2 / \|e^{AT} z_0\|_2$. “Small p^* ” is the library-default slice three grid points above zero. For the untransformed system above threshold, $p^* = \lambda^+ T + 1$ and the post-selection penalty is e^{2p^*} ; entries marked “_” exceed the p -domain ($\lambda^+ T + 1 > p_{\max}$). The symmetrized column uses the ϵ -shifted transform at $\epsilon = 10^{-3}$ recovered at the same small p^* , with the T -independent penalty $\kappa_2^2(S)$ of Table 3.2. The 16×16 run uses a reduced p register ($n_p = 8$), which sets its higher error floor.

Configuration	T	Untransformed			Symmetrized ($\epsilon = 10^{-3}$)	
		err. (small p^*)	$p^* \geq \lambda^+ T$	err. (above thr.)	penalty e^{2p^*}	err. (small p^*)
8×8 ($\sigma_{\max} = 0.5$, $n_p = 11$)	10	3.4	7.8	1.2×10^{-4}	6.3×10^6	1.2×10^{-6}
	20	8.3	14.7	2.4×10^{-1}	5.4×10^{12}	2.4×10^{-6}
	40	2.6×10^1	—	—	—	4.4×10^{-6}
8×8 ($\sigma_{\max} = 1.0$, $n_p = 11$)	10	7.2	10.3	2.2×10^{-2}	8.4×10^8	2.1×10^{-5}
	20	1.6×10^1	19.6	3.0×10^2	1.0×10^{17}	2.3×10^{-5}
	40	2.9×10^1	38.1	7.3×10^{15}	1.2×10^{33}	3.5×10^{-5}
16×8 ($\sigma_{\max} = 0.5$, $n_p = 10$)	10	2.5	7.9	1.1×10^{-4}	7.4×10^6	4.2×10^{-6}
	20	4.8	14.8	4.5×10^{-1}	6.7×10^{12}	6.0×10^{-6}
	40	1.3×10^1	—	—	—	1.1×10^{-5}
16×16 ($\sigma_{\max} = 0.5$, $n_p = 8$)	10	2.0	8.0	1.3×10^{-2}	9.2×10^6	1.1×10^{-4}
	20	5.6	15.0	7.5×10^1	1.0×10^{13}	5.2×10^{-4}
	40	2.5×10^1	—	—	—	3.8×10^{-3}

conditioning is ϵ -controlled, not grid-controlled, and the symmetrizer cost does not blow up under mesh refinement. The eigenvector symmetrizer $W = (VV^\dagger)^{-1}$ is four to six orders of magnitude worse conditioned ($\kappa_2 \sim 3 \times 10^8$) and fails outright on near-defective configurations—the Lyapunov route is the right construction.

- *Tradeoff*: $T^* = \ln \kappa^2 / (2\lambda^+) = 6.7\text{--}16.0$ across all configurations; at $T = 20$ the symmetrizer wins by 4–9 orders of magnitude in post-selection cost, at $T = 40$ by 16–26 orders. T^* grows only logarithmically with κ , hence very weakly with grid size.

Structure and compressibility of S . We measured the structure of S directly and tested 25 approximate symmetrizers (distance-banded, magnitude-thresholded, block-pattern, identity-plus-low-rank, hybrid, and inverse-side-banded families) for dissipativity and end-to-end recovery on the $n = 512$ and $n = 1024$ systems. Three findings. (i) Identity-plus-low-rank is *ruled out*: the correction $E = S - \alpha I$ is numerically full rank (995 of 1024 singular values above 10^{-2} of the top), and $\lambda_{\max}(\tilde{H}_1)$ stays at the *untransformed* $\lambda^+ = 0.67$ for ranks 8–64—no few-term LCU implementation of S exists. (ii) Strict sparsification preserving $\lambda_{\max}(\tilde{H}_1) \leq \epsilon$ requires essentially the full active support ($\approx 48\text{--}51\%$ of entries; on the remaining “dead” indices—interior memory and padding, 30–50% of the space— S is exactly a multiple of the identity, which is free structure). (iii) *Moderate compression works* once the residual λ_b of an approximate S_b is absorbed as an additional shift (undone by $e^{\lambda_b T} \leq e^{2.6} \approx 14$ at the worst measured $\lambda_b = 0.066$, $T = 40$): magnitude thresholding at 35% density recovers to 8.7×10^{-6} ($T = 10$) and 5.4×10^{-5} ($T = 40$); a rank-32-plus-band-8 hybrid at 26% reaches 1.3×10^{-5} and 6.6×10^{-4} (all $n = 1024$; the conditioning of every usable S_b stays $\sim 5 \times 10^2$, equal to the exact S). The energy of E is 99.6% concentrated on layer-strip index pairs. The implementation route is therefore block-encoding of a $\sim 35\%$ -dense operator (times the free dead-index identity factor), not a local circuit or a short LCU; at current theory the symmetrizer is best used as classical pre/post-processing, and a structured-ansatz Lyapunov construction (solving for W within a polynomially parametrised operator family) is the open problem these measurements sharpen.

Honest limitations. The construction requires a dense $O(n^3)$ classical precomputation of W , S , S^{-1} ; the fate of the sparse quantised-tensor-train (QTT) term structure under the dense similarity (Trotter circuits for $\tilde{A}$) is not assessed; the κ scaling beyond $n = 4096$ is extrapolation (though saturating); the compression study is at $n \leq 1024$.

3.5 Explicit circuits

3.5.1 Projector-extended Bell-basis term evolution

The circuit of Sato *et al.* [3] exponentiates ladder-string terms (alphabet $\{I, \sigma_{01}, \sigma_{10}\}$) via a Bell-basis change followed by one multi-controlled R_z ; Hu, Jin, Liu, and Zhang [17] extended the framework to

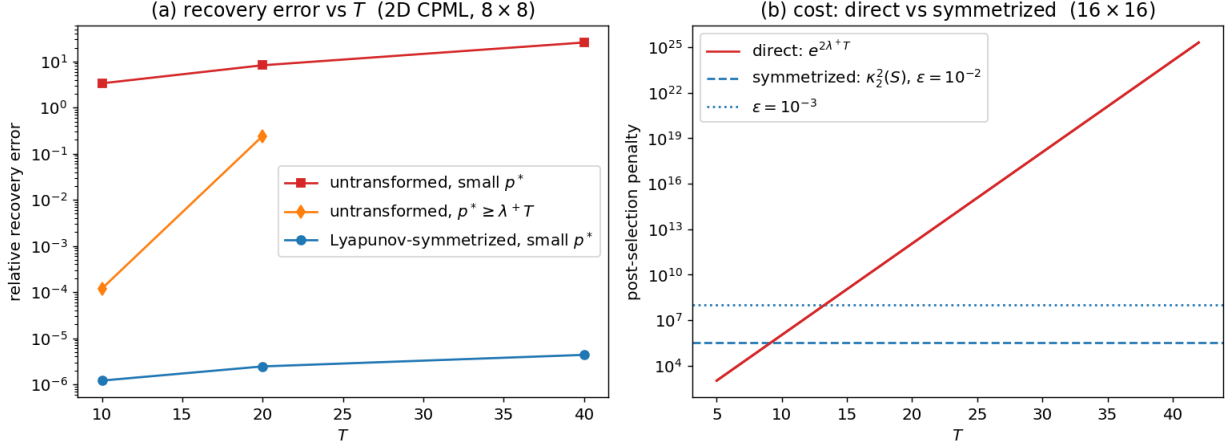


Figure 3.4 The Lyapunov symmetrizer. (a) Relative recovery error versus evolution time T for the 2D CPML system (8×8 grid, $\sigma_{\max} = 0.5$): untransformed recovery at the default small p^* (squares) is $O(1)$ garbage; untransformed recovery above the threshold $p^* \geq \lambda^+ T$ (diamonds) degrades with T as e^{p^*} amplifies the p -grid floor and leaves the admissible p -domain entirely by $T = 40$; the Lyapunov-symmetrized recovery at the same small p^* (circles, $\epsilon = 10^{-3}$, shifted transform) stays at the 10^{-6} level, essentially flat in T . (b) Post-selection cost on the 16×16 grid: the direct $e^{2\lambda^+ T}$ penalty (solid) against the T -independent symmetrizer penalty $\kappa_2^2(S)$ at $\epsilon = 10^{-2}$ (dashed) and 10^{-3} (dotted); the crossovers are the T^* values of Table 3.2.

non-unitary PDE dynamics via Schrödingerisation, and Jin, Liu, and Yu [18] built gate-level circuits for the heat equation with physical boundaries. Relative to these we add the *projector* extension: strings containing $z = |0\rangle\langle 0|$ and $o = |1\rangle\langle 1|$ (block projectors of separated multi-field systems) enter as plain 0-/1-controls; the Bell rotation outside the projected subspace is undone exactly, so the circuit implements $e^{-i \delta t (cP + \text{h.c.})}$ for projector-valued P . Figure 3.5(a) shows the compiled circuit for a representative projector-containing term of the separated 2D wave Hamiltonian, and Figure 3.5(b) the full Schrödingerised pipeline it slots into.

Due diligence (full-text check of both papers): neither Ref. [17] nor Ref. [18] circuitises projector-containing strings—both use only the $\{I, \sigma_{01}, \sigma_{10}\}$ alphabet. We confirmed this at code level in the published companion notebook of Ref. [17]: its term circuit (W_j) implements the contiguous ladder pattern with rotations controlled uniformly on all lower qubits (an all-or-nothing X -conjugation flag serving the periodic corner terms), with no per-qubit 0-/1-control selection (the nilpotent shift needs no projector guards in scalar single-field problems, and Ref. [18] routes boundary data into source terms precisely to avoid non-Toeplitz structure). Ref. [5] comes closest: its symbolic operator algebra over $\{I, \sigma_{00}, \sigma_{11}, \sigma_{01}, \sigma_{10}\}$ produces projector factors from boundary-truncated shifts and piecewise-constant coefficients, and poly-depth Trotterisation is asserted by citation to Ref. [3]; but no gate-level compilation rule for the projector factors and no projector-term circuit is given. A caveat we state explicitly: the

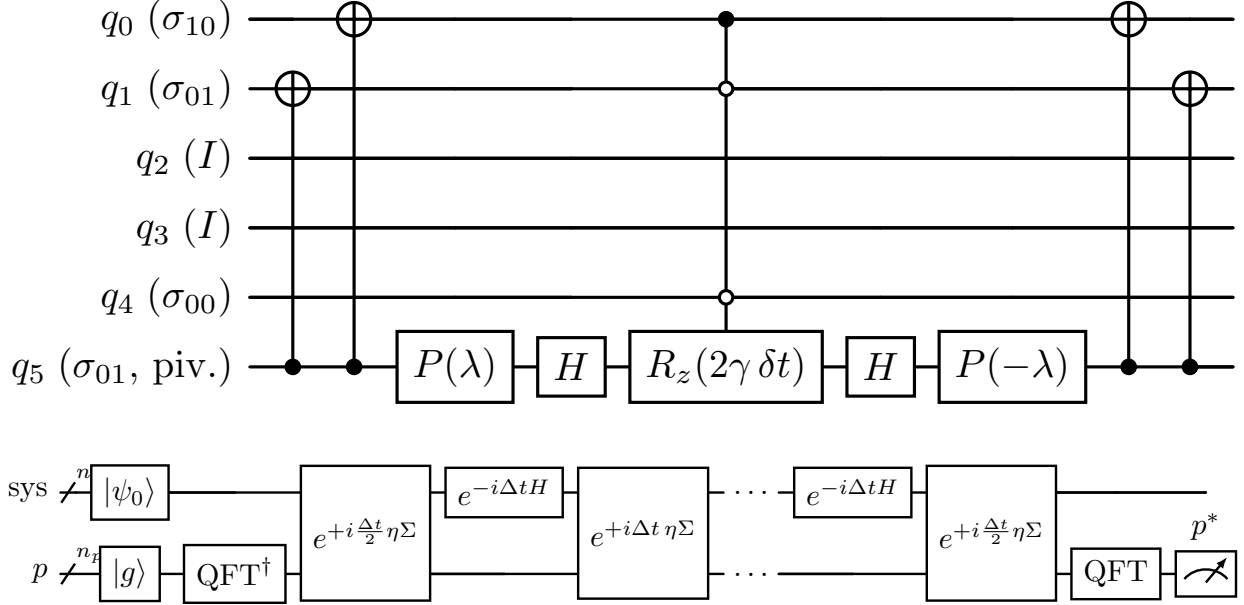


Figure 3.5 Explicit circuits. (a) Bell-basis evolution $e^{-i\delta t(cP+\text{h.c.})}$ of the projector-containing term $P = \sigma_{01} \otimes \sigma_{00} \otimes I \otimes I \otimes \sigma_{01} \otimes \sigma_{10}$ of the separated 2D wave Hamiltonian (string `mzIImp`, qubit q_k carrying string character $5 - k$; $\lambda = \arg c$, $\gamma = |c|$; $\lambda = 0$ for the real wave-equation coefficients): CNOTs rotate the ladder qubits to the Bell basis, and the single multi-controlled R_z carries the block projector σ_{00} (q_4) as an *open-circle* 0-control—the entire compilation rule of Section 3.5.1. The gate sequence is unitarily identical to the library-compiled circuit (verified to machine precision). (b) The Schrödingerised pipeline (Section 3.5.2) on bundled registers: product-state preparation $|g\rangle \otimes |\psi_0\rangle$, $\text{QFT}^\dagger$ on the p register, Strang-split steps alternating the Bell wave circuit $e^{-i\Delta t H}$ with the diagonal damping phases $e^{+i\Delta t\eta\Sigma}$ (half steps at the ends), closing QFT , and recovery by measuring the p register, post-selecting the slice at p^* , and rescaling classically by e^{p^*} .

abstract class $|a\rangle\langle b| + |b\rangle\langle a|$ over basis states in Lemma 1 of Refs. [3, 17] implicitly *contains*

Hermitian-paired projector strings as an existence statement; our contribution is the explicit alphabet

extension and compilation rule (projector factor $\rightarrow$ 0-/1-control) and its application to multi-field

block-coupled systems, not a new diagonalisation principle. The extension is backward compatible.

Unitary-level validation: the error is pure per-step Trotter error $0.50\delta t^2$ with no phase offset. This unlocks the separated 2D wave Hamiltonian (`zm/mz` block couplings) needed for per-direction absorbing layers.

3.5.2 The Schrödingerised circuit

The full pipeline is

$$\begin{aligned} \text{init}(g \otimes \psi_0) &\rightarrow \text{QFT}_p^\dagger \rightarrow [\text{Trotter steps}] \\ &\rightarrow \text{QFT}_p \rightarrow \text{slice at } p^*. \end{aligned} \quad (3.15)$$

Each step (diagonal H_1) factorises into the wave circuit (the Bell construction of Section 3.5.1 on the system register) and a diagonal damping phase with entries $e^{+i\delta t\eta_k\sigma_j}$. Both Trotter levels are symmetrised (Strang outer; palindromic term order inner), with measured clean second order: errors $4 \times 10^{-2} \rightarrow 10^{-2} \rightarrow 2.7 \times 10^{-3}$ at 50/100/200 steps on 17 qubits.

Fitted orders [1D benchmark, error versus the exact Schrödingerised evolution to isolate the splitting; Figure 3.2(d)]: slope -1.02 (order 1), -2.01 (order 2 with both levels symmetrised); against the full matrix-exponential reference the order-2 error saturates at the p -discretisation floor, as expected.

Warping-profile order [recovery error versus n_p , fixed 1D PML case; Figure 3.2(c)]: $e^{-|p|}$ slope ~ -1 in Δp ($4.1 \times 10^{-2} \rightarrow 6.3 \times 10^{-4}$ over $n_p = 6-11$); C^1 cubic ~ -2 ($1.3 \times 10^{-2} \rightarrow 2.4 \times 10^{-6}$), confirming Remark 4.9 of Ref. [12] in this setting.

3.5.3 End-to-end demonstrations

- 1D CPML, 14 qubits (6 system + 8 p), 60 steps, $T = 30$: recovered solution within 7×10^{-4} of the exact non-unitary evolution.
- 2D sponge, 8×8 grid, 17 qubits, $T = 10$: 2.7×10^{-3} at 200 steps (Figure 3.6).
- 2D sponge, 32×32 grid, 21 qubits (12 system + 9 p), up to 240 steps (983 native operations per step): errors 7.7×10^{-3} / 1.7×10^{-2} / 1.4×10^{-2} at $T = 8/16/24$; energy decays from 0.97 to 1.25×10^{-3} (absorption complete); nine minutes of statevector simulation (Figure 3.7).

Qubit accounting: the absorbing overhead is +1 (sponge) or +2 (CPML) block qubits plus $n_p \approx 8-10$ for the p register; n_p depends only logarithmically on T , the damping rate, and the target accuracy.

3.6 Complexity

Term counts inherit the QTT structure of the difference operators: $2n_x + 2$ terms in 1D and $4n_x + 4$ in 2D ($n_x = n_y$), exactly linear in the register size (fitted slopes 4.000/2.000, intercepts 4/2). Transpiled to the $\{CX, u\}$ basis, one Strang wave step costs 217/459/834/1343 operations for $n_{\text{sys}} = 6/8/10/12$; a power-law fit gives exponent 2.63 with $r^2 = 0.99997$ (Figure 3.8)—convincingly polynomial, and in fact asymptotically quadratic: the term count is linear in n and the ancilla-free V-chain multi-controlled- R_z is linear in its control count, so ops/step = $O(n^2)$. The constants are fixed by the measurements. The per-step cost in the $\{CX, U\}$ basis is reproduced to 0.03% ($r^2 = 0.9999999$) by the quadratic

$$G_{\text{step}}(n) = 16.7n^2 - 112.7n + 293, \quad (3.16)$$

whose structure is transparent: a Strang step exponentiates $2n$ Hermitian-paired terms ($4n_x + 4$ at $n = 2n_x + 2$), each one Bell basis change plus one multi-controlled R_z whose ancilla-free V-chain cost is

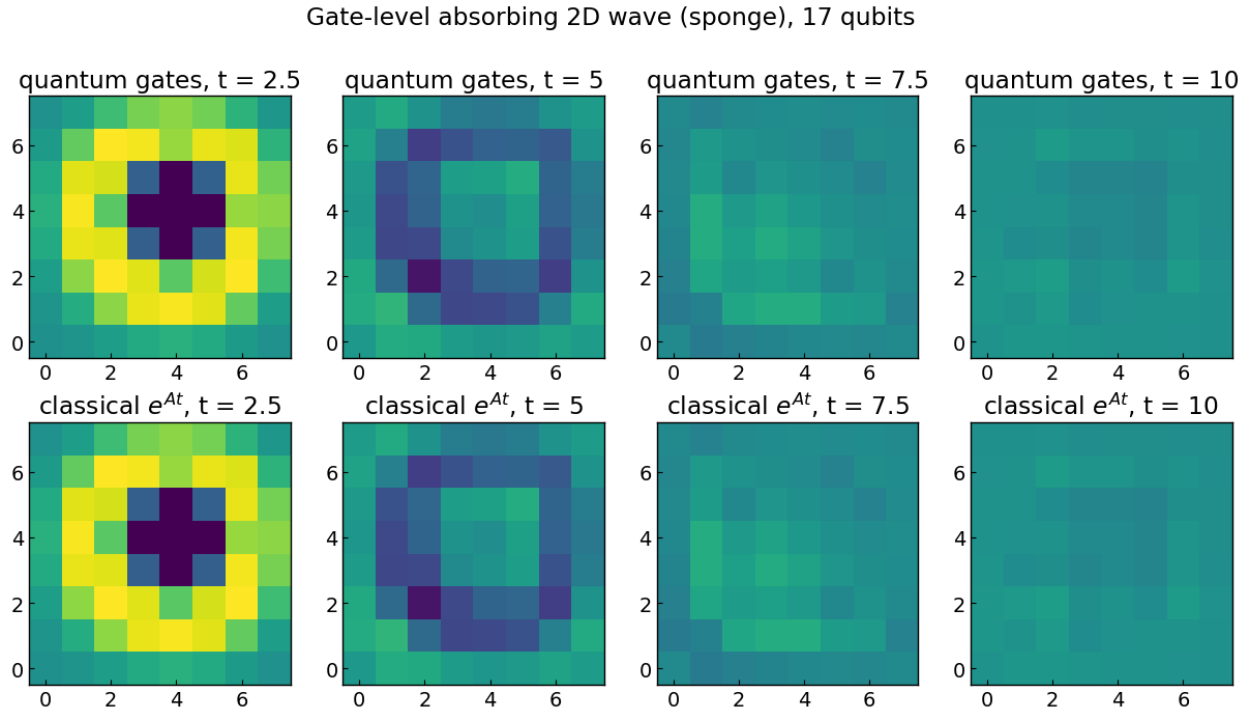


Figure 3.6 Gate-level absorbing 2D wave evolution (sponge layer, 8×8 grid, 17 qubits = 6 grid + 2 block + 9 p -register). Top row: snapshots of the recovered field from the Trotterised quantum circuit at $t = 2.5, 5, 7.5, 10$. Bottom row: the exact non-unitary reference $e^{At}\psi_0$. The expanding pulse is absorbed in the boundary layer with no visible reflection; snapshot errors are at the 10^{-3} level (Section 3.5.3).

Gate-level absorbing 2D wave, 32x32 grid (21 qubits, 240 Trotter steps at T=24)

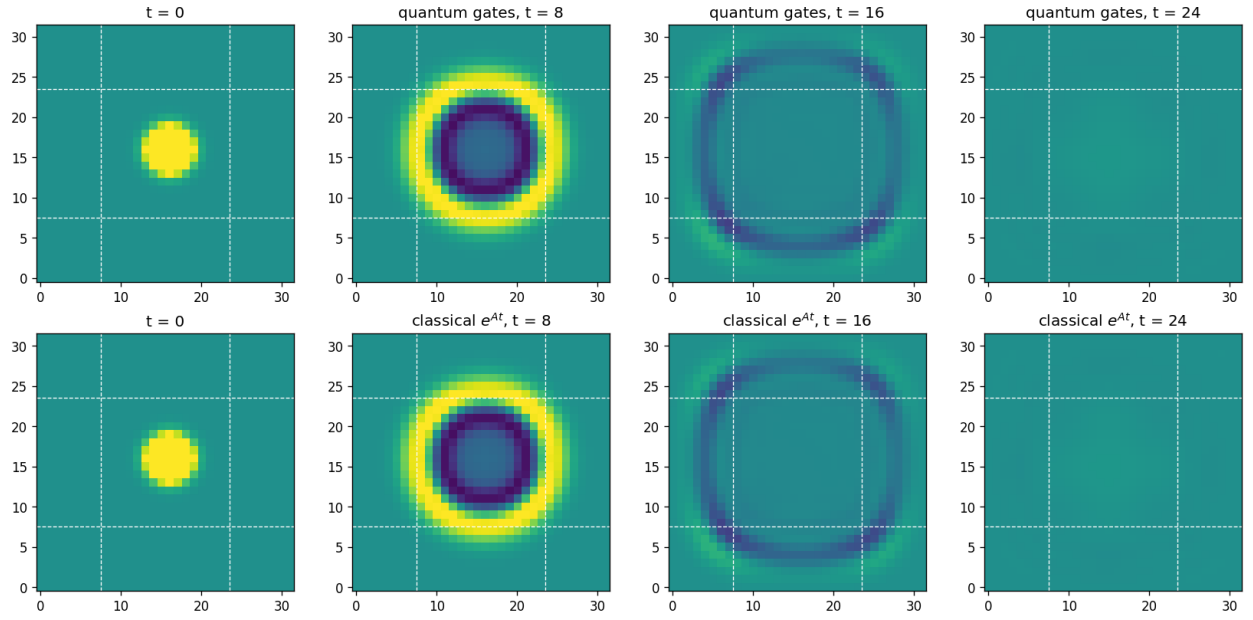


Figure 3.7 Gate-level absorbing 2D wave evolution at scale: 32×32 grid, 21 qubits (12 system + 9 p -register), up to 240 second-order Trotter steps (983 native operations per step). Top row: recovered field from the quantum circuit at $t = 0, 8, 16, 24$ (dashed lines mark the absorbing-layer strips). Bottom row: exact reference $e^{At}\psi_0$. The outgoing ring is absorbed across the layer; the total energy decays from 0.97 to 1.25×10^{-3} , and the recovered field matches the reference to 7.7×10^{-3} – 1.7×10^{-2} .

exactly $32k - 31$ elementary operations for $k \geq 8$ controls (measured), linear in k ; with the QTT ladder structure averaging $\bar{k} \approx n/4$ controls per term, the leading coefficient is $2n \cdot 32 \bar{k}/n^2 \approx 16.7$. The total gate count is steps $\times$ ops/step with steps $= T/\delta t$ and δt set by the second-order bound $\delta t = \sqrt{\varepsilon_{\text{tr}}/(C_2 T)}$, where C_2 is the Strang error constant of the slope-(-2) fits of Figure 3.2(d):

$$G_{\text{total}} = \sqrt{\frac{C_2 T^3}{\varepsilon_{\text{tr}}}} G_{\text{step}}(n) = O\left(\frac{T^{3/2}}{\sqrt{\varepsilon_{\text{tr}}}} n^2\right), \quad (3.17)$$

plus the damping diagonal per step and two QFT(n_p); the power-law fit across the measured range reads $n^{2.63}$ because the subleading terms of Eq. (3.16) have not yet decayed at $n \leq 12$.

Caveat (honest). The damping factor is a single native **diagonal** instruction on the simulator, but a *generic* diagonal on n qubits synthesises to $\Theta(2^n)$ elementary gates: we measure a CX count of exactly $2^n - 2$ and totals within 15% of the $2^{n+1} - 3$ synthesis bound. For hardware the structure must be exploited: the diagonal is $D_\eta \otimes \Sigma$ with η linear in the p index and σ a degree- m polynomial profile on the layer strips, so phase-polynomial constructions give $\text{poly}(n)$ circuits, and the logic-minimisation synthesis of piecewise-constant diagonal operators of Ref. [5] (Espresso-compressed projector strings with matrix-product-state coefficient oracles) provides an existing $\text{poly}(n)$ toolbox directly applicable to the strip-wise polynomial σ profiles; making this explicit is deferred to future work.

3.7 Discussion and outlook

The λ^+ obstruction of Lemma 3.1 is generic for auxiliary-field absorbers—memory variables, auxiliary-differential-equation formulations, split fields—because the injection/source asymmetry is structural rather than an artefact of one discretisation. The Lyapunov symmetrizer is the corresponding general remedy, and its quantum-implementable variants (block-encoding S , structured or banded approximate symmetrizers) are open. Natural next steps include a hardware experiment in the style of Ref. [3]; quantum-signal-processing or qubitisation in place of Trotter (and the query-optimal Schrödingerisation formulation of Ref. [19]); CPML for Maxwell’s equations [14, 15]; and smoother warping profiles (error-function or numerically optimal) beyond the C^1 cubic.

Code and data availability. The implementation (`lib/absorbing.py`, `lib/schrodingerisation.py`), all experiment scripts and JSON result files (`paper/experiments/`), and the executed demonstration notebook are available in the accompanying repository at <https://github.com/x-repos/qawave-cpml>.

3.8 Proof of Lemma 3.1

Fix a layer node j with $\sigma_j > 0$ and a stencil neighbour i such that the difference entry satisfies $|D_{ji}| = 1/h$ (wave speed $c = 1$; the general case restores a factor c in Eq. (3.11)). Write $\beta = \sigma_j/\kappa_j + \alpha_j \geq 0$

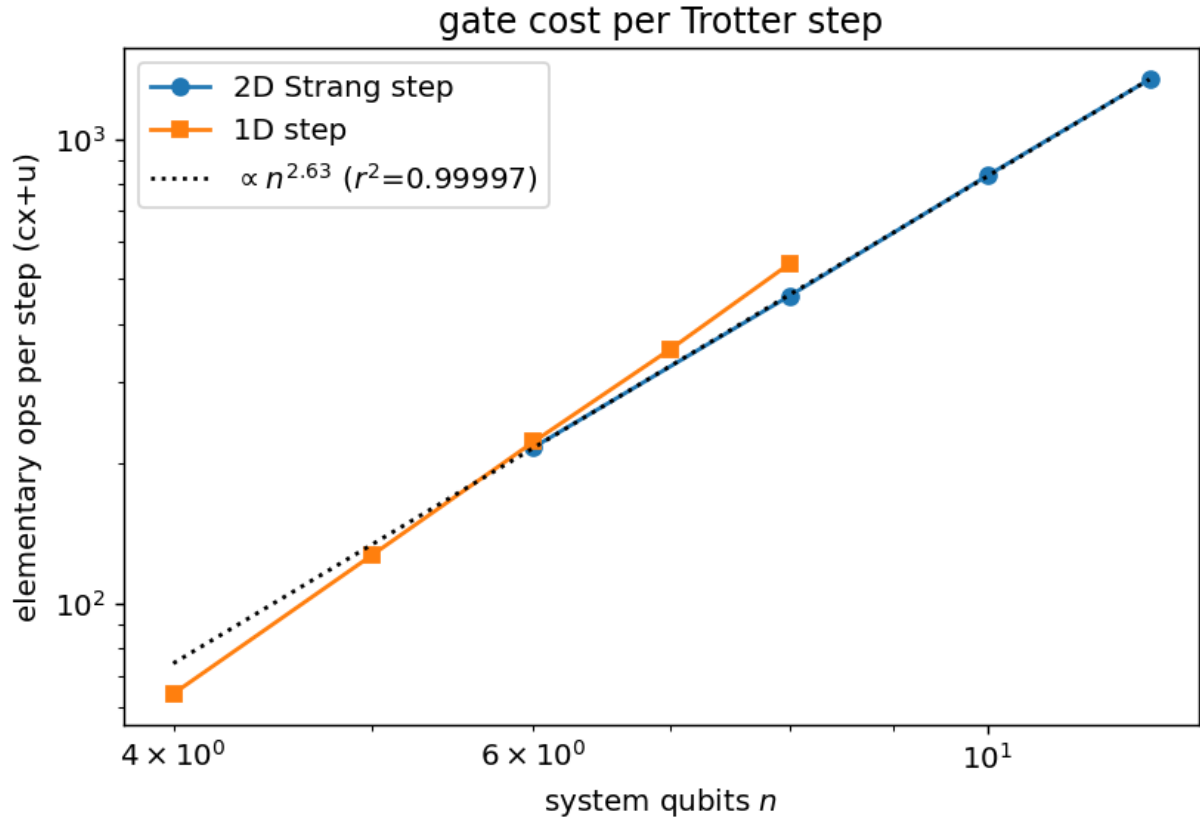


Figure 3.8 Gate cost per Trotter step after transpilation to the $\{CX, u\}$ basis, versus the number of system qubits n : 1D steps (squares) and 2D Strang steps (circles). The dotted line is the power-law fit $\propto n^{2.63}$ ($r^2 = 0.99997$) to the 2D series; the term count itself is exactly linear in n (Section 3.6).

for the CFS decay rate, γ for the injection coefficient, and $s = \sigma_j |D_{ji}| / (\kappa_j^2 \gamma)$ for the memory-source magnitude. Let $G = \text{diag}(g) > 0$ be *any* positive diagonal similarity, and set

$$u = \frac{g_{v_j}}{g_{\phi_j}}, \quad t = \frac{g_{\phi_j}}{g_{w_i}}, \quad \rho = u t = \frac{g_{v_j}}{g_{w_i}}. \quad (3.18)$$

The transformed generator $\tilde{A} = GAG^{-1}$ has unchanged diagonal, and the three relevant coupling families scale as follows.

- (i) *Injection pair* (v_j, ϕ_j) : entry $u\gamma$, reverse entry 0, so $\tilde{H}_1$ contains the principal 2×2 submatrix

$$\begin{bmatrix} 0 & u\gamma/2 \\ \bar{u}\bar{\gamma}/2 & -\beta \end{bmatrix} \implies \lambda_{\max}(\tilde{H}_1) \geq f(|u|\gamma, \beta), \quad (3.19)$$

with $f(x, \beta) := (\sqrt{\beta^2 + x^2} - \beta)/2$.

- (ii) *Memory-source pair* (w_i, ϕ_j) : entry ts analogously, giving $\lambda_{\max}(\tilde{H}_1) \geq f(|t|s, \beta)$.

- (iii) *Physical wave pair* (v_j, w_i) : elementwise, $\tilde{A}[v, w] = -iD_{ji}\rho$ and $\tilde{A}[w, v] = +iD_{ji}/\rho$ (using that D is real and $D^\dagger = -D_b$), so

$$\begin{aligned} \tilde{H}_1[v_j, w_i] &= \frac{iD_{ji}}{2}(\rho^{-1} - \rho) \\ \implies \lambda_{\max}(\tilde{H}_1) &\geq \frac{|D_{ji}|}{2}|\rho^{-1} - \rho|. \end{aligned} \quad (3.20)$$

All three are lower bounds on $\lambda_{\max}(\tilde{H}_1)$ by Cauchy interlacing (the eigenvalues of any principal submatrix interlace those of the full Hermitian matrix).

Case analysis. If $|\rho| \leq 1/2$ then $|\rho^{-1} - \rho| \geq 3/2$, so (iii) gives $\lambda_{\max}(\tilde{H}_1) \geq 3/(4h)$. Otherwise $|\rho| > 1/2$ implies

$$|u\gamma| \cdot |ts| = |\rho|\gamma s > \frac{\gamma s}{2} = \frac{\sigma_j |D_{ji}|}{2\kappa_j^2}, \quad (3.21)$$

hence $\max(|u|\gamma, |t|s) \geq \sqrt{\sigma_j/(2h\kappa_j^2)}$ and (i)–(ii) give $\lambda_{\max}(\tilde{H}_1) \geq f(\sqrt{\sigma_j/(2h\kappa_j^2)}, \beta)$. Therefore, for every positive diagonal scaling,

$$\lambda_{\max}(\tilde{H}_1) \geq \min \left\{ \frac{3}{4h}, \frac{\sqrt{\beta^2 + \sigma_j/(2h\kappa_j^2)} - \beta}{2} \right\}, \quad (3.22)$$

which is $\Theta(\sqrt{\sigma_j/h})$ as $h \rightarrow 0$ (the second branch is the minimum once $\sigma_j/(2h) \gg \beta^2$, i.e. for any fixed profile and small h). The bound is independent of γ and of the scaling G . The explicit balanced choice $\gamma^* = \sqrt{2\sigma_{\max}/h}$ attains the same rate from above: measured $\lambda^+ = 0.68\text{--}0.97$ across our configurations, versus the lower bound 0.18 at the 8×8 , $\sigma_{\max} = 0.5$ parameters—the bound is a rate, not a sharp constant. ■

Consistency check. The collapsed 1D form (Section 3.4.1) has no injection pair: the memory is eliminated and the damping is diagonal, so the lemma does not apply—and indeed $H_1 = -\text{diag}(\sigma) \preceq 0$ there.

CHAPTER 4

SEISMIC TRAVELTIME INVERSION WITH QUANTUM ANNEALING

A paper published in *Scientific Reports*⁴.

Hoang Anh Nguyen^{5,6} and Ali Tura⁵

4.1 Abstract

This study demonstrates the application of quantum computing based quantum annealing to seismic traveltime inversion, a critical approach for inverting highly accurate velocity models. The seismic inversion problem is first converted into a Quadratic Unconstrained Binary Optimization problem, which the quantum annealer is specifically designed to solve. We then solve the problem via quantum annealing method. The inversion is applied on a synthetic velocity model, presenting a carbon storage scenario at depths of 1000-1300 meters. As an application example, we also show the capacity of quantum computing to handle complex, noisy data environments. This work highlights the emerging potential of quantum computing in geophysical applications, providing a foundation for future developments in high-precision seismic imaging.

4.2 Introduction

Quantum computing is an emerging field with significant promise for various scientific and engineering disciplines. As we stand at the frontier of this technological revolution, early-stage research in quantum computing is crucial for the advancement of geophysics. Numerous studies have begun to explore the integration of quantum computing within this field, highlighting its immense and revolutionary potential [25]. For instance, quantum annealers can perform well in solving tomography optimization problems [26]. The quantum computing is applied for binary-value full waveform inversion, addressing issues related to velocity variations [27]. In the frequency domain, the seismic wave equation can be reduced to a system of linear equations, allowing for the application of quantum annealing [28]. Furthermore, it has been shown that quantum annealing impedance inversion with L1 norm regularization can dramatically enhance accuracy and anti-noise capabilities [29].

A quantum annealer is a specific type of quantum computer designed to solve optimization problems [30]. The quantum annealing process in quantum annealers can find the minimum energy state of a system,

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corresponding to the optimal solution of a given problem [31]. This process is achieved by utilizing quantum fluctuations, allowing the system to tunnel through energy barriers [32]. While there are various types of models in quantum computing [33, 34], this particular feature allows quantum annealing to efficiently explore complex energy landscapes, making them particularly well-suited for solving optimization problems.

Most previous attempts to address seismic problems using quantum annealers have primarily involved relatively simple models [35, 36]. For conventional approach by classical computers, the cross-well seismic inversion between boreholes can be computationally expensive [37], necessitating the development of new methods to tackle these challenges. Therefore, in this study, we aim to advance this line of research by applying quantum annealing to a complex problem: Seismic traveltime inversion of the velocity model between two boreholes. Our focus is on developing an inversion strategy that can accurately invert the velocity model with noisy data despite the limitation of the quantum hardware, specifically targeting carbon storage scenarios at depths of 1000-1300 meters. We use quantum annealer at D-Wave Advantage System, which has at least 5000 qubits [38]. Clearly, this travel-time inversion method can be applied to other acquisition geometries and data such as surface seismic, vertical seismic profile (VSP), earthquake or micro seismic data.

4.3 Results

We start the quantum annealing inversion process with exact traveltime data without noise and constant initial velocity model v_{ini} of 3475 m/s. The initial model and the results of the inverted model v_{inv} at each iteration obtained after the first 9 iterations indicate rapid convergence (Figure 4.1). Notably, in the first iteration, the carbon storage area is immediately identified with high precision.

The component-wise relative errors e_{ij} between the true $v_{true,ij}$ and final inverted velocity model $v_{final,ij}$ after 10 iterations is shown in Figure 4.2. The component-wise relative errors are calculated by $e_{ij} = |v_{inv,ij} - v_{true,ij}| / |v_{true,ij}|$. The most significant errors occurs in the shallow and deep regions with weakest constraints, yielding a maximum relative error value of about 0.326%. In contrast, the carbon storage area, spanning depths from 1100 to 1200 m, demonstrates exceptionally low errors due to the high ray coverage. The high-accuracy result underscores the effectiveness of the quantum annealing approach to the traveltime inversion.

For seismic traveltime inversion problem, the quantum annealing method and the classical linear least squares approach produce similar levels of error under ideal conditions, where data is free of noise [39]. However, real-world data often contain random noise, making it essential to assess the robustness of these methods under realistic conditions. Since we use the first-arrival traveltime, the data is relatively clean

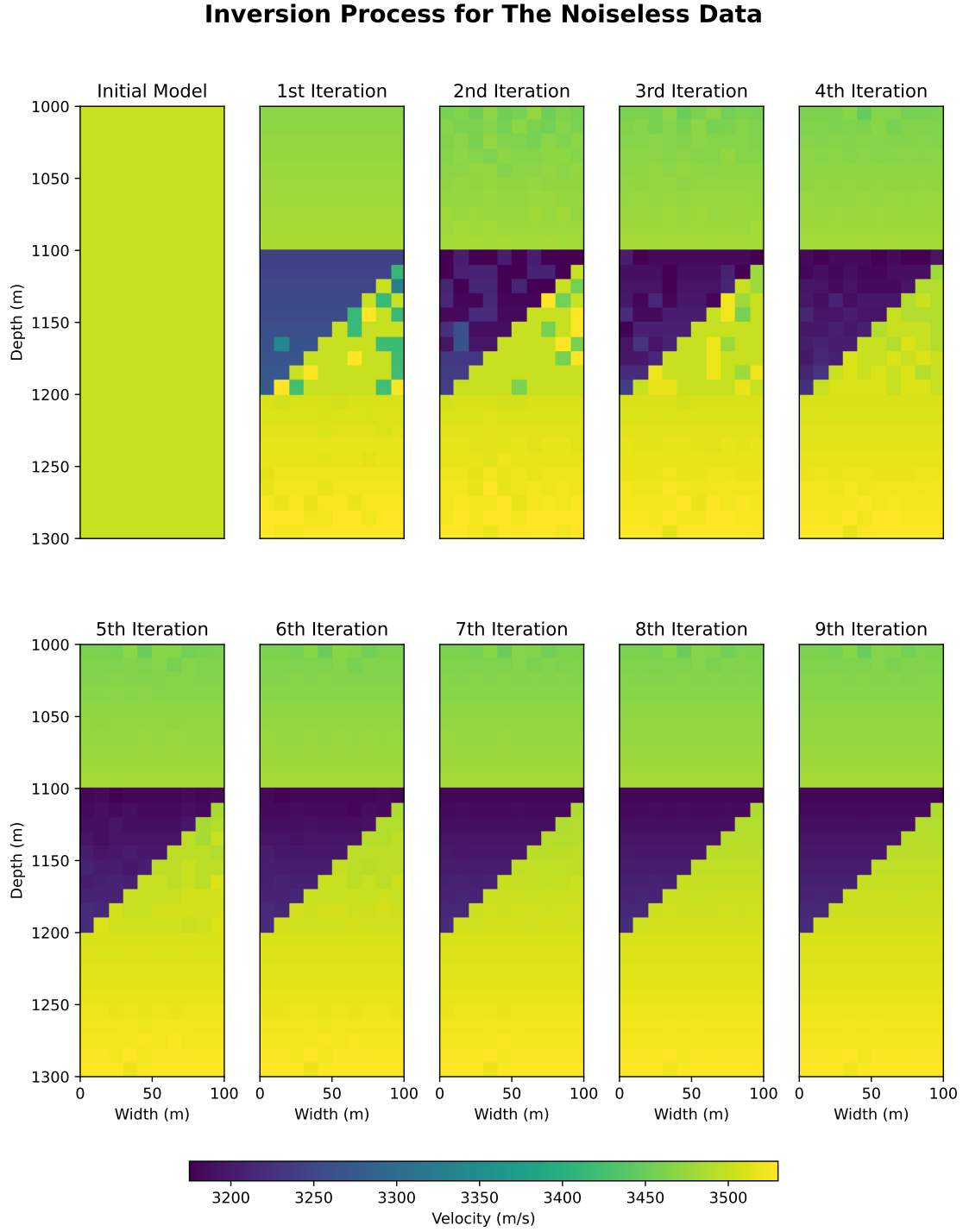


Figure 4.1 The starting model v_{ini} and the inverted velocity model v_{inv} over the first 9 iterations with exact, noise-free traveltimes data.

[40]. Therefore, we introduce the random noise in a range from 1% to 5% into the synthetic data. We compare the outcomes of the Tikhonov regularization least squares, serving as the standard method, with

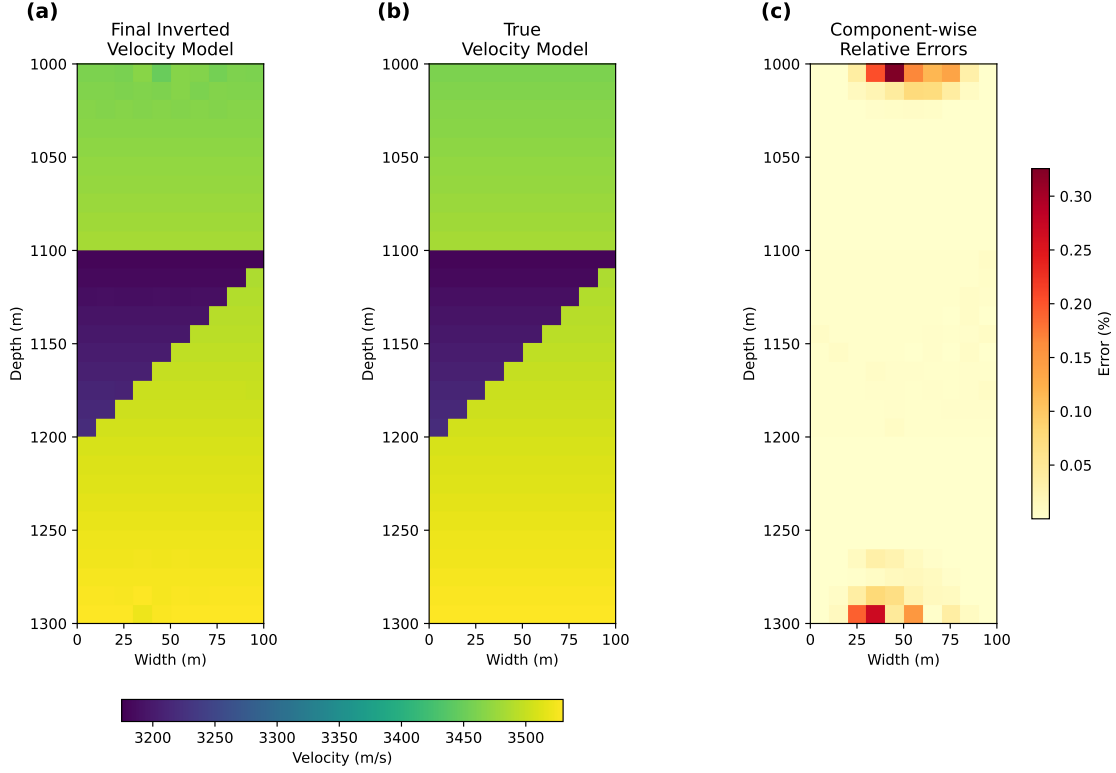
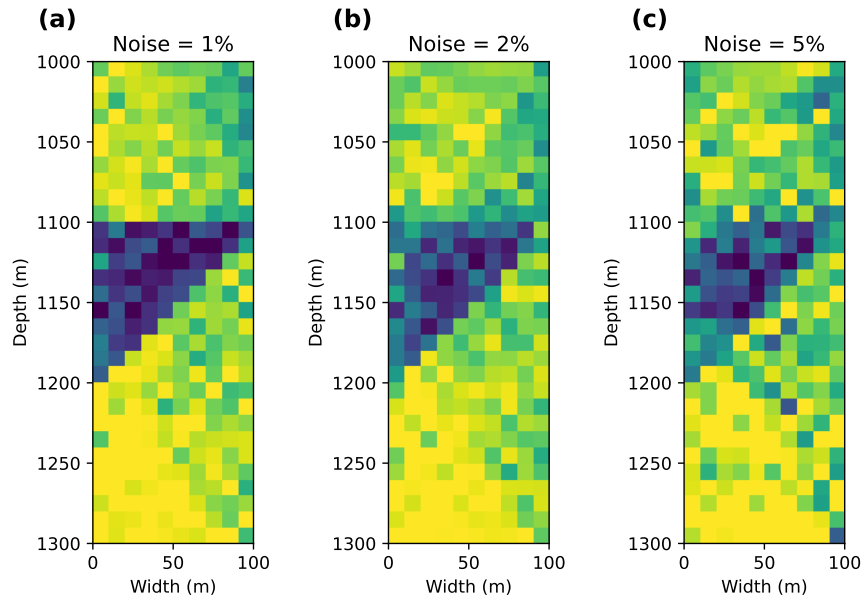


Figure 4.2 Velocity models and error: (a) final noiseless inverted model after 10 iterations $v_{inv,ij}$, (b) true model $v_{true,ij}$, and (c) the component-wise relative errors e_{ij} between the final inverted and true velocity model.

that of the quantum annealing method.

The results reveal a stark contrast in the sensitivity of these methods to noise. In this problem, the Tikhonov regularization linear least squares method is sensitive to noise (Figure 4.3a, b, c). At the noise level of 1% of the traveltime, while this method can identify the region of carbon storage, the deviation of the inverted model from true model is considerable. As the noise level increases to 2% and 5% of the traveltime, the linear least squares method almost fails to accurately determine the velocity model. This significant sensitivity limits its effectiveness in processing noisy seismic data, posing challenges for practical applications. In contrast, with the same noise, the quantum annealing method is more robust (Figure 4.3d, e, f). At the 1% noise level, the differences between the noise-free model and the results obtained are small. Differences start to appear primarily in shallow and deep areas where there is less constraint. Remarkably, at the 5% noise level, the quantum annealing method still effectively reproduces the velocity model. In areas with high ray coverage, these differences are small. This analysis underscores the potential of quantum annealing method for handling noisy seismic data more effectively than the classical linear least squares method.

Tikhonov Regularization Linear Least Squares



Quantum Annealing

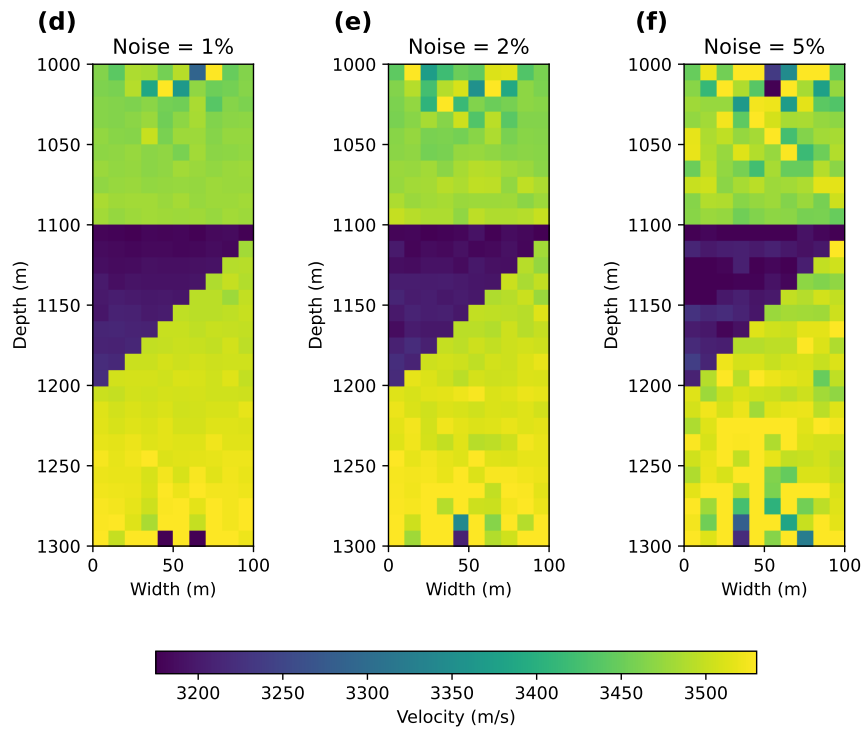


Figure 4.3 Traveltime inversion results with different noisy data using linear least squares (a, b, c) and quantum annealing method (d, e, f)

4.4 Discussion

In this study, we demonstrate several advancements to enhance the robustness and accuracy of seismic traveltime inversion results using the quantum annealing method. A significant improvement involves incorporating non-uniform source and receiver spacing. By using non-uniform spacing, we achieve better ray coverage of both shallow and deep regions, thereby increasing the overall accuracy of the final inverted velocity model.

The initial problem is broken down into smaller sub-problems, we also introduce the boundary L during the inversion process for better constraint and accommodating hardware limitations. While this paper does not utilize parallel processing, solving these sub-problems independently enables parallel execution, which can significantly reduce computational time.

For practical applications where noise is an unavoidable factor, our method demonstrates a remarkable ability to handle noisy data effectively and is highly suitable for the ill-conditioned problems, maintaining high efficiency and accuracy. The key advantage of the quantum annealing method is its ability to locate the global minimum of the objective function more effectively than classical methods, which are often trapped in local minima. Quantum tunneling allows the quantum system to explore a broader solution space and tunnel barriers that would hinder classical optimization methods [41]. This method can result in more accurate and reliable inversion outcomes, particularly in complex and heterogeneous environments. The quantum annealer solves problems using principles of quantum mechanics, which inherently depend on the operating environment of the machine. Consequently, running the same problem multiple times may yield slightly different results because of the probabilistic nature of quantum computations [42, 43]. However, we expect the advancements in quantum technology will decrease this variability and also reduce computational time.

4.5 Conclusion

The integration of quantum annealing in seismic traveltime inversion represents a potentially major advancement in geophysics. We utilise the open-access D-Wave Advantage quantum annealer to solve the inversion problem for a synthetic velocity model, representing a carbon storage scenario. The quantum annealing recursive method can give a good inverted velocity result after 10 iterations. Notably, the quantum method outperforms the classical linear least squares method in dealing with noisy data, where classical methods sometimes struggle. Despite promising results, the current state of quantum computing is not without its challenges. The slight variability in the results due to quantum noise [44–46] requires multiple runs to ensure reliable results. However, these challenges are expected to diminish as advances in quantum technology continue. This study suggests that quantum annealing could revolutionize seismic

inversion processes, offering more accurate solutions in scenarios where traditional methods are computationally intensive or even infeasible. As quantum computing matures, its applications in geophysics are likely to expand, encompassing more complex and larger-scale problems. This research not only underscores the potential of quantum annealing in seismic inversion but also sets the stage for future exploration into other areas of inverse problems. We expect that continued development of quantum computing technologies promises to unlock new capabilities, possibly making it an indispensable tool for new challenges.

4.6 Methods

4.6.1 Quantum Annealing

Quantum computing is rapidly emerging as a pivotal area of scientific and technological advancement, attracting considerable investment and interest due to its profound potential [47–49]. Unlike classical computers that use bits, which exist only in states of 0 or 1, quantum computers employ quantum bits, or qubits. Qubits possess unique properties such as superposition, entanglement, and interference [50–52], enabling them to perform certain complex computations beyond the capabilities of classical computers [53, 54]. Qubits can be constructed from various physical systems such as photons, trapped atoms, nuclear magnetic resonance, quantum dots, dopants in solids, and superconductors [55]. Previous research [56–60] has provided evidence that quantum computing possibly surpasses classical computers in terms of processing speed and efficiency for certain problems.

The quantum annealing process facilitates the finding of the global minimum of a cost function efficiently. This process can be described using the real-time Schrödinger equation [61]:

$$i\hbar \frac{d}{dt} |\Psi(t)\rangle = H(t) |\Psi(t)\rangle \quad (4.1)$$

where $|\cdot\rangle$ is the ket of the Dirac notation [62], i is the imaginary unit, t is time, $\hbar$ is the reduced Planck's constant, $\Psi(t)$ is the wave function, $|\Psi(t)\rangle$ is the quantum state vector, H is the Hamiltonian representing the total energy of the quantum system [63, 64]. If $\hbar$ is set as 1, Equation 4.1 becomes:

$$i \frac{d}{dt} |\Psi(t)\rangle = H(t) |\Psi(t)\rangle \quad (4.2)$$

The Hamiltonian in quantum annealing can be composed of two components [65, 66]:

$$H(t) = A(t)H_0 + B(t)H_1 \quad (4.3)$$

where H_0 is the initial Hamiltonian, representing a system with an initial ground state. H_1 is the final Hamiltonian, whose ground state encodes the solution to the optimization problem. $A(t)$ and $B(t)$ are time-dependent coefficients. $A(t)$ and $B(t)$ are set in a range of 0 to 1 so that $A(t_0) \gg B(t_0)$ at the initial

time t_0 and $B(t_1) \gg A(t_1)$ at the final time t_0 . During the process, $A(t)$ monotonically decreases, while $B(t)$ monotonically increases. At the start of the annealing process, $H(t) \approx H_0$. At the end of the annealing process, $H(t) \approx H_1$. Thus, the system transitions from the ground state of H_0 to the ground state of H_1 . If $H(t)$ changes sufficiently slowly, the state evolves adiabatically [67].

The problems are then often mapped onto a Quadratic Unconstrained Binary Optimization (QUBO) or Ising model [68]:

$$\text{QUBO: } \min_{x_j=0,1} \left(\sum_{j \leq k} x_j Q_{jk} x_k + C_1 \right) \quad (4.4)$$

$$\text{Ising: } \min_{s_j=\pm 1} \left(\sum_j h_j s_j + \sum_{j < k} J_{jk} s_j s_k + C_2 \right) \quad (4.5)$$

where j, k are indices, ranging over all qubits. In the QUBO model (Equation 4.4), Q_{jk} is the QUBO matrix with values $Q_{jk} \in \mathbb{R}$. The binary variable vector is $\mathbf{x}$ with $x_j \in \{0, 1\}$. In the Ising model (Equation 4.5), the problem is defined by the biases $h_j \in \mathbb{R}$ and the couplers $J_{jk} \in \mathbb{R}$, and the binary variable vector is $\mathbf{s}$ with $s_j \in \{-1, 1\}$. C_1 and C_2 are constants which do not affect the solution of the optimization problem. The Ising model and the QUBO model are mathematically equivalent, allowing them to be translated into each other. This equivalence provides a flexible approach to problem-solving, enabling the conversion of problems between these models based on the requirements and available tools. There are also tools, such as `ToQUBO.jl`, designed to convert standard problems into the QUBO format [69]. In this paper, we utilize the quantum annealer from D-Wave Advantage Systems [38] to employ direct quantum processing unit (QPU) methods for seismic travel inversion.

4.6.2 Seismic Data Acquisition

We construct a velocity model representing carbon storage applications, as shown in Figure 4.4a. This model spans a depth range from 1000 m to 1300 m and extends 100 m horizontally. The size of the grid cell for this model is 10 x 10 m. Within this model, the carbon storage structure is depicted as a wedge, starting from 1100 m and reaching a maximum depth of 1200 m. The velocity model is constructed with varying velocities to reflect real-world geological conditions. The average velocity within the carbon storage area ranges from 3180 to 3220 m/s, which is about 11% lower than the surrounding background velocity, which ranges from 3530 to 3640 m/s. Furthermore, the velocities increase with depth.

The uniform placement (Figure 4.4b) is commonly used in seismic data acquisition for simplicity in boreholes. However, this approach results in significantly lower ray coverage in the shallow and deep sections compared to the middle section. To address these limitations, in our survey, 20 pairs of sources and receivers are non-uniformly deployed within two boreholes (Figure 4.4c). The non-uniform deployment

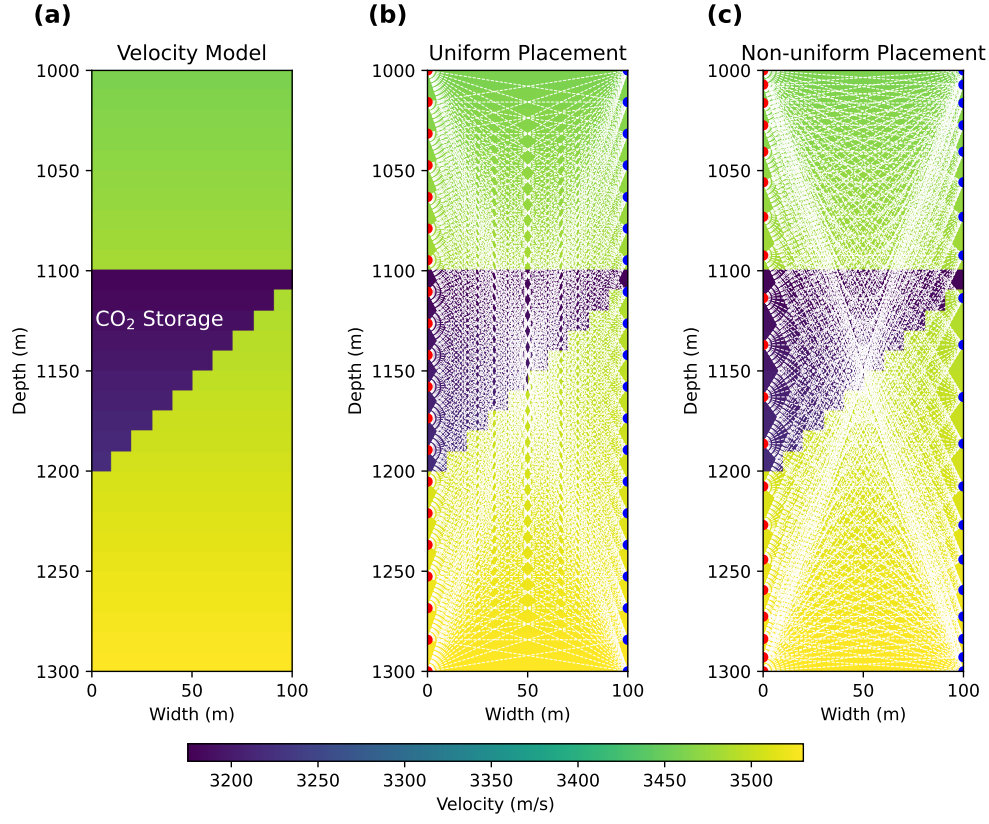


Figure 4.4 The carbon storage velocity model and ray coverage patterns. Red dots are sources, blue dots are receivers, and white lines represent the ray paths. (a) The synthetic velocity model with a wedge-shaped low-velocity carbon storage formation. (b) Ray coverage from sources and receivers placed in a uniform grid within two boreholes. (c) Ray coverage from sources and receivers placed in a non-uniform pattern, enhancing coverage and constraints for the quantum annealing process.

is designed to introduce more constraints for the quantum annealing process, thereby improving the overall accuracy of the seismic inversion. The sources and receivers of non-uniform placement are distributed according to a quadratic polynomial distribution.

4.6.3 Transforming Ray Equations to QUBO

The set of ray equations can be represented as [37]:

$$\mathbf{D}\mathbf{s} = \mathbf{T}, \quad (4.6)$$

where $\mathbf{D}$ is the matrix of distance increments d_j , $\mathbf{s}$ is the slowness vector, and $\mathbf{T}$ is the travel time vector. The size of $\mathbf{D}$ can be very large, therefore solving for $\mathbf{s}$ through matrix operations on $\mathbf{D}$ is computationally intensive. This challenge is exacerbated by the relatively sparse and random distribution of elements within $\mathbf{D}$. Consequently, alternative methods are required to solve these problems efficiently while maintaining accuracy. Quantum annealers can provide quantum metaheuristic algorithms to address this issue.

Equation 4.6 can be solved by minimizing the objective function:

$$f(\mathbf{s}) = \|\mathbf{D}\mathbf{s} - \mathbf{T}\|_2^2. \quad (4.7)$$

The objective function $f(\mathbf{s})$ computes the difference between the observed travel times and those predicted by the model given a slowness vector $\mathbf{s}$. Minimizing this function ensures that the model's predictions align as closely as possible with the observed data, thus achieving an optimal fit. Quantum annealers offer a direct approach to solving binary objective functions [70]:

$$f(\mathbf{q}) = \|\mathbf{A}^d \mathbf{q} - \mathbf{b}\|_2^2. \quad (4.8)$$

In this formulation, $\mathbf{q}$ is a binary vector, $\mathbf{A}^d$ is a real-valued matrix, and $\mathbf{b}$ is a real-valued vector. Because quantum computers are designed to solve QUBO problems, transforming real-valued variables to binary values is essential. However, the number of binary variables n_{binary} increases with the number of bits R used for fixed-point approximation, and it is related to the number of real-valued variables n_{real} as $n_{\text{binary}} = n_{\text{real}} \times R$. Higher values of R yield greater accuracy in representing the floating-point numbers, but the current limitations of quantum computer hardware restrict the number of qubits available. To address this issue without excessively increasing the number of binary variables, the initial velocity guess, variable boundaries [39] and recursive methods [71] are employed. The initial guess and the boundaries are used to rescale the range of the slowness vector $\mathbf{s}$ to a new vector $\mathbf{x}$ such that $x_i \in [0, 2)$, facilitating easier binary representation. Recursive methods are then applied to enhance the precision of floating-point divisions. These methods iteratively refine the estimate of $\mathbf{s}$, reducing the error at each step. The initial objective function Equation 4.7 can be reformulated as:

$$f(\mathbf{x}) = \|\mathbf{D}\mathbf{x} - \mathbf{b}\|_2^2, \quad (4.9)$$

where $\mathbf{b} = (\mathbf{T} + L\mathbf{I} - \mathbf{D}\mathbf{s}_0)/L$, L is the variable boundaries, $\mathbf{s}_0$ is the initial guess of the slowness vector, and $\mathbf{I}$ is the identity vector. The slowness vector $\mathbf{s}$ is then in the range of $[\mathbf{s}_0 - L\mathbf{I}; \mathbf{s}_0 + L\mathbf{I}]$. This range ensures that the solution space is adequately covered. To express this as a binary objective function, x_i is represented in binary form using the R bit fixed-point approximation:

$$x_i = \sum_{r=0}^{R-1} 2^{-r} q_r, \quad (4.10)$$

where $q_r \in \{0, 1\}$ is the value of the r -th bit. This transformation is essential for harnessing the computational power of quantum annealers, which are inherently designed to solve binary optimization problems. The new matrix $\mathbf{A}^d$ in Equation 4.8 is derived from $\mathbf{D}$ and R such that $\mathbf{D}\mathbf{x} = \mathbf{A}^d \mathbf{q}$. The QUBO matrix Q_{ij} in Equation 4.4 is then constructed from the given matrix $\mathbf{D}$ and the calculated vector $\mathbf{b}$ [70, 72]:

$$Q_{jj} = \sum_i A_{ij}(A_{ij} - 2b_j), \quad (4.11)$$

$$Q_{jk} = 2 \sum_i A_{ij}A_{ik}. \quad (4.12)$$

The QUBO matrix is directly input into the Quantum Processing Unit (QPU). The system utilizes `DWaveSampler()` to employ a D-Wave system as the sampler. Subsequently, `EmbeddingComposite()` manages the mapping between the problem and the D-Wave system's numerically indexed qubits, a process known as minor-embedding [73].

In this study, we perform traveltime inversion using $R = 3$ for 10 iterations with quantum annealing. The total number of real-valued variables of the problem is 300. Due to quantum hardware limitations, we break down the model into 30 layers with 10 variables each. This division reduces the complexity of each sub-problem, making it manageable for the quantum processor and allowing for better control of the boundary L . Since the problem from Equation 4.6 is depth-independent, we simplify the process by adjusting the system's coordinates at each iteration. The approach ensures that each layer is treated independently, reducing the overall complexity of the inversion. By implementing these techniques, we can efficiently solve large-scale traveltime inversion problems using quantum annealers.

4.6.4 Tikhonov Regularization Linear Least Squares Inversion

For ill-conditioned problems, small changes in $\mathbf{D}$ or $\mathbf{T}$ can lead to significant variations in the results [74]. To mitigate the effects of noise in the data, we employ Tikhonov regularization methods. The new objective function (Equation 4.7) can be expressed as a general regularized form [75]:

$$f_\lambda(\mathbf{s}) = \|\mathbf{Ds} - \mathbf{T}\|_2^2 + \lambda g(\mathbf{s}), \quad (4.13)$$

where λ is the regularization parameter controlling the trade-off between the data fidelity term $\|\mathbf{Ds} - \mathbf{T}\|_2^2$ and the regularization term $g(\mathbf{s})$. The Tikhonov regularization is flexible and allows different types of regularization functions. The standard Tikhonov regularization with L_2 -norm is in the form:

$$f_\lambda(\mathbf{s}) = \|\mathbf{Ds} - \mathbf{T}\|_2^2 + \lambda \|\mathbf{s}\|_2^2 \quad (4.14)$$

where $\|\mathbf{s}\|_2^2 = \mathbf{s}^T \mathbf{s}$ penalizes large values in the solution. Another form is the first-order Tikhonov regularization with a smoothness regularization:

$$f_\lambda(\mathbf{s}) = \|\mathbf{Ds} - \mathbf{T}\|_2^2 + \lambda \|D_1 \mathbf{s}\|_2^2 \quad (4.15)$$

where D_1 is the first-order difference operator which enforces smooth variation in $\mathbf{s}$ by penalizing large first derivatives. Similarly, second-order Tikhonov regularization penalizes the curvature of the solution:

$$f_\lambda(\mathbf{s}) = \|\mathbf{D}\mathbf{s} - \mathbf{T}\|_2^2 + \lambda\|D_2\mathbf{s}\|_2^2 \quad (4.16)$$

where D_2 is the second-order difference operator which enforces smooth curvature by penalizing large second derivatives. In general, $g(\mathbf{s})$ can be $g(\mathbf{s}) = \|\mathbf{s}\|_2^2$ for standard L_2 -norm regularization, $g(\mathbf{s}) = \|D_1\mathbf{s}\|_2^2$ for first-order smoothness, or $g(\mathbf{s}) = \|D_2\mathbf{s}\|_2^2$ for second-order smoothness. The choice of $g(\mathbf{s})$ depends on prior knowledge and the desired properties of the solution. Here, we use a custom Tikhonov regularization $g(\mathbf{s}) = \|\mathbf{s} - \mathbf{s}_0\|_2^2$, where $\mathbf{s}_0$ is the initial guess for the slowness and is chosen as the input for the quantum annealing process. The objective function is now expressed as:

$$f_\lambda(\mathbf{s}) = \|\mathbf{D}\mathbf{s} - \mathbf{T}\|_2^2 + \lambda\|\mathbf{s} - \mathbf{s}_0\|_2^2. \quad (4.17)$$

The solution to this regularized problem is given by:

$$\mathbf{s} = (\mathbf{D}^T\mathbf{D} + \lambda\mathbf{I})^{-1} (\mathbf{D}^T\mathbf{T} + \lambda\mathbf{s}_0). \quad (4.18)$$

4.6.5 Cost Analysis

Quantum annealing directly solves $f(\mathbf{q}) = \|\mathbf{A}^d\mathbf{q} - \mathbf{b}\|_2^2$ (Equation 4.8) by providing the solution binary vector $\mathbf{q}$. The quantum annealing process involves three sources of computational cost: preparing the binary problem, executing the annealing on the quantum hardware, and post-processing the results. The cost of preparing and post-processing is $\mathcal{O}(mn^2 + mnc + n^2c^2)$. For a single iteration of the loop, the cost of executing the annealing process on the quantum hardware is $\mathcal{O}(\mathcal{P}(cn))$. Here, m and n represent the rows and columns of the matrix $\mathbf{D}$, respectively. $\mathcal{P}(cn)$ is a polynomial term. The parameter $c = |\Theta| + 1$, where Θ used for representing the variables x_j as a fixed-point approximation in terms of power of 2 (Equation 4.10). The general form of the set Θ is defined as [72]:

$$\Theta = \{2^l : l \in [o, p] \wedge l, o, p \in \mathbb{Z}\}, \quad (4.19)$$

where l is contiguous integer values within the interval $[o, p]$, and o and p are the lower and upper bounds of the interval. In our scheme, instead of using a large range for o and p , we refine the precision of x_i iteratively over multiple loops to achieve higher accuracy. For a 3-bit fixed-point approximation with $x_i \in [0, 2)$, $o = 0$ and $p = -2$. For the K iterations, the total cost for process is:

$$\mathcal{O}(mn^2 + mnc + n^2c^2 + K \cdot \mathcal{P}(cn)), \quad (4.20)$$

For Tikhonov regularization methods with the direct solver (e.g., LU decomposition), the computational cost primarily depends on solving the modified linear system with the solution presented in Equation 4.18. The cost for finding $\mathbf{s}$ with a given λ is $\mathcal{O}(mn^2 + n^3)$. Selecting the optimal λ is crucial for achieving a balance between data fidelity and regularization. One commonly used technique is the L-curve

method, which plots the norm of the regularization term $\|R(\mathbf{s})\|$ against the residual norm $\|\mathbf{Ds} - \mathbf{T}\|$ on a log-log scale [76, 77]. For N_λ values of λ , the cost of solving the Tikhonov-regularized system for each λ is $\mathcal{O}(N_\lambda \cdot (mn^2 + n^3))$. In addition, the computation of the residual norm $\|\mathbf{Ds} - \mathbf{T}\|$ for each λ involves matrix-vector multiplications, contributing a cost of $\mathcal{O}(N_\lambda \cdot mn)$. If we want to find the optimal λ automatically, the curvature of the L-curve must be computed. This requires additional operations such as numerical differentiation and curvature estimation, which are negligible compared to the cost of solving the regularized systems. Thus, the total cost for the L-curve method, including the automatic selection of λ , is:

$$\mathcal{O}(N_\lambda \cdot (mn^2 + n^3)). \quad (4.21)$$

From Equations 4.20 and 4.21, achieving a computational cost of $\mathcal{O}(\mathcal{P}(cn)) < \mathcal{O}(n^3)$ would enable quantum annealing to significantly accelerate problem-solving. Research efforts, such as those utilizing multi-qubit correction techniques [78], aim to realize this improvement. These approaches can potentially achieve a substantial reduction in computational complexity. This advancement would facilitate rapid convergence to the global minimum and unlock considerable performance gains.

4.7 Acknowledgments

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4.8 Author Contributions

H. A. Nguyen contributed to the methodology, formal analysis, writing of the manuscript, providing professional oversight and manuscript revision. A. Tura was responsible for conceptualization and manuscript revision.

4.9 Competing Interests

The authors declare no competing interests.

4.10 Data Availability

Access to the D-Wave System in this study is available at cloud.dwavesys.com. The code related to this research be accessed at <https://github.com/x-repos/traveltime-inversion-annealing>.

CHAPTER 5

ACCELERATING PHYSICS-INFORMED NEURAL NETWORKS FOR FULL WAVEFORM INVERSION USING A HYBRID QUANTUM-CLASSICAL FINITE-BASIS ARCHITECTURE

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5.1 Abstract

Full waveform inversion (FWI) reconstructs heterogeneous material properties from receiver data but remains computationally demanding. Physics-informed neural networks (PINNs) and their domain-decomposed variants (FBPINNs) offer a mesh-free alternative but face convergence challenges when representing complex velocity fields. We present a hybrid quantum-classical FBPINN for acoustic FWI, bringing together quantum computing and classical machine learning, in which the decomposed wavefield network and the global velocity network are implemented as classical-to-quantum pipelines terminating in parameterized quantum circuits (PQCs). The PQCs are realized as differentiable JAX statevector simulators, enabling end-to-end automatic differentiation through the classical PINN, the quantum circuit, and the physics-informed loss. On a geophysical anomaly benchmark, the quantum hybrid reaches a lower L1 velocity error than the primary classical FBPINN baseline in approximately $8\times$ fewer training iterations, despite using approximately 33% fewer trainable parameters, and it outperforms all 15 classical hyperparameter variants tested. A second benchmark (checkerboard) demonstrates the generality of the inversion pipeline, confirming that the quantum hybrid architecture can recover structured spatial variations beyond the localized anomaly benchmark. Our framework is broadly applicable to wave-based inverse problems beyond geophysics, including medical ultrasound tomography and non-destructive evaluation.

Index Terms: Quantum Computing, Full Waveform Inversion, Wave Propagation, Parameterized Quantum Circuit, PINNs.

5.2 Introduction

Full waveform inversion (FWI) is a powerful technique for reconstructing material properties from recorded waveform data by exploiting the full information content of wave signals, including amplitude,

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phase, and frequency. Originally developed in the context of exploration geophysics for subsurface velocity imaging [79], FWI has since become central to a wide range of disciplines. In global seismology, it enables high-resolution imaging of crustal and mantle structures [80, 81]; in hydrocarbon exploration and CO₂ monitoring, it provides detailed velocity models for reservoir characterization [82] and is increasingly applied to high-density datasets acquired via distributed acoustic sensing [83]. Beyond geophysics, FWI principles have found growing applications in medical imaging, where ultrasound computed tomography employs waveform inversion to reconstruct tissue properties for breast cancer detection and characterization [84, 85], and in photoacoustic tomography, where wave-based inversion recovers optical absorption maps from acoustic measurements [86]. Non-destructive evaluation of engineering structures similarly leverages full waveform methods to detect defects and characterize material properties [87]. Despite its versatility across these domains, FWI remains computationally demanding, requiring iterative solutions of the forward wave equation and the computation of gradients through adjoint-state methods, often at substantial cost for large-scale and high-frequency problems. These computational demands have further motivated machine-learning surrogates for rapid waveform modeling and inversion [88].

Physics-informed neural networks (PINNs) have emerged as a mesh-free alternative for solving partial differential equations (PDEs) by minimizing PDE residuals at collocation points [89], approximating solutions without traditional numerical discretization. This paradigm has been extended to inverse problems, where unknown model parameters are optimized alongside the solution. Recent works have applied PINNs to wave-related problems [90–92], offering a fundamentally different approach to the classical adjoint-state framework. In particular, Rasht-Behesht *et al.* [93] demonstrated the application of PINNs to FWI, achieving accurate velocity reconstruction from receiver data by jointly optimizing the wavefield and velocity networks. Physics-informed formulations have since been applied to seismic inversion through the acoustic wave equation [94], while purely data-driven networks instead learn a direct mapping from seismic data to subsurface velocity models [95, 96]. However, standard PINNs face well-documented challenges when applied to problems involving high-frequency wave propagation, multi-scale physics, and large computational domains. The spectral bias of neural networks, their tendency to learn low-frequency features before high-frequency ones, can severely limit convergence in wave propagation problems. To address these limitations, finite-basis physics-informed neural networks (FBPINNs) were introduced by Moseley *et al.* [97], employing a domain decomposition strategy in which the global solution is represented as a sum of locally supported basis functions, each parameterized by an independent neural network. This approach effectively mitigates spectral bias, improves parallel scalability, and enables the solution of problems that are intractable for standard PINNs. To date, however, no work has applied the domain decomposition advantages of FBPINNs to waveform inversion.

In parallel with advances in classical scientific machine learning, quantum computing has attracted growing interest for its potential to accelerate computational tasks across diverse domains. Parameterized quantum circuits (PQCs), also known as variational quantum circuits (VQCs), have emerged as a near-term approach to quantum machine learning, operating within the constraints of current noisy intermediate-scale quantum (NISQ) devices [98]. PQCs encode classical data into quantum states through feature maps, process the information through parameterized unitary operations, and extract classical outputs via measurement. Several studies have explored the integration of quantum circuits with neural networks, forming hybrid quantum-classical architectures that leverage the expressive power of quantum Hilbert spaces while maintaining the trainability and flexibility of classical networks [99–101].

The intersection of quantum computing and PINNs has begun to receive attention, with recent works investigating quantum physics-informed neural networks (QPINNs) for solving differential equations [102]. Leong *et al.* [103] benchmarked hybrid quantum physics-informed neural networks (HQPINNs) for high-speed flow problems, demonstrating that these architectures offer competitive accuracy and stability with a reduced parameter cost compared to classical PINNs. While they found that pure quantum models struggle with non-harmonic features like shocks, the hybrid approach successfully mitigates these artifacts by leveraging classical sub-networks. Similarly, Berger *et al.* [104] proposed trainable embedding quantum physics-informed neural networks (TE-QPINNs) for solving nonlinear PDEs, including the Navier-Stokes equations. Their approach achieved superior results compared to classical PINNs while maintaining the same number of parameters, highlighting the potential for more efficient optimization in high-dimensional parameter spaces. These studies suggest that quantum circuits can serve as powerful function approximators within the PINN framework, potentially offering advantages in expressivity and parameter efficiency for forward PDE problems. Complementing these variational formulations, direct gate-based quantum algorithms have also been developed to solve PDEs without a learned model, with explicit circuit constructions for the acoustic wave equation and related operators [105] and a demonstration of one-dimensional wave-equation simulation on near-term quantum hardware [106]. When extending quantum methods to geophysical inverse problems, research has thus far primarily relied on alternative computing paradigms. For instance, quantum annealing has recently been applied to seismic traveltime tomography [107] and to pre-stack and post-stack seismic inversion [108], with the inverse problem cast as a least-squares objective, mapped to a quadratic unconstrained binary optimization (QUBO) or Ising Hamiltonian formulation, and solved using a quantum annealer. More recently, hybrid quantum neural networks have also been explored for pre-stack and post-stack seismic inversion within physics-informed encoder-decoder architectures [109]. However, the application of gate-based hybrid quantum-classical neural architectures to highly nonlinear inverse problems like FWI, where both the wavefield and the

velocity model are represented by learned networks and coupled through the acoustic wave equation, remains largely unexplored.

In this work, we present, to the best of our knowledge, the first application of gate-based quantum computing to full waveform inversion: a hybrid quantum-classical FBPINN for acoustic FWI. The framework represents both the decomposed wavefield network and the global velocity network by classical-to-quantum pipelines: classical pre-networks map the input coordinates into low-dimensional latent spaces matched to the number of qubits, the PQC’s process the encoded features through parameterized rotations and entangling gates, and classical output layers rescale the bounded circuit outputs into physical units. To isolate the architectural benefits of the quantum integration from the hardware noise of current near-term devices, the PQC’s are implemented as JAX-native statevector simulators. This approach enables exact, end-to-end automatic differentiation through the classical PINN, the quantum circuit, and the physics-informed loss, avoiding the prohibitive computational overhead of parameter-shift rules required on physical hardware. This design combines the scalability advantages of FBPINN domain decomposition with the structured, bounded expressivity of PQC-based parameterizations. We evaluate the architecture on two geophysical benchmarks and compare it against a purely classical FBPINN baseline as well as 15 classical hyperparameter variants. In the following sections, we present the comparative performance of our hybrid architecture, discuss its broader implications for scientific machine learning, and detail the underlying mathematical and algorithmic methodology.

5.3 Methodology

5.3.1 Acoustic wave equation

Two-dimensional acoustic wave propagation in a heterogeneous medium with spatially varying wave speed $\alpha(\mathbf{x})$ and constant density is governed by the acoustic wave equation

$$\frac{\partial^2 \phi(\mathbf{x}, t)}{\partial t^2} = \alpha^2(\mathbf{x}) \nabla^2 \phi(\mathbf{x}, t) + s(\mathbf{x}, t), \quad (5.1)$$

where $\phi(\mathbf{x}, t)$ is the scalar displacement potential, ∇^2 is the spatial Laplacian, and $s(\mathbf{x}, t)$ is the seismic source. The spatial gradient of ϕ yields the physical displacement field.

In the context of FWI, the objective is to reconstruct the unknown wave speed $\alpha(\mathbf{x})$ from surface observations. Within our PINN architecture, to avoid the numerical challenges of explicitly modeling the source singularity $s(\mathbf{x}, t)$, the source is instead encoded through the initial conditions. This is achieved by specifying the wavefield at two closely spaced early times, derived from a spectral-element forward simulation [110]. The top boundary of the domain acts as a free surface where the acoustic pressure vanishes, which is mathematically equivalent to $\nabla^2 \phi = 0$ in this displacement-potential formulation.

Accordingly, the source-free form of Equation 5.1 is enforced as a PDE residual at interior collocation points, while the initial states and free-surface boundary conditions are imposed as soft constraints within the training loss. The measured receiver data are incorporated as an additional data-misfit term, penalizing the discrepancy between the predicted wavefield at receiver locations and the observed seismograms. The complete composite loss function used to train the FBPINN is detailed in the Hybrid architecture section.

5.3.2 Finite-basis physics-informed neural network

PINNs [89] approximate solutions of PDEs by a neural network $u_\theta(\mathbf{x})$ and train θ to minimize the PDE residual at a set of collocation points $\{\mathbf{x}_i\}_{i=1}^N$ drawn from the domain. Writing the governing PDE in residual form $\mathcal{N}[u](\mathbf{x}) = 0$, the PDE loss is

$$\mathcal{L}_{\text{PDE}} = \frac{1}{N} \sum_{i=1}^N |\mathcal{N}[u_\theta(\mathbf{x}_i)]|^2, \quad (5.2)$$

with spatial and temporal derivatives computed by automatic differentiation of the network with respect to its inputs. Boundary conditions, initial conditions, and observational data are added as additional weighted loss terms (the full composite loss used for FWI is given in the Hybrid architecture subsection below).

Standard PINNs, however, are known to suffer from spectral bias, a tendency to learn low-frequency features before high-frequency ones, which severely limits their effectiveness in multi-scale wave problems such as FWI, where both the velocity field and the wavefield exhibit sharp transitions and oscillatory structure.

To overcome this limitation, we adopt the FBPINN framework of Moseley *et al.* [97]. The global domain Ω is partitioned into K overlapping rectangular subdomains $\{\Omega_k\}$, each equipped with an independent neural network $u_{\theta_k}(\mathbf{x})$. The global solution is reconstructed as a partition-of-unity sum,

$$u(\mathbf{x}) = \sum_{k=1}^K w_k(\mathbf{x}) u_{\theta_k}(\mathbf{x}), \quad (5.3)$$

where the window functions w_k are smooth cosine bumps supported on Ω_k with $\sum_k w_k(\mathbf{x}) = 1$ for all $\mathbf{x} \in \Omega$. Each local network only needs to represent the solution's behavior on its own subdomain, with two consequences: (i) the local frequency content each network must learn is bounded by the subdomain size, alleviating spectral bias; and (ii) training is embarrassingly parallel across subdomains. The physics-informed loss is evaluated on the reconstructed global solution, so gradients flow through the window-weighted sum back to the individual subdomain networks.

For the wavefield $\phi(x, z, t)$ we use a three-dimensional subdomain decomposition (n_x, n_z, n_t) , $5 \times 3 \times 5$ in the anomaly baseline with subdomain overlap 0.35. The velocity network $\alpha(x, z)$ is not decomposed; it is a single global network, since the velocity field typically varies more smoothly than the wavefield and does

not require local representation.

5.3.3 Parameterized quantum circuits

We first summarize the quantum-computing concepts used by the hybrid architecture. A single qubit is a two-level quantum system with state

$$|\psi\rangle = c_0 |0\rangle + c_1 |1\rangle, \quad |c_0|^2 + |c_1|^2 = 1, \quad (5.4)$$

where $c_0, c_1 \in \mathbb{C}$ are probability amplitudes. An n -qubit register is represented by a normalized statevector in $\mathbb{C}^{2^n}$,

$$|\psi\rangle = \sum_{k=0}^{2^n-1} c_k |k\rangle, \quad \sum_{k=0}^{2^n-1} |c_k|^2 = 1. \quad (5.5)$$

The parameterized single-qubit rotations used in this work are

$$R_k(\theta) = e^{-i\theta \sigma_k/2}, \quad k \in \{x, y, z\}, \quad (5.6)$$

with explicit matrices

$$\begin{aligned} R_x(\theta) &= \begin{pmatrix} \cos \frac{\theta}{2} & -i \sin \frac{\theta}{2} \\ -i \sin \frac{\theta}{2} & \cos \frac{\theta}{2} \end{pmatrix}, \\ R_y(\theta) &= \begin{pmatrix} \cos \frac{\theta}{2} & -\sin \frac{\theta}{2} \\ \sin \frac{\theta}{2} & \cos \frac{\theta}{2} \end{pmatrix}, \\ R_z(\theta) &= \begin{pmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{pmatrix}. \end{aligned} \quad (5.7)$$

The composition $R_z(\theta_3)R_y(\theta_2)R_x(\theta_1)$ spans $\text{SU}(2)$ and can prepare any single-qubit pure state up to a physically irrelevant global phase. Entanglement between neighbouring qubits is introduced by the controlled-NOT (CNOT) gate,

$$\text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad (5.8)$$

which flips the target qubit if and only if the control qubit is in state $|1\rangle$.

A PQC [100] applies a trainable unitary $U(\boldsymbol{\theta}_q)$ to the initial state $|0\rangle^{\otimes n}$ and returns a scalar expectation value of a Hermitian observable, yielding a smooth, differentiable function of the circuit parameters $\boldsymbol{\theta}_q$. Its expressibility depends on the circuit depth, qubit connectivity, and gate set [111].

The PQC employed in this work (Figure 5.1) operates on n qubits with L variational layers. In the data-embedding layer, starting from $|0\rangle^{\otimes n}$, each qubit i receives an $R_y(\beta_i z_i)$ rotation that encodes the i -th component of the input vector $\mathbf{z} \in \mathbb{R}^n$ as a rotation angle (angle encoding [112]), scaled by a trainable basis parameter β_i :

$$|\psi_{\text{enc}}(\mathbf{z}; \boldsymbol{\beta})\rangle = \bigotimes_{i=0}^{n-1} R_y(\beta_i z_i) |0\rangle. \quad (5.9)$$

This is followed by L variational layers. Each layer $l = 1, \dots, L$ applies a general single-qubit rotation $R_z(\theta_{l,i}^{(3)}) R_y(\theta_{l,i}^{(2)}) R_x(\theta_{l,i}^{(1)})$ to every qubit i , introducing $3n$ trainable parameters per layer, and then applies a linear CNOT ladder $\text{CNOT}(i, i+1)$ for $i = 0, \dots, n-2$ to entangle neighboring qubits. The total number of trainable gate parameters is $3nL$, plus n basis parameters from the embedding layer. In the measurement layer, the Pauli- Z expectation values $\langle Z_i \rangle$ on each qubit are averaged to produce a single scalar output

$$f(\mathbf{z}; \boldsymbol{\beta}, \boldsymbol{\theta}_q) = \frac{1}{n} \sum_{i=0}^{n-1} \langle \psi(\mathbf{z}; \boldsymbol{\beta}, \boldsymbol{\theta}_q) | Z_i | \psi(\mathbf{z}; \boldsymbol{\beta}, \boldsymbol{\theta}_q) \rangle \in [-1, 1]. \quad (5.10)$$

The three design choices of this output head, angle encoding with trainable scales, a hardware-efficient variational block, and a local averaged read-out, are each motivated below; their representational consequences are taken up quantitatively in the Discussion and Section 5.8. The encoding rotation $R_y(\beta_i z_i)$ rotates qubit i by an angle linear in the latent component z_i , so each coordinate enters the state through a bounded, smooth map whose dependence on z_i is periodic in the encoded angle. Making the per-qubit scale β_i a trainable parameter lets the circuit adapt the frequencies through which the latent variables are read out, a role analogous to learned input scalings in classical coordinate networks [112]; for angle-encoded circuits this produces a finite-frequency (Fourier) structure in $\mathbf{z}$ whose accessible frequencies are set by $\boldsymbol{\beta}$ [113].

Each variational layer is a hardware-efficient ansatz [114, 115]: a universal single-qubit rotation $R_z R_y R_x$ on every qubit, which spans $\text{SU}(2)$ and so imposes no a priori restriction on the local state ($3n$ angles per layer), interleaved with a linear nearest-neighbour CNOT ladder. We adopt this structure because it is built only from one- and two-qubit gates on adjacent wires, the native low-overhead operations on most physical architectures [98, 115], and because the expressibility and entangling capacity of this gate family have been characterised at modest depth [100, 111]. The averaged observable $f = \frac{1}{n} \sum_i \langle Z_i \rangle$ is correspondingly a sum of single-qubit (weight-one) Pauli terms rather than a global, high-weight observable; as a network head each $\langle Z_i \rangle \in [-1, 1]$ yields a single scalar bounded in $[-1, 1]$ that the subsequent affine layer rescales to physical units. The compactness of this $3nL + n$ -angle parameterization comes with a known trainability caveat: deep or globally measured hardware-efficient circuits can exhibit barren plateaus (vanishingly small gradients) [116], and the severity of this effect depends on the locality of the cost [117]. The small width ($n = 4$), shallow depth ($L = 2$), and local read-out used here are the regime in which such gradients are least suppressed, and the differentiable statevector simulation of backpropagates through the circuit directly, without parameter-shift estimators.

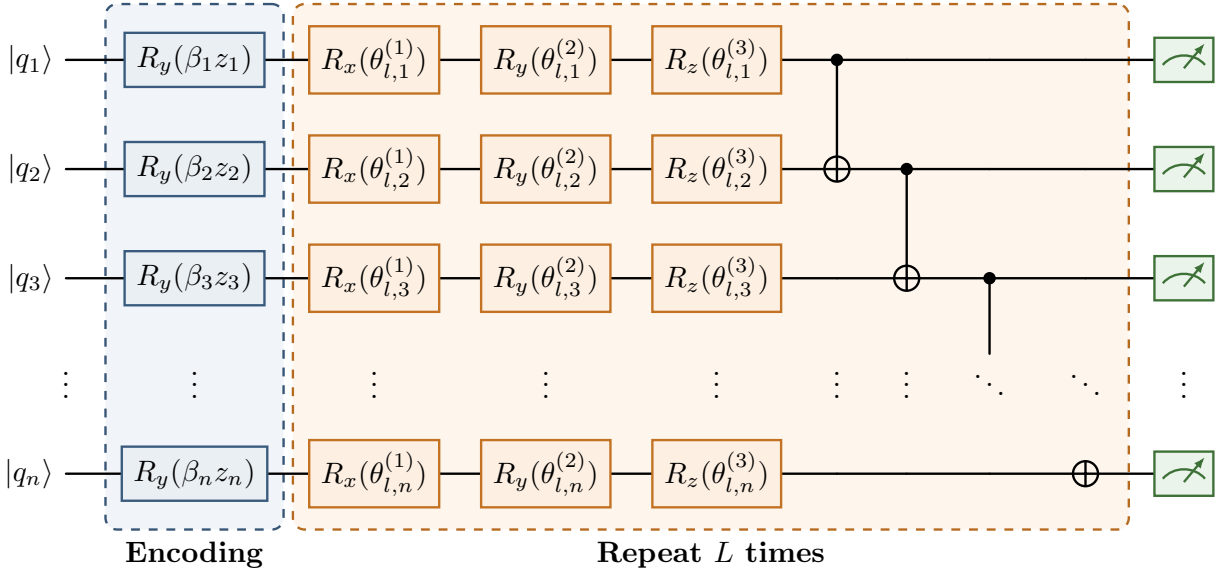


Figure 5.1 Multilayer PQC $\mathcal{Q}$. Each input feature z_i is angle-encoded via an $R_y(\beta_i z_i)$ rotation (blue encoding layer), where β_i is a trainable scaling parameter. Variational layers of single-qubit rotations R_x , R_y , R_z followed by a linear CNOT ladder are repeated L times (orange block). Each qubit is measured in the Pauli-Z basis (green meters), and the outputs are averaged to yield a scalar in $[-1, 1]$.

These arguments concern inductive bias, parameter efficiency, and trainability, not any computational speedup. The circuit is evaluated by exact JAX statevector simulation and, because its output is a finite-frequency function of $\mathbf{z}$, the realized function class is in principle classically representable; we make no claim of quantum advantage. We further note that capacity is not concentrated in one wide entangled register: the architecture instantiates 76 independent four-qubit PQCs, one per wavefield subdomain $(5 \times 3 \times 5)$ plus one for the global velocity network, each carrying its own angles $(\boldsymbol{\beta}, \boldsymbol{\theta}_q)$, so the per-circuit width stays fixed at four qubits as the subdomain decomposition is refined.

5.3.4 JAX-based statevector simulator

We implement the PQC as a lightweight statevector simulator in JAX [118] rather than relying on an external framework such as PennyLane [119] or Qiskit [120]. The simulator represents the quantum state as a 2^n -dimensional complex vector and applies gate operations via tensor contractions: single-qubit rotation gates (R_x , R_y , R_z) are constructed as 2×2 unitary matrices and applied with `jnp.tensordot` on the reshaped state tensor, and CNOT gates are realized through index permutation and conditional flipping on the control-target subspace. Expectation values are computed analytically from the statevector amplitudes, without stochastic sampling, as in PennyLane’s `default.qubit` backend.

On near-term quantum hardware, gradients of PQC outputs with respect to gate parameters are typically computed via the parameter-shift rule [121, 122], which evaluates the circuit at two shifted

parameter values per gate and therefore requires $2p$ circuit evaluations per gradient for a circuit with p parameters. In classical statevector simulation, each gate application is already a standard differentiable operation, so embedding the circuit in JAX offers two concrete advantages. First, `jax.grad` and `jax.jvp` backpropagate through the full hybrid pipeline (classical PINN layers, quantum circuit, and loss function) without parameter-shift rules or cross-framework bridging. Second, the entire training loop can be compiled into a single optimized XLA program via `jax.jit`, eliminating Python-level dispatch and framework-bridging overhead. The trade-off is the exponential cost of representing the quantum state explicitly: the simulator holds all 2^n complex amplitudes, giving $\mathcal{O}(2^n)$ memory, and applies $\mathcal{O}(Ln)$ gates that each sweep the full state, giving $\mathcal{O}(Ln2^n)$ computation. This limits the approach to moderate qubit counts; for the circuit sizes used here ($n \leq 8$, $L \leq 6$), this cost is negligible compared with the classical network and loss evaluation. We verify numerical correctness by comparing parameter gradients $\partial f / \partial \theta_q$ against PennyLane’s `default.qubit` backend [119] across eight circuit configurations (2–8 qubits, 1–6 layers); the maximum absolute deviation is below 2×10^{-7} , consistent with `float32` machine precision (Figure 5.2).

5.3.5 Hybrid quantum-classical architecture

The hybrid architecture (Figure 5.3) couples two neural networks with physics-informed constraints: a wavefield network $\phi(x, z, t)$, decomposed across subdomains, and a velocity network $\alpha(x, z)$, which is a single global network. In the classical baseline, both networks are fully-connected tanh networks. In the quantum hybrid, both networks are classical-to-quantum pipelines.

Defining the latent vector

$$\mathbf{z}(\mathbf{x}) = \mathcal{N}(\mathbf{x}; \boldsymbol{\theta}_c), \quad (5.11)$$

each classical-to-quantum pipeline has the form

$$g(\mathbf{x}) = w_{\text{out}} \mathcal{Q}(\mathbf{z}(\mathbf{x}); \boldsymbol{\beta}, \boldsymbol{\theta}_q) + b_{\text{out}}, \quad (5.12)$$

where $\mathcal{N}(\mathbf{x}; \boldsymbol{\theta}_c)$ is a classical fully-connected tanh sub-network with parameters $\boldsymbol{\theta}_c$, $\mathcal{Q}(\cdot; \boldsymbol{\beta}, \boldsymbol{\theta}_q)$ is the PQC with encoding scales $\boldsymbol{\beta}$ and variational circuit parameters $\boldsymbol{\theta}_q$ (returning a scalar in $[-1, 1]$), and $(w_{\text{out}}, b_{\text{out}})$ are learnable output weights. For the anomaly benchmark experiments we use $n = 4$ qubits and $L = 2$ variational layers in both networks. The velocity network output is further mapped to physical velocity units via

$$\alpha(x, z) = \alpha_{\text{bg}} + A \tanh(g(x, z)) \cdot m(x, z), \quad (5.13)$$

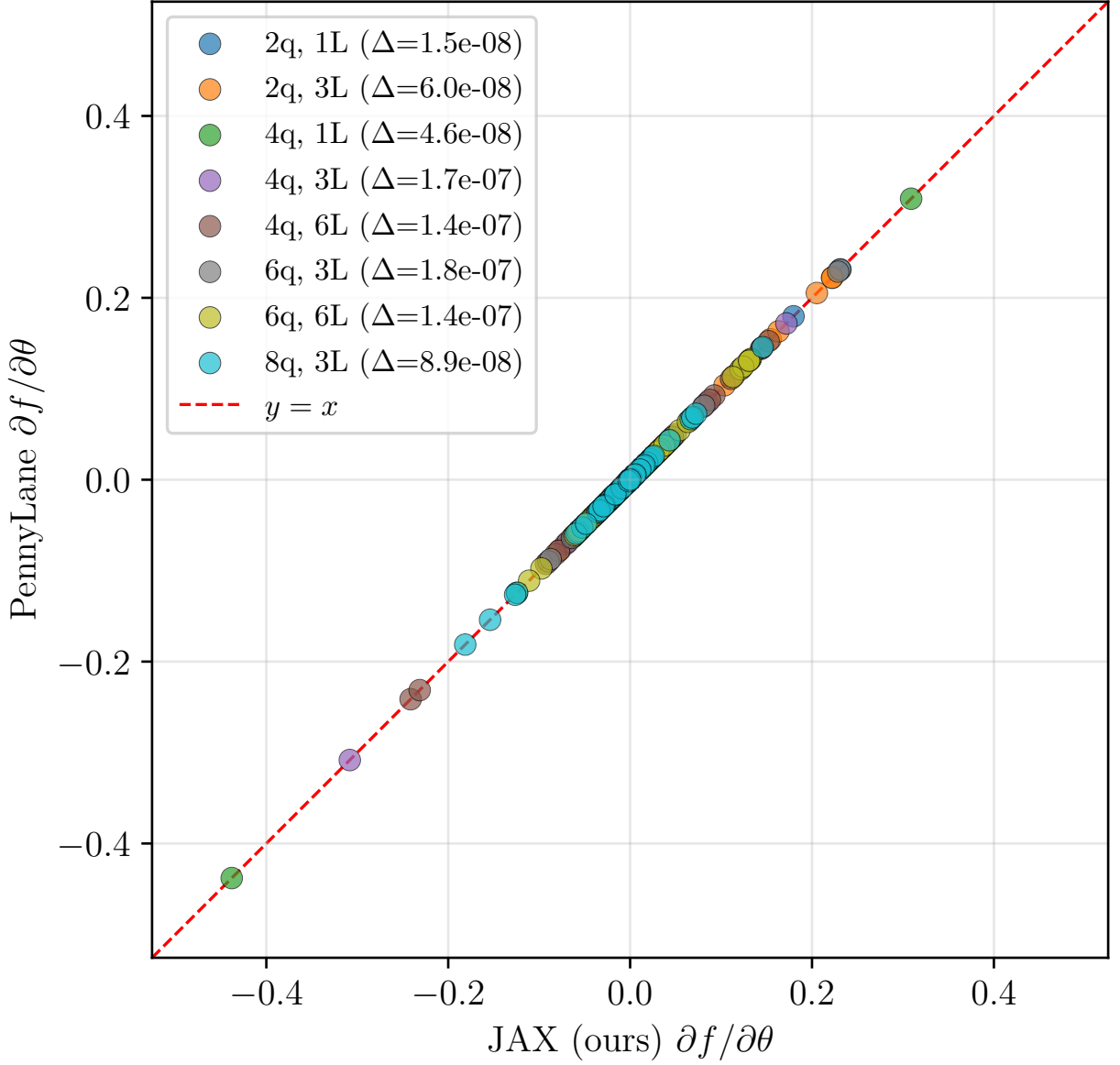


Figure 5.2 Comparison of parameter gradients $\partial f / \partial \theta_q$ computed by our JAX statevector implementation and PennyLane’s `default.qubit` backend across eight circuit configurations (2–8 qubits, 1–6 layers). All values fall on the identity line $y = x$ with maximum absolute deviation $\Delta < 2 \times 10^{-7}$, confirming numerical equivalence of the two implementations.

with background velocity $\alpha_{\text{bg}} = 3.0 \text{ km/s}$, amplitude $A = 2.0 \text{ km/s}$, and a smooth spatial mask $m(x, z) \in [0, 1]$ (the product of four logistic cutoffs) that confines the velocity perturbation to the interior of the physical domain. This construction hard-codes physical bounds ($|\alpha - \alpha_{\text{bg}}| \leq A$) and zero perturbation at the domain edges, acting as a mild regularization of the inverse problem.

The full training objective is

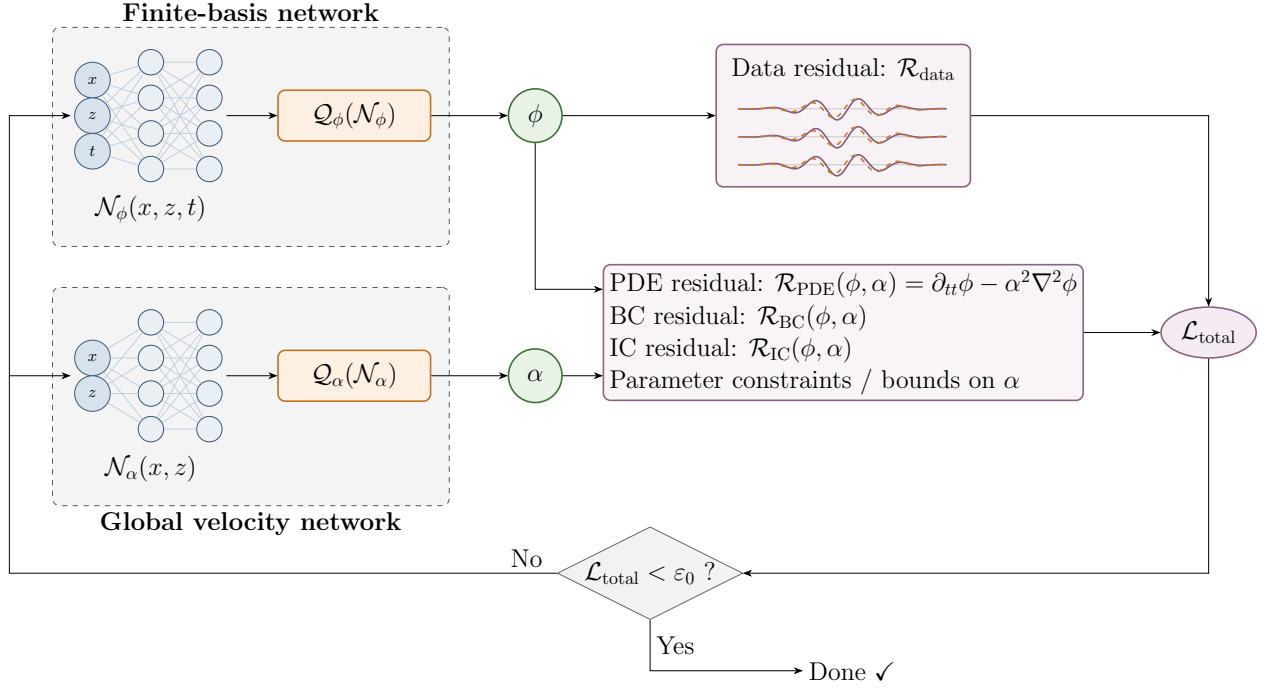


Figure 5.3 Schematic of the finite-basis classical-quantum hybrid architecture, combining classical neural networks and quantum circuits with physics-informed constraints. Blue boxes denote classical networks $\mathcal{N}$, orange boxes quantum circuits $\mathcal{Q}$, green circles the predicted fields ϕ and α , and purple nodes the physics- and data-based residuals contributing to $\mathcal{L}_{\text{total}}$. The inset depicts the observed (solid purple) versus predicted (dashed orange) seismograms at the receivers, whose misfit defines the data residual $\mathcal{R}_{\text{data}}$.

$$\begin{aligned} \mathcal{L}_{\text{total}} = & \lambda_{\text{data}} \mathcal{L}_{\text{data}}(\phi) + \lambda_{\text{PDE}} \mathcal{L}_{\text{PDE}}(\phi, \alpha) \\ & + \lambda_{\text{BC}} \mathcal{L}_{\text{BC}}(\phi) + \lambda_{\text{IC}} \mathcal{L}_{\text{IC}}(\phi), \end{aligned} \quad (5.14)$$

where $\mathcal{L}_{\text{data}}$ is the data misfit between the predicted and observed displacement fields at the receiver locations, $\mathcal{L}_{\text{PDE}}$ enforces the source-free form of the acoustic wave equation Equation 5.1 via automatic differentiation, and $\mathcal{L}_{\text{IC}}, \mathcal{L}_{\text{BC}}$ are the snapshot-based initial-condition and free-surface boundary-condition constraints. All parameters (classical-network weights θ_c , embedding scales β , variational circuit angles θ_q , and output scalings) are optimized jointly by Adam gradient descent through the full differentiable pipeline.

5.3.6 Training setup

The training configuration described in this subsection is specific to the anomaly benchmark. All training uses the Adam optimizer at learning rate 10^{-4} (baseline; see Table 5.1 for variants). Each training step uses a collocation batch of shape $(n_x, n_z, n_t) = (40, 40, 25)$ across the subdomains (40,000 PDE collocation points in total), together with $40 \times 40 = 1,600$ points per displacement snapshot for the two initial-condition constraints, approximately 1,000 receiver observations (20 receivers subsampled every 100 simulation timesteps), and 5,000 randomly sampled boundary-condition points on the top surface. The

composite loss uses fixed weights $\lambda_{\text{PDE}} = 0.1$, $\lambda_{\text{IC}} = 1.0$ (applied equally to the two snapshot constraints), $\lambda_{\text{data}} = 1.0$ (receiver data), and $\lambda_{\text{BC}} = 0.1$; the per-component loss decomposition for the classical baseline over training is shown in Figure 5.4. The classical baseline is trained for 500,000 iterations; the quantum hybrid is trained for 65,000 iterations, at which point it has already surpassed the classical baseline’s 500,000-step L1 velocity error. Recovering higher-frequency velocity structure (as in the checkerboard benchmark of Section 5.4.2, which is driven at a higher source frequency) generally requires a higher collocation-point density to sample the shorter wavelengths, and may additionally call for a finer subdomain decomposition and larger subdomain networks.

Training uses JAX’s just-in-time compilation and vectorized batching over subdomains to compile the update step into a single XLA program. The complete update step (classical-network forward/backward passes, PQC statevector evolution, Jacobian computation by automatic differentiation, and the composite loss) executes as a single GPU kernel. For the quantum hybrid, the statevector footprint is negligible: the 4-qubit PQC requires only $2^4 = 16$ complex amplitudes per forward pass, so memory cost is dominated by the classical wavefield subdomain networks. Spatial coordinates were nondimensionalized by a reference length $\ell_0 = 3$ km, applied identically to x and z , in all training computations; this length is chosen equal to the background wave speed (in km/s) so that the scaled PDE coefficient α^2/ℓ_0^2 is $\mathcal{O}(1)$, balancing the spatial and temporal derivative terms of the residual. The reference length is a normalization constant rather than the domain size: the physical inversion domain is still 1.15×0.45 km.

5.3.7 Network architectures and experimental setup

To establish a rigorous comparison, we evaluate two distinct configurations applied to the identical FWI problem: a purely classical FBPINN baseline and our proposed hybrid quantum-classical architecture. In the classical baseline, both the decomposed wavefield network and the global velocity network are standard fully-connected networks with tanh activations. In the hybrid model, both are replaced by classical-to-quantum pipelines that terminate in four-qubit parameterized quantum circuits (PQCs).

Beyond the network heads, both configurations share the same FBPINN subdomain decomposition strategy ($5 \times 3 \times 5$ grid with a 0.35 overlap for the wavefield), the same use of a single global velocity network, and the same physics-informed composite loss function incorporating the PDE residual, wavefield snapshot initial conditions (ICs), data misfit, and free-surface boundary constraints. Notably, the quantum hybrid model achieves its expressivity using significantly fewer trainable parameters. We note that the two configurations are not matched parameter-for-parameter: the wavefield and velocity pre-networks differ in width and depth in addition to the quantum head (Table 5.1), so the comparison reflects the combined classical-to-quantum substitution rather than the quantum circuit in isolation, and a parameter-matched

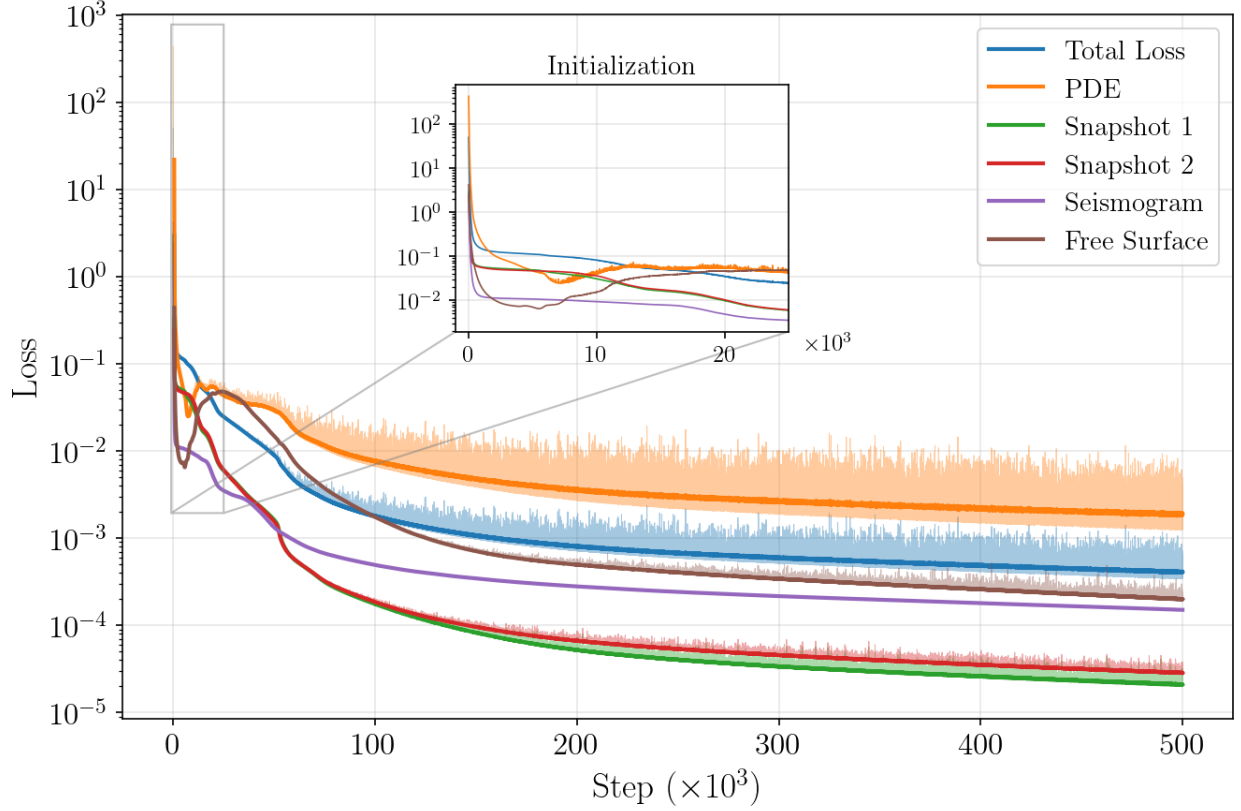


Figure 5.4 Training loss decomposition for the classical FBPINN baseline on the anomaly benchmark: total loss, PDE residual, displacement-snapshot initial conditions (two components), seismogram misfit, and free-surface boundary condition, plotted versus training iteration. Inset: early-iteration behaviour showing rapid initial loss reduction.

classical control with the same backbone and a plain output head is not included here. Schematic representations of the hybrid framework (Figure 5.3) and the PQC topology (Figure 5.1) are provided alongside a summary of the network architectures in Table 5.1. Methodological details, including the underlying acoustic wave equations and loss derivations, are detailed in the preceding subsections.

5.4 Results

We evaluate the proposed hybrid architecture on two wave-propagation benchmarks. The primary benchmark reconstructs a localized low-velocity anomaly from recordings acquired by a one-sided receiver array, providing a direct quantitative comparison between the quantum hybrid and a purely classical FBPINN baseline under matched FBPINN, loss, and optimizer settings. A second benchmark tests a checkerboard velocity pattern using the exact same network configuration, with only the collocation point density increased, to assess whether the proposed inversion framework can recover more spatially structured velocity variations without additional architectural tuning.

5.4.1 Anomaly benchmark

Our primary evaluation uses the velocity model established by Rasht-Behesht *et al.* [93], which defines a 2D acoustic domain ($1.15 \text{ km} \times 0.45 \text{ km}$) featuring a localized low-velocity anomaly (approximately 2.5 km/s embedded within an approximately 3.0 km/s background; Figure 5.5a). Ground-truth receiver data are generated by a spectral-element forward simulation [123] driven by a Gaussian source at $(0.35, 0.19) \text{ km}$ with dominant frequency $f_0 = 20 \text{ Hz}$. The inversion process is constrained by 500 ms of wavefield displacements recorded by a vertical array of 20 receivers positioned along the right boundary ($x = 1.15 \text{ km}$), supplemented by two early displacement snapshots ($t_0 = 0.100 \text{ s}$ and $t_1 = 0.115 \text{ s}$). This transmission-dominated configuration, featuring a single source and a one-sided receiver array, represents a severely ill-posed inverse problem. The limited angular illumination inherently restricts the recovery of high-wavenumber spatial components, rigorously testing the regularizing capacity of the chosen neural architecture.

Starting from a purely homogeneous initial guess (Figure 5.5b), both the classical and hybrid architectures successfully localize the low-velocity anomaly without suffering from the cycle-skipping artifacts that frequently plague classical adjoint-state FWI under poor starting models, which conventional FWI typically mitigates through multiscale or envelope-based strategies [124]. As illustrated by the inverted velocity field of the quantum hybrid model (Figure 5.5c), the recovered anomaly closely tracks the true model’s spatial distribution and peak amplitude. The minor lateral smearing observed in the reconstruction is consistent with the theoretical limits of resolving power imposed by the single-source geometry, confirming that the network has extracted the maximum available information from the receiver data.

The clearest distinction between the classical and quantum approaches emerges in their optimization dynamics. We quantify inversion accuracy by the L1 velocity error, the mean absolute difference between the inverted velocity field α and the true model α_{true} evaluated over an $N = 40 \times 40$ grid spanning the domain,

$$\text{L1} = \frac{1}{N} \sum_{i=1}^N |\alpha(x_i, z_i) - \alpha_{\text{true}}(x_i, z_i)|, \quad (5.15)$$

reported in km/s . Figure 5.5d tracks the L1 velocity error against training iterations, revealing structurally different convergence trajectories. The quantum hybrid model undergoes a rapid initial descent, achieving an L1 error of 2.4×10^{-2} in merely 65,000 iterations. By contrast, the classical baseline exhibits prolonged plateauing behavior, requiring 500,000 iterations to reach a comparable, albeit slightly higher, final error of 2.9×10^{-2} . This corresponds to an approximately $8\times$ reduction in the number of optimization iterations

required to reach comparable or lower L1 velocity error, achieved despite the hybrid model using significantly fewer trainable parameters (101,812 vs. 151,836). We emphasize that this comparison concerns optimization iterations rather than wall-clock runtime, since the per-iteration cost differs between the classical and hybrid architectures. We also note that the traces in Figure 5.5d begin at different L1 errors: each model’s velocity (α) network initializes to a slightly different field about the homogeneous background, so the starting error reflects the random initialization of the differing α -network architectures rather than any quantum effect (one classical velocity-network variant initializes to essentially the same L1 as the quantum hybrid). The comparison should accordingly be read as the iteration count needed to reach a given error, not as progress from a shared starting point.

To test whether this performance gap could be explained by sub-optimal classical tuning, we conducted a systematic classical hyperparameter sweep comprising 15 distinct FBPINN variants. These baselines spanned a wide spectrum of configurations, including severely over-parameterized velocity networks (up to 60×3 hidden layers), alternative subdomain partitions ($3 \times 2 \times 3$ and $8 \times 4 \times 8$), varying learning rates, and adjusted physical loss weightings (detailed comprehensively in Table 5.1). As depicted by the thin colored traces in Figure 5.5d, simply increasing classical capacity or tuning hyperparameters failed to bridge the gap; none of the 15 classical configurations matched both the rapid early descent and the final L1 error achieved by the quantum hybrid within the tested training window (500,000 iterations).

5.4.2 Checkerboard benchmark

To further probe the resolution limits of the inversion pipeline, we test a checkerboard velocity model: a 3×3 grid of alternating ± 0.4 km/s perturbations on a 3.0 km/s background. The acquisition geometry is identical to the anomaly benchmark (same domain, source position, and receiver layout; Figure 5.5a), as is the homogeneous initial model (Figure 5.5b). The checkerboard forward simulation, however, uses a higher dominant source frequency, $f_0 = 60$ Hz, compared with $f_0 = 20$ Hz for the anomaly benchmark, so that the shorter wavelengths can resolve the finer spatial structure of the 3×3 checkerboard. The checkerboard experiment thus tests recovery of higher spatial-frequency velocity structure under the same limited acquisition geometry.

Both the classical and quantum hybrid FBPINNs are run with the exact same network architectures and hyperparameters established for the anomaly benchmark (Table 5.1). The training setup differs only in the use of a higher collocation point density, required to adequately constrain the smaller-scale spatial structures of the checkerboard pattern. Both models successfully resolve the checkerboard (Figure 5.6b). As expected from the single-source, one-sided receiver geometry, structural resolution degrades for blocks situated further from the receiver array. The successful inversion under this otherwise unchanged

Table 5.1 Network configurations and full hyperparameter sweep for the anomaly benchmark. The **classical** baseline (first row) and **quantum** hybrid (last row) are the two compared configurations; the intervening rows are the 15 classical hyperparameter variants. $\mathcal{Q}(n_q^{\times n_L})$ denotes a parameterized quantum circuit with n_q qubits and n_L layers. Values marked “–” are identical to the **classical** baseline. Subdomains column shows $\phi(\cdot)$ for wavefield decomposition.

Model	ϕ network $(x, z, t) \rightarrow \phi$	α network $(x, z) \rightarrow \alpha$	LR	Subs	Overlap	λ_{PDE}	λ_{IC}	λ_{data}	λ_{BC}	# Pars
classical	$\mathcal{N}: 32^{\times 2} \rightarrow 16^{\times 2}$	$\mathcal{N}: 20^{\times 5}$	10^{-4}	$\phi(5 \times 3 \times 5)$	0.35	0.1	1.0	1.0	0.1	151,836
<i>ϕ-network size variants</i>										
phi16x2	$\mathcal{N}: 16^{\times 2}$	–	–	–	–	–	–	–	–	25,836
phi64x3	$\mathcal{N}: 64^{\times 3}$	–	–	–	–	–	–	–	–	649,836
<i>ϕ subdomain decomposition variants</i>										
sub3x2x3	–	–	–	$\phi(3 \times 2 \times 3)$	–	–	–	–	–	37,779
sub8x4x8	–	–	–	$\phi(8 \times 4 \times 8)$	–	–	–	–	–	514,017
<i>α-network size variants</i>										
alpha10x2	–	$\mathcal{N}: 10^{\times 2}$	–	–	–	–	–	–	–	150,226
alpha10x3	–	$\mathcal{N}: 10^{\times 3}$	–	–	–	–	–	–	–	150,336
alpha20x3	–	$\mathcal{N}: 20^{\times 3}$	–	–	–	–	–	–	–	150,996
alpha40x3	–	$\mathcal{N}: 40^{\times 3}$	–	–	–	–	–	–	–	153,516
alpha40x5	–	$\mathcal{N}: 40^{\times 5}$	–	–	–	–	–	–	–	156,796
alpha60x3	–	$\mathcal{N}: 60^{\times 3}$	–	–	–	–	–	–	–	157,636
<i>Training hyperparameter variants</i>										
lr5e4	–	–	5×10^{-4}	–	–	–	–	–	–	–
lr5e5	–	–	5×10^{-5}	–	–	–	–	–	–	–
overlap50	–	–	–	–	0.50	–	–	–	–	–
wpde01_wbc01	–	–	–	–	–	0.01	–	–	0.01	–
wpde1_wbc1	–	–	–	–	–	1.0	–	–	1.0	–
<i>Quantum hybrid</i>										
quantum	$\mathcal{N}: 32^{\times 2}$ $\rightarrow \mathcal{Q}: 4^{\times 2}$	$\mathcal{N}: 20^{\times 3}$ $\rightarrow \mathcal{Q}: 4^{\times 2}$	–	–	–	–	–	–	–	101,812

configuration confirms that the hybrid architecture generalizes beyond the anomaly benchmark without additional tuning.

5.5 Discussion

The quantum hybrid’s central result, a lower final L1 velocity error in approximately $8\times$ fewer training iterations with approximately 33% fewer trainable parameters, raises the question of what in the architecture produces it. We identify three contributing properties of the classical-to-PQC pipeline [114], while noting that our experiments do not isolate their individual effects.

First, the PQC is a parameter-efficient nonlinear map. Its output is governed by only $3nL + n$ trainable parameters, yet each enters through the joint action of non-commuting rotations rather than as an independent weight, so the same angles perform both feature mixing and output generation. This shared parameterization is consistent with the hybrid model matching or exceeding the classical baseline while using fewer parameters. We do not invoke the dimension of the underlying Hilbert space as a source of advantage: the measured output is a single bounded scalar, and for the circuits used here ($n = 4$) it realizes

only the low-order trigonometric family derived below.

Second, angle encoding endows the PQC output with a finite-frequency Fourier structure in the variables presented to the encoder. Because the PQC acts on the learned latent vector $\mathbf{z}(\mathbf{x}) = \mathcal{N}(\mathbf{x})$ rather than on the spatial coordinates directly, the relevant specialization of the result of Schuld *et al.* [113] is

$$f(\mathbf{x}; \boldsymbol{\beta}, \boldsymbol{\theta}_q) = \sum_{\boldsymbol{\omega} \in \Omega} c_{\boldsymbol{\omega}}(\boldsymbol{\theta}_q) e^{i \boldsymbol{\omega} \cdot (\boldsymbol{\beta} \odot \mathbf{z}(\mathbf{x}))}, \quad (5.16)$$

where the coefficients $c_{\boldsymbol{\omega}}$ are set by the trainable circuit angles and the accessible index set Ω is fixed by the encoding. For the $R_y(\beta_i z_i)$ encoding used here, each coordinate contributes only the indices $0, \pm 1$, so that for a single encoded variable

$$f(s) = A + B \cos(\beta s) + C \sin(\beta s), \quad (5.17)$$

with physical frequencies $0, \pm \beta_i$ that the trainable scales β_i adapt during optimization; the derivation is given in Section 5.8. This finite-frequency structure is not unique to quantum circuits: classical Fourier-feature mappings [125], and to a looser degree the nested sinusoidal activations of SIREN networks [126], build the representation from a trigonometric basis and are an established remedy for the spectral bias of plain tanh networks. Relative to a fixed Fourier-feature layer with prescribed frequencies [125], the PQC differs in two respects: the encoding scales $\boldsymbol{\beta}$ are trainable, so the physical frequency grid adapts rather than being fixed a priori (a property it shares with SIREN, whose first-layer frequencies are likewise learned), and the entangling CNOT layers couple the per-coordinate features, populating cross-coordinate terms of Ω that an axis-aligned (additive) Fourier-feature layer does not produce on its own (an isotropic random Fourier mapping with a dense frequency matrix mixes coordinates and would). Two caveats temper this argument: the frequency bound applies to f as a function of the learned latent $\mathbf{z}$ and not of the spatial coordinates, and our classical sweep contains only tanh architectures. A matched classical control with an explicit Fourier-feature or sinusoidal embedding is therefore the most informative comparison still missing, and we flag it as a priority below.

Third, the velocity output map hard-codes the range constraint $|\alpha - \alpha_{\text{bg}}| \leq A$. This bound is applied identically in both architectures, so it cannot by itself explain the gap; we note it only because, together with the bounded PQC output, it limits the latent signal entering the final map and may reduce large early-training excursions. Separately, the entangling CNOT layers provide a parameterization in which correlations among the latent variables are generated by the circuit dynamics rather than by stacked affine-plus-nonlinearity layers. Whether this yields a more favorable optimization trajectory, rather than a function class beyond the reach of a comparable MLP, is an empirical question we do not resolve here.

We stress that these properties bear mainly on representational capacity and parameter efficiency; none of them, on its own, predicts a faster convergence rate, so the link between the PQC parameterization and the observed iteration-count reduction remains to be established. The consistency of the improvement across the 15 classical hyperparameter variants (Figure 5.5d) argues against the result being solely due to an unlucky baseline, though the models also begin from different initial L1 errors set by their velocity-network initializations. We emphasize, finally, that these gains are in optimization iterations on classically simulated circuits and do not constitute a quantum computational advantage; the simulated PQC realizes a function class that is, by construction, classically representable.

Our work complements recent hybrid quantum-PINN studies in other PDE-constrained settings. Leong *et al.* [103] benchmarked hybrid quantum PINNs for high-speed compressible flow, finding competitive accuracy at reduced parameter cost compared with classical PINNs. Berger *et al.* [104] introduced trainable-embedding quantum PINNs for nonlinear PDEs including the Navier-Stokes equations, again observing efficiency gains at matched parameter count. The present work differs in two ways. First, we target a coupled forward-inverse problem (waveform inversion) rather than a pure forward PDE; the velocity field is learned jointly with the wavefield. Second, we integrate the PQC within a domain-decomposed FBPINN [97], extending the PINN-based FWI formulation of Rasht-Behesht *et al.* [93] to handle larger and higher-frequency inversion problems via overlapping subdomain networks.

Several limitations should be noted: (i) The present experiments are two-dimensional acoustic and extending to three-dimensional elastic inversion, as pursued with classical physics-guided deep networks [127], is the natural next step but is constrained by the $\mathcal{O}(2^n)$ memory cost of statevector simulation as qubit counts grow. Scaling to larger circuits on real hardware further raises the concern of barren plateaus (vanishingly small gradients in high-depth parameterized circuits [116]), which would require careful initialization and ansatz choice. (ii) Inversions use a single source; production FWI pipelines incorporate a large number of sources with different illuminations, and PINN-based FWI in particular is known to be sensitive to loss-landscape pathologies under data sparsity [128], which learned generative priors have been proposed to regularize [129]. (iii) The quantum circuit is simulated classically. On real NISQ hardware, gradients would be computed via the parameter-shift rule (two circuit evaluations per parameter), and device noise could erode the iteration-level advantage observed here.

Looking forward, the small qubit counts used here ($n = 4$, $L = 2$) are compatible with near-term quantum hardware, which suggests that genuine hardware demonstrations are feasible once noise-mitigation techniques reach sufficient maturity. Systematic studies at matched parameter counts across circuit depths and widths would clarify how the convergence advantage scales with quantum-circuit capacity. A matched classical control with an explicit Fourier-feature or sinusoidal embedding, added to

the hyperparameter sweep, would in particular isolate how much of the observed advantage reflects the finite-frequency inductive bias rather than the quantum parameterization itself. Because the framework targets the general structure of physics-informed wave inversion rather than any geophysics-specific feature, we expect it to transfer directly to the broader wave-based inverse problems, such as medical ultrasound tomography, photoacoustic imaging, and non-destructive evaluation.

5.6 Conclusion

We introduced a hybrid quantum-classical FBPINN for acoustic FWI, in which the decomposed wavefield network and the global velocity network are represented by PQC-based classical-to-quantum pipelines within a domain-decomposed physics-informed framework, implemented as an end-to-end differentiable JAX program that avoids parameter-shift-rule overhead and jointly optimizes the classical PINN parameters and quantum circuit angles. On the anomaly benchmark, the hybrid model reaches a comparable or lower L1 velocity error in approximately $8\times$ fewer training iterations than the primary classical FBPINN baseline, despite using approximately 33% fewer trainable parameters, and it outperforms all 15 classical hyperparameter variants tested within the same 500,000-iteration window. We emphasize that these gains are in optimization iteration count on classically simulated circuits and do not constitute a claim of quantum computational advantage, and that the mechanism behind the iteration-count reduction remains to be established. Key limitations include reliance on classical statevector simulation, a two-dimensional acoustic setup, a single-source configuration, and a comparison that is not parameter-matched and spans only tanh classical architectures. A matched classical control with an explicit Fourier-feature or sinusoidal embedding is the most informative experiment still needed. It would isolate how much of the advantage reflects a finite-frequency inductive bias rather than the quantum parameterization. Extension to elastic, three-dimensional, and multi-source inversion, and validation on near-term quantum hardware, are the natural next steps. Because the framework targets the general structure of physics-informed wave inversion rather than any geophysics-specific feature, we expect it to transfer to other wave-based inverse problems, such as medical ultrasound tomography and non-destructive evaluation.

5.7 Data and Code Availability

The data and code that support the findings of this study are available in the project repository at <https://github.com/x-repos/quFWI>.

5.8 Appendix: Fourier Structure of the Angle-Encoded PQC

This note derives the finite-frequency structure of the PQC output used in the Discussion, specializing the general result of Schuld *et al.* [113] to the angle encoding adopted in this work.

Consider first a single encoded scalar s with Hermitian generator G and encoding unitary $U_{\text{enc}}(s) = e^{-i\beta s G}$, acting on the reference state $|\psi_0\rangle = |0\rangle^{\otimes n}$. Because the variational block $V(\boldsymbol{\theta}_q)$ is applied after the encoding, the averaged measurement observable $O = \frac{1}{n} \sum_{i=0}^{n-1} Z_i$ can be folded into the Heisenberg-picture operator $M(\boldsymbol{\theta}_q) = V(\boldsymbol{\theta}_q)^\dagger O V(\boldsymbol{\theta}_q)$, so the measured output is

$$f(s) = \langle \psi_0 | U_{\text{enc}}(s)^\dagger M(\boldsymbol{\theta}_q) U_{\text{enc}}(s) | \psi_0 \rangle. \quad (5.18)$$

Writing the spectral decomposition $G = \sum_a \lambda_a \Pi_a$ gives $U_{\text{enc}}(s) = \sum_a e^{-i\beta s \lambda_a} \Pi_a$, and therefore

$$f(s) = \sum_{a,b} e^{i\beta s(\lambda_a - \lambda_b)} \langle \psi_0 | \Pi_a M(\boldsymbol{\theta}_q) \Pi_b | \psi_0 \rangle, \quad (5.19)$$

so the accessible frequency indices are the pairwise eigenvalue differences, $\Omega \subseteq \{\lambda_a - \lambda_b\}$. For the $R_y(\beta s)$ encoding used here, $G = Y/2$ has eigenvalues $\pm \frac{1}{2}$, so the differences are $0, \pm 1$ and

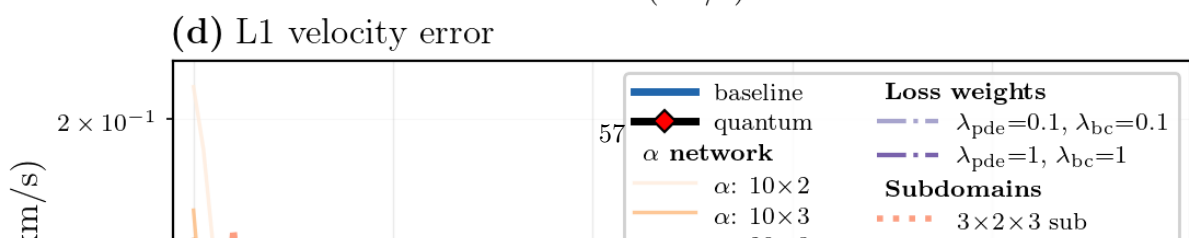
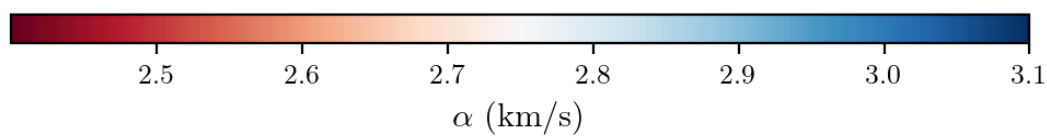
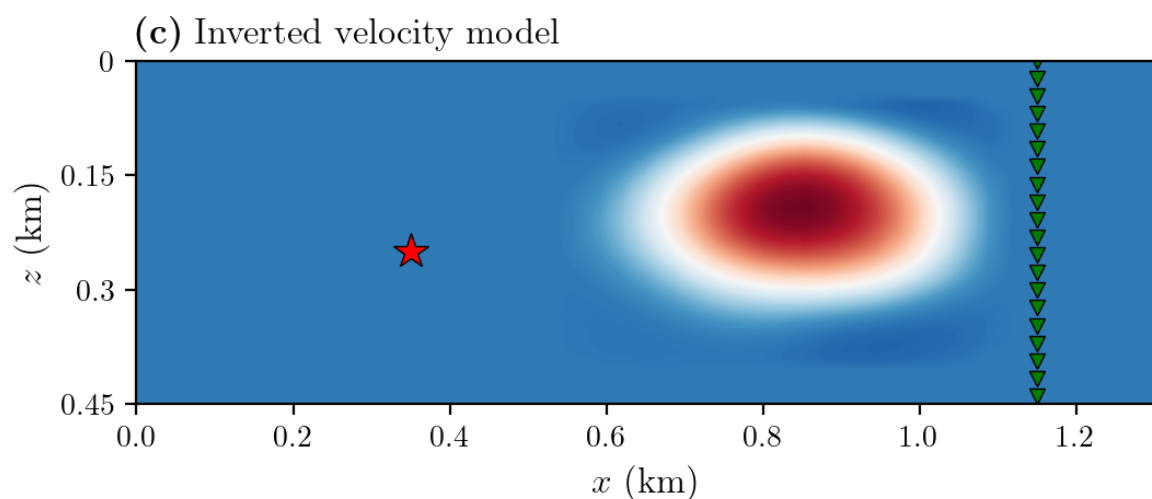
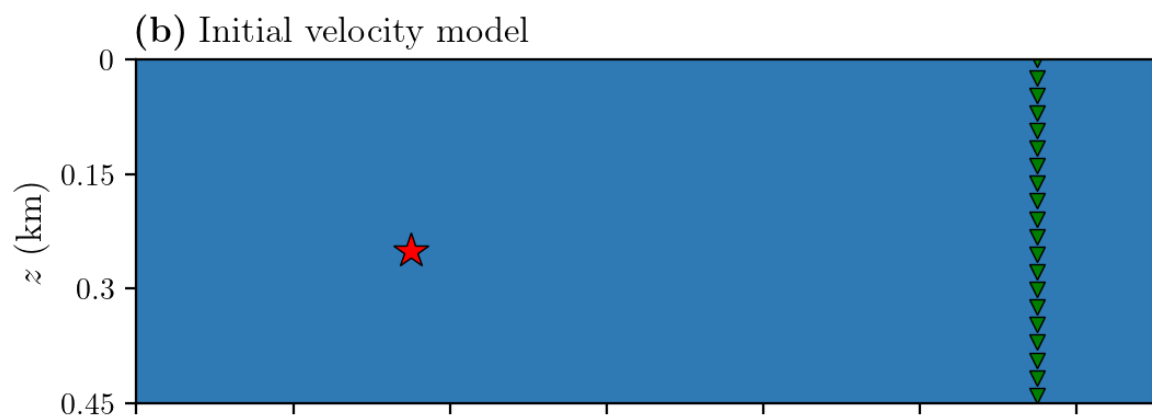
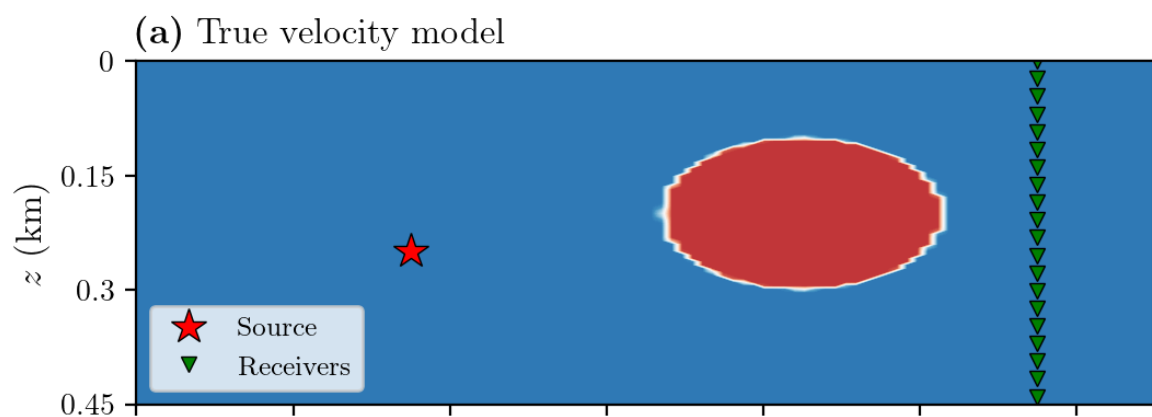
$$f(s) = A + B \cos(\beta s) + C \sin(\beta s). \quad (5.20)$$

The trainable angles change A , B , and C but cannot create frequencies outside this set. This single-frequency (degree-one) spectrum is a consequence of encoding each variable exactly once: the L variational layers act through the trainable parameters $\boldsymbol{\theta}_q$ and do not re-encode the data, so they enrich the coefficients but not the frequency set. Repeated (“data re-uploading”) encodings would extend the accessible spectrum to higher harmonics, but are not used here.

For the full register, each coordinate z_i is encoded on its own qubit through $R_y(\beta_i z_i)$, so the encoding generators $\{Y_i/2\}$ act on disjoint tensor factors and mutually commute. The encoding unitary then factorizes as $\prod_i e^{-i\beta_i z_i Y_i/2}$, the per-coordinate spectra combine additively, and the accessible index set becomes $\Omega \subseteq \{-1, 0, 1\}^n$, with physical frequencies $\boldsymbol{\beta} \odot \boldsymbol{\omega}$. This is the multivariate Fourier form quoted in the Discussion. The entangling CNOT layers and variational angles determine which coefficients $c_{\boldsymbol{\omega}}$ are nonzero, in particular populating cross-coordinate indices, but they do not enlarge Ω .

5.9 Acknowledgment

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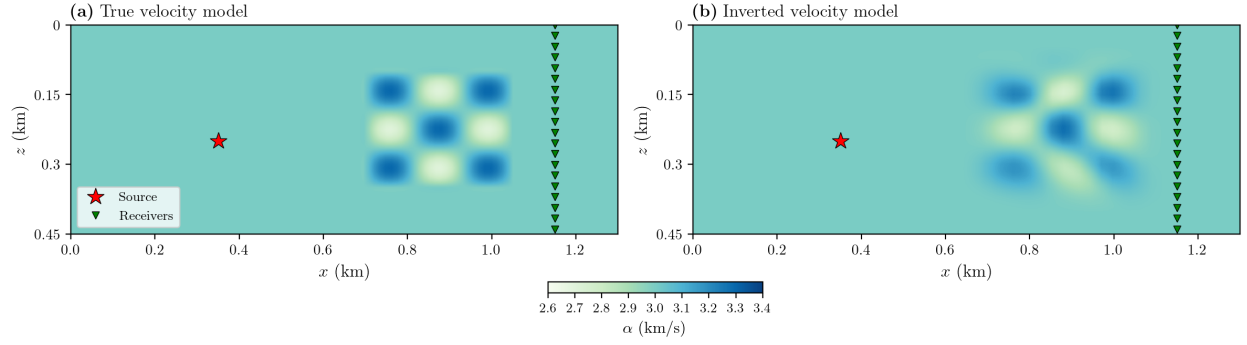


Figure 5.6 Checkerboard benchmark inversion. (a) True velocity model: a 3×3 checkerboard of alternating ± 0.4 km/s perturbations on a 3.0 km/s background, with source (red star) and receiver array (green triangles). (b) Inverted velocity model from the quantum hybrid FBPINN, using the identical network configuration from the anomaly benchmark with only the collocation point density increased.

CHAPTER 6

CONCLUSIONS

This chapter is a placeholder for the conclusions.

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APPENDIX A

THE CONVOLUTIONAL PML IN THE HAMILTONIAN-SIMULATION VARIABLES

This appendix expands the one-line statements of Sections 3.3.2, 3.4.1, and 3.4.2: the causal kernel and memory ODE from the CFS stretch (Appendix A.1), the one-dimensional memory generator and the frequency-domain collapse (Appendix A.2), the proof that the collapse is exact (Appendix A.3), the two-dimensional eight-block generator (Appendix A.4), the staggered profile-sampling rule (Appendix A.5), and the profile formulas (Appendix A.6). Throughout, D is the forward-difference matrix and $D^\dagger = -D_b$ its negative backward counterpart (Section 3.3.1); all operators are exactly those constructed in the accompanying code.

A.1 From the CFS stretch to the memory ODE

We fix the Fourier convention $\hat{f}(\omega) = \int dt e^{-i\omega t} f(t)$, $f(t) = (2\pi)^{-1} \int d\omega e^{+i\omega t} \hat{f}(\omega)$, so that $\partial_t \mapsto i\omega$ and the causal exponential has the transform $\int_0^\infty dt e^{-\beta t} e^{-i\omega t} = (\beta + i\omega)^{-1}$ for $\beta > 0$. A PML stretches the coordinate into the complex plane, $\tilde{x} = \int_0^x s(x', \omega) dx'$, which in the frequency domain replaces every spatial derivative by $\partial_x \mapsto s(\omega)^{-1} \partial_x$ inside the layer. For the CFS factor of Eq. (3.2), partial fractions in the variable $i\omega$ give

$$\frac{1}{s(\omega)} = \frac{\alpha + i\omega}{\kappa(\alpha + i\omega) + \sigma} = \frac{1}{\kappa} + \hat{\zeta}(\omega), \quad (\text{A.1})$$

with

$$\hat{\zeta}(\omega) = -\frac{\sigma}{\kappa^2} \frac{1}{\beta + i\omega}, \quad \beta := \frac{\sigma}{\kappa} + \alpha \quad (\text{A.2})$$

the CFS decay rate: indeed $1/s - 1/\kappa = (\kappa - s)/(\kappa s)$ with $\kappa - s = -\sigma/(\alpha + i\omega)$ and $\kappa s = \kappa^2(\beta + i\omega)/(\alpha + i\omega)$. The single pole at $i\omega = -\beta$ inverts, by the causal transform above, to the exponential kernel of Eq. (3.3), $\zeta(t) = -(\sigma/\kappa^2) e^{-\beta t} \theta(t)$, and frequency multiplication is time convolution: $s^{-1} \partial_x f = \kappa^{-1} \partial_x f + \zeta * \partial_x f$.

The convolution becomes local in time once it is named. Let $\phi := \zeta * \partial_x f$. Multiplying Eq. (A.1) by $(\beta + i\omega)$ gives the polynomial identity $(\beta + i\omega) \hat{\zeta} = -\sigma/\kappa^2$, hence $i\omega \hat{\phi} = -\beta \hat{\phi} - (\sigma/\kappa^2) \widehat{\partial_x f}$, which transforms back to the memory ODE of Eq. (3.4). Equivalently, in the time domain, $\partial_t \phi = \zeta(0^+) \partial_x f + \zeta' * \partial_x f$ with $\zeta(0^+) = -\sigma/\kappa^2$ and $\zeta' = -\beta \zeta$. One first-order ODE per stretched derivative thus carries the convolution exactly; this is the recursive-convolution CPML of Roden and Gedney [6].

A.2 One dimension: memory generator and collapse

In the variables of Section 3.3.1 the undamped 1D system is $\dot{v} = -iDw$, $\dot{w} = -iD^\dagger v$ [Eq. (3.1)]. Each equation contains exactly one spatial derivative, so the stretch generates one memory field per physical field. Sampling the profiles per field (Appendix A.5) defines the diagonal matrices $\Sigma_v = \text{diag}(\sigma_{v,j})$, $K_v = \text{diag}(\kappa_{v,j})$, $B_v = \text{diag}(\sigma_{v,j}/\kappa_{v,j} + \alpha_{v,j})$, and the layer projector $\Pi_v = \text{diag}(\mathbf{1}[\sigma_{v,j} > 0])$, and likewise for w . Applying the stretched-derivative rule to $-iDw$ and carrying the convolution by the rescaled memory field $\phi_v := \gamma^{-1} \zeta_v * (-iDw)$ —where $\gamma > 0$ is a free constant rescaling under which the physical trajectory is invariant, set to the balanced value $\gamma = \sqrt{2\sigma_{\max}/h}$ of Appendix 3.8—gives, by Eq. (3.4),

$$\dot{v} = -iK_v^{-1}Dw + \gamma\phi_v, \quad \dot{\phi}_v = -B_v\phi_v + \frac{i}{\gamma}\Sigma_v K_v^{-2}Dw, \quad (\text{A.3})$$

and the analogous pair for (w, ϕ_w) with $D \rightarrow D^\dagger$, $v \leftrightarrow w$. The source of ϕ_v carries the factor Σ_v , so ϕ_v vanishes identically outside the layer strips for memory-free data; masking the injection column by Π_v therefore changes no such trajectory while removing the spurious $O(\gamma)$ interior entries that an unmasked column would contribute to the Hermitian part H_1 . On the four-block state $[v; w; \phi_v; \phi_w]$ (two extra block qubits) the generator is

$$A_{\text{mem}} = \begin{bmatrix} 0 & -iK_v^{-1}D & \gamma\Pi_v & 0 \\ -iK_w^{-1}D^\dagger & 0 & 0 & \gamma\Pi_w \\ 0 & \frac{i}{\gamma}\Sigma_v K_v^{-2}D & -B_v & 0 \\ \frac{i}{\gamma}\Sigma_w K_w^{-2}D^\dagger & 0 & 0 & -B_w \end{bmatrix}. \quad (\text{A.4})$$

Collapse for $\kappa = 1$, $\alpha = 0$. In this case the stretch degenerates: $s(\omega) = (i\omega + \sigma)/(i\omega)$, so the stretched v equation, $i\omega\hat{v} = s_v(\omega)^{-1}(-iD\hat{w})$ per Fourier mode, multiplies out to

$$(i\omega + \sigma_v)\hat{v} = -iD\hat{w}, \quad (\text{A.5})$$

because the symbol $s(\omega) i\omega = i\omega + \sigma$ is a *first-degree polynomial* in $i\omega$. Transforming back,

$\dot{v} = -iDw - \sigma_v v$, and likewise $\dot{w} = -iD^\dagger v - \sigma_w w$: the convolution collapses to the local damping of Eq. (3.8), $A = -iH - \text{diag}(\sigma_v, \sigma_w)$, with no memory fields at all.

A.3 Proof of the exact collapse

The frequency-domain argument above is formal (it commutes the stretch with the time-domain initial-value problem); the following statement makes the equivalence exact at the semi-discrete level.

Lemma A.2 (exact collapse) *Let $\kappa \equiv 1$ and $\alpha \equiv 0$ in the generator of Eq. (A.4), so that $B_v = \Sigma_v$ and $B_w = \Sigma_w$, and define the defects*

$$e_v := \phi_v + \gamma^{-1}\Sigma_v v, \quad e_w := \phi_w + \gamma^{-1}\Sigma_w w. \quad (\text{A.6})$$

Then $\dot{e}_v = \dot{e}_w = 0$ along every solution of $\dot{\Psi} = A_{\text{mem}}\Psi$. Consequently a memory-form trajectory projects onto a trajectory of the collapsed generator Eq. (3.8) if and only if $e_v(0) = e_w(0) = 0$; for the standard embedding $\phi_v(0) = \phi_w(0) = 0$ this is the condition $\Sigma_v v(0) = 0$ and $\Sigma_w w(0) = 0$, i.e. initial data supported outside the layer.

Proof. Using both rows of Eq. (A.3) with $B_v = \Sigma_v$,

$$\begin{aligned}\dot{e}_v &= \left[-\Sigma_v \phi_v + \frac{i}{\gamma} \Sigma_v D w \right] + \gamma^{-1} \Sigma_v [-i D w + \gamma \Pi_v \phi_v] \\ &= (\Sigma_v \Pi_v - \Sigma_v) \phi_v = 0,\end{aligned}$$

since $\Sigma_v \Pi_v = \Sigma_v$ (the projector is the support indicator of Σ_v); identically for e_w . Equivalently, the block-row operator $E = \begin{bmatrix} \gamma^{-1} \Sigma_v & 0 & I & 0 \\ 0 & \gamma^{-1} \Sigma_w & 0 & I \end{bmatrix}$ annihilates the generator from the left, $E A_{\text{mem}} = 0$ (verified to round-off in our implementation). On the invariant slice $e \equiv 0$ the memory ansatz $\phi_v = -\gamma^{-1} \Sigma_v v$ of Section 3.4.1 holds at all times, the injection term becomes $\gamma \Pi_v \phi_v = -\Sigma_v v$, and (v, w) obeys Eq. (3.8); conversely, every collapsed trajectory lifts to the memory form through the ansatz. ■

When $e(0) \neq 0$, the conserved defect acts on the physical equations as the static in-layer forcing $\gamma \Pi_v e_v(0)$, $\gamma \Pi_w e_w(0)$, so the discrepancy between the two forms scales with the layer content $\|\Sigma_v v(0)\|$, $\|\Sigma_w w(0)\|$ of the initial data—and vanishes exactly for data supported outside the layer. Numerically, propagating the same interior pulse with Eq. (A.4) and with Eq. (3.8) agrees to 6×10^{-13} (Section 3.4.1).

Remark (the roles of α and κ). The cancellation uses only $\alpha \equiv 0$: for a graded κ the modified defect $e_v = \phi_v + \gamma^{-1} \Sigma_v K_v^{-1} v$ is conserved by the same computation, and the convolution collapses to the local form $\dot{v} = -i K_v^{-1} D w - K_v^{-1} \Sigma_v v$ (both statements again verified to round-off). For $\alpha > 0$, instead, $\dot{e}_v = -\text{diag}(\alpha_v) \phi_v \neq 0$ —equivalently, $s(\omega) i\omega$ is no longer polynomial in $i\omega$ —and the memory fields are irreducible: the four-block form is then the genuine content of the CFS layer.

A.4 Two dimensions: the eight-block generator

In 2D the damping acts per direction: the x stretch replaces every ∂_x and the y stretch every ∂_y . On the separated-block state of Section 3.3.1 the undamped equations are $\dot{v} = -i D_x w_x - i D_y w_y$, $\dot{w}_x = -i D_x^\dagger v$, $\dot{w}_y = -i D_y^\dagger v$, with $D_x = D \otimes I_{N_y}$ and $D_y = I_{N_x} \otimes D$ on the flattened $N_x N_y$ grid register (x register more significant). The v equation contains one derivative of each direction and acquires two memory fields ϕ_{vx} , ϕ_{vy} ; each w equation contains a single derivative and acquires one, ϕ_x for w_x and ϕ_y for w_y . The four memory fields plus the three physical fields are padded by one zero block to the eight blocks of Eq. (3.9) (three block qubits).

Because σ_x depends on x only, its diagonal samples factorise on the grid register: $\Sigma_v^x = S_x \otimes I_{N_y}$ and $\Sigma_w^x = S_x^{\text{st}} \otimes I_{N_y}$, where $S_x = \text{diag } \sigma(x_j)$ holds the nodal samples and $S_x^{\text{st}} = \text{diag } \sigma(x_{j-1/2})$ the half-cell

samples (Appendix A.5); symmetrically $\Sigma_v^y = I_{N_x} \otimes S_y$ and $\Sigma_w^y = I_{N_x} \otimes S_y^{\text{st}}$. With $\kappa \equiv 1$ (our 2D configurations), the decay rates are $B_v^x = \Sigma_v^x + \text{diag } \alpha_x(x_j)$ etc., and the masks $\Pi_v^x = \text{diag}(\mathbf{1}[\Sigma_v^x > 0])$, ... are the layer-strip projectors. Repeating the derivation of Appendix A.2 once per (field, stretched direction) pair gives the generator, in the block ordering of Eq. (3.9),

$$A = \begin{bmatrix} 0 & -iD_x & -iD_y & \gamma\Pi_v^x & \gamma\Pi_v^y & 0 & 0 & 0 \\ -iD_x^\dagger & 0 & 0 & 0 & 0 & \gamma\Pi_w^x & 0 & 0 \\ -iD_y^\dagger & 0 & 0 & 0 & 0 & 0 & \gamma\Pi_w^y & 0 \\ 0 & \frac{i}{\gamma}\Sigma_v^x D_x & 0 & -B_v^x & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{i}{\gamma}\Sigma_v^y D_y & 0 & -B_v^y & 0 & 0 & 0 \\ \frac{i}{\gamma}\Sigma_w^x D_x^\dagger & 0 & 0 & 0 & 0 & -B_w^x & 0 & 0 \\ \frac{i}{\gamma}\Sigma_w^y D_y^\dagger & 0 & 0 & 0 & 0 & 0 & -B_w^y & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad \text{on } [v; w_x; w_y; \phi_{vx}; \phi_{vy}; \phi_x; \phi_y; 0], \quad (\text{A.7})$$

which is, entry for entry, the matrix assembled by the accompanying implementation.

Strip masking. The source of each memory field carries its own profile factor (Σ_v^x for ϕ_{vx} , ...), so the field remains identically zero outside its layer strip for memory-free initial data; the corresponding injection column is masked by the strip projector, exactly as in 1D, so that H_1 carries no spurious interior coupling.

Corners. In a corner cell both $\sigma_x > 0$ and $\sigma_y > 0$ hold simultaneously: both directions' memory fields are active there, the same generator entries of Eq. (A.7) apply unchanged, and no corner-specific terms arise. This is the structural advantage of the unsplit convolutional PML over split-field formulations [6]. The measured stability, $\max \text{Re eig}(A) = 0$ to round-off, is reported in Section 3.4.2.

A.5 The staggered sampling rule

For the forward-difference convention, $(Dw)_j = (w_{j+1} - w_j)/h$. The component v_j is updated through $(Dw)_j$ and represents the field at the integer node $x_j = jh$; the component w_j is updated through $(D^\dagger v)_j \propto v_j - v_{j-1}$, a difference centered at the half-cell point $x_{j-1/2}$, so the w field lives on the staggered grid (the backward convention shifts it to $x_{j+1/2}$). The sampling rule follows: *every profile factor in an evolution equation is sampled on the grid of the field that the equation updates*,

$$\sigma_{v,j} = \sigma(x_j), \quad \sigma_{w,j} = \sigma(x_{j\mp 1/2}), \quad (\text{A.8})$$

with the upper (lower) sign for the forward (backward) convention, and per direction in 2D (Table A.1): only the x profile of the w_x family and the y profile of the w_y family are staggered; all other dependencies are nodal.

Ablation. Sampling all profiles at the integer nodes instead—the only change being the half-cell shift of the w profiles—degrades the measured gold-standard reflection of Section 3.4.2 by factors of 120–235. At the reflection floors of Figure 3.2(a,b) the half-cell stagger is load-bearing, not a refinement.

Table A.1 Staggered sampling of the damping profiles in 2D (forward-difference convention; the backward convention flips the half-cell shifts). Each profile is sampled on the grid of the field its equation updates.

Field	Lives at	Profile samples
v	(x_j, y_k)	$\sigma(x_j), \sigma(y_k)$
w_x	$(x_{j-1/2}, y_k)$	$\sigma(x_{j-1/2})$
w_y	$(x_j, y_{k-1/2})$	$\sigma(y_{k-1/2})$

A.6 Profile formulas

Within a layer of n_{pml} points (width $L = n_{\text{pml}}h$) the damping is polynomially graded in the depth d measured from the interface,

$$\sigma(d) = \sigma_{\text{max}} (d/L)^m, \quad 0 \leq d \leq L, \quad (\text{A.9})$$

and zero outside; the interface node itself is undamped, $\sigma(0) = 0$, the outermost node attains σ_{max} , and staggered samples beyond the wall are clipped to $d = L$. The amplitude follows from the normal-incidence round trip: a wave crossing the layer, reflecting off the terminating hard wall, and crossing back accumulates the attenuation

$$R = \exp\left[-\frac{2}{c} \int_0^L \sigma(x) dx\right] = \exp\left[-\frac{2\sigma_{\text{max}}L}{(m+1)c}\right], \quad (\text{A.10})$$

so prescribing a design reflection R_0 gives the Collino–Tsogka amplitude [8]

$$\sigma_{\text{max}} = -\frac{(m+1)c \ln R_0}{2L}, \quad (\text{A.11})$$

used with the standard recipe $m = 2$, $R_0 = 10^{-3}$ [7]. The nominal R_0 is realised only for layers of $\gtrsim 8$ points; for thinner layers the discrete transition error dominates (a four-point layer reflects at the 0.5–2% level for grid-scale wavelengths), consistent with the saturation of Figure 3.2(a).

When the CFS parameters are used they are graded with the same geometry,

$$\kappa(d) = 1 + (\kappa_{\text{max}} - 1)(d/L)^m, \quad \alpha(d) = \alpha_{\text{max}}(1 - d/L), \quad (\text{A.12})$$

κ rising from 1 at the interface to κ_{max} at the wall with the same polynomial as σ , and α tapering linearly [7] from α_{max} just inside the interface to zero at the wall (and vanishing identically outside the layer); in the implementation $\alpha(d) = \alpha_{\text{max}}[1 - (\sigma(d)/\sigma_{\text{max}})^{1/m}]$, which is identical. A nonzero α improves grazing-incidence and evanescent absorption but unmatched the layer at low frequencies (the stretch is matched only for $\omega \gg \alpha$), so the fixed-horizon absorption benchmarks of Section 3.4.2 use $\alpha_{\text{max}} = 0$. Finally, the memory rescaling defaults to $\gamma = \sqrt{2\sigma_{\text{max}}/h}$, the choice that balances the injection and memory-source contributions to $\lambda_{\text{max}}(H_1)$ (Appendix 3.8).

APPENDIX B

SCHRÖDINGERISATION: DISCRETE FORMULAS AND RECOVERY

This appendix records the complete discrete recipe behind the summary of Section 3.3.3, with the exact sign, grid, and ordering conventions of our implementation; these are the conventions validated end to end by the demonstrations of Section 3.5.3.

B.1 Hermitian split and warped transform

The semi-discrete absorbing dynamics $\dot{z} = Az$ is split as

$$A = H_1 + iH_2, \quad H_1 = \frac{A + A^\dagger}{2}, \quad H_2 = \frac{A - A^\dagger}{2i}, \quad (\text{B.1})$$

with the dissipative part H_1 and the Hamiltonian part H_2 both Hermitian. The warped phase-space transform of Ref. [10] appends one auxiliary variable p and sets $w(t, p) = e^{-p}z(t)$ for $p \geq 0$ (extended to $p < 0$ by the profile g of Section B.3). The sign structure follows in one line: along the exact solution, $\partial_t w = e^{-p}Az = Aw$ and $\partial_p w = -e^{-p}z = -w$, so $H_1 w = -H_1 \partial_p w$ and

$$\partial_t w = -H_1 \partial_p w + iH_2 w, \quad (\text{B.2})$$

a transport equation in p : the non-unitary part of A has become advection of the auxiliary variable.

B.2 Discrete setup and per-mode evolution

The p register holds $N_p = 2^{n_p}$ grid points on the periodic domain $[-p_{\max}, p_{\max})$,

$$p_j = -p_{\max} + j \Delta p, \quad \Delta p = \frac{2p_{\max}}{N_p}, \quad (\text{B.3})$$

$j = 0, \dots, N_p - 1$, with dual frequencies $\eta_k = \pi k / p_{\max}$ for $k = -N_p/2, \dots, N_p/2 - 1$ in FFT ordering [i.e. $\eta = 2\pi \times \text{fftfreq}(N_p, \Delta p)$]. We fix the Fourier *synthesis* convention $w(\cdot, p) = \sum_k \hat{w}_k e^{+i\eta_k p}$ (NumPy: $w = \text{ifft}(\hat{w})$ along the p axis, $\hat{w} = \text{fft}(w)$). Substituting $\partial_p \rightarrow i\eta_k$ in Eq. (B.2) decouples the modes into the unitary evolutions of Eq. (3.6),

$$\begin{aligned} \frac{d\hat{w}_k}{dt} &= -i(\eta_k H_1 - H_2) \hat{w}_k \\ \implies \hat{w}_k(T) &= e^{-iT(\eta_k H_1 - H_2)} \hat{w}_k(0). \end{aligned} \quad (\text{B.4})$$

Stacking the modes, the whole evolution is one Hamiltonian simulation,

$$H_{\text{tot}} = D_\eta \otimes H_1 - I_p \otimes H_2, \quad D_\eta = \text{diag}(\eta_k), \quad (\text{B.5})$$

acting on the tensor order (p register, system register) with the p register most significant: on the circuit, qubits $0, \dots, n_s - 1$ carry the system register (little-endian) and qubits $n_s, \dots, n_s + n_p - 1$ the p register, so the flattened statevector index is $j 2^{n_s} + s$. The opening QFT[†] on the p register maps the grid amplitudes to $\hat{w}/\sqrt{N_p}$ (the unitarily normalised FFT), $e^{-iT H_{\text{tot}}}$ acts in the η basis, and the closing QFT returns to the p grid [Figure 3.5(b)].

B.3 Initial data and the C^1 profile

The initial condition $w(0, p_j) = g(p_j) z(0)$ is a product state across the two registers,

$$|W(0)\rangle = \frac{|g\rangle \otimes |z_0\rangle}{N_0}, \quad N_0 = \|g\|_2 \|z_0\|_2, \quad (\text{B.6})$$

with the normalisation N_0 recorded classically and undone at recovery. The plain profile $g(p) = e^{-|p|}$ is kinked at $p = 0$ and limits the p discretisation to $O(\Delta p)$; the C^1 cubic of Ref. [12, Remark 4.9] replaces it on $(-1, 0)$,

$$g(p) = \begin{cases} e^{-p}, & p \geq 0, \\ \left(\frac{3}{e} - 3\right)p^3 + \left(\frac{4}{e} - 5\right)p^2 - p + 1, & -1 < p < 0, \\ e^p, & p \leq -1, \end{cases} \quad (\text{B.7})$$

restoring $O(\Delta p^2)$. The junctions are C^1 by construction: at $p = 0^-$ the cubic gives $g = 1$, $g' = -1$, matching e^{-p} ; at $p = -1^+$ it gives $g = g' = e^{-1}$, matching e^p . Because $g(p) = e^{-p}$ *exactly* for $p \geq 0$, the recovery formula below is unchanged by the smoothing. Measured on the 1D CPML benchmark of Figure 3.2(c) (2^5 spatial points, $n_{\text{pml}} = 8$, $\sigma_{\text{max}} = 1$, $T = 30$, $p_{\text{max}} = 18$): the recovery error at the default slice falls from 4.1×10^{-2} to 6.3×10^{-4} over $n_p = 6$ –11 for the kinked profile and from 1.3×10^{-2} to 2.4×10^{-6} for the cubic, with fitted slopes of $\log_2(\text{error})$ versus n_p of -1.2 and -2.6 (-1.6 and -3.4 on the pre-floor points $n_p \leq 9$)—at or above the nominal first- and second-order rates, confirming the one-order upgrade of Ref. [12] in this setting.

B.4 Recovery: threshold, wraparound, plateau

The solution is recovered from a single grid slice,

$$z(T) = e^{p^*} w(T, p^*), \quad p^* \geq \lambda_{\text{max}}^+(H_1) T, \quad (\text{B.8})$$

where $\lambda_{\text{max}}^+ = \max\{0, \lambda_{\text{max}}(H_1)\}$ and the validity threshold on p^* is Theorem 3.1 of Ref. [12]. When $H_1 \preceq 0$ (the collapsed 1D CPML and the sponge) any $p^* > 0$ is admissible, and the implementation defaults to the slice three grid points above $p = 0$, i.e. $p^* = 3\Delta p$. For indefinite H_1 (the 2D CPML memory form of Lemma 3.1) the slice must be placed explicitly at $p^* \geq \lambda^+ T$; below the threshold the recovered state is $O(1)$ wrong (Figure 3.3).

Two further discrete conditions guard the slice. *Wraparound*: in Eq. (B.2) an H_1 -eigencomponent with eigenvalue λ is advected in p with velocity λ , so the most strongly damped components travel a distance $\lambda^- T$ *leftwards*, with $\lambda^- = |\lambda_{\min}(H_1)|$ ($= \sigma_{\max}$ for the diagonal absorbers). The discrete p domain is periodic, so $p_{\max}$ must be wide enough that this leftward-transported mass does not wrap through $p = -p_{\max}$ back into the recovery region. *Plateau*: since $e^p w(T, p) = z(T)$ for every admissible p , the function $p \mapsto e^p \|w(T, p)\|$ must be flat across the recovery window. The implementation tabulates it over the first twelve grid points right of $p = 0$ and reports the relative peak-to-peak spread; a flat plateau certifies that $p_{\max}$, N_p , and the profile are adequate (our regression tests require a spread below 5×10^{-2}), while a tilted or stepped plateau flags wraparound or an under-resolved profile.

B.5 Measurement interpretation

The pipeline is unitary, so the final statevector is exactly $W(T)/N_0$, holding $w_s(T, p_j)/N_0$ at index $j 2^{n_s} + s$. Measuring the p register returns outcome p_j with probability $\|w(T, p_j)\|^2/N_0^2$ and collapses the system register onto $w(T, p_j)/\|w(T, p_j)\| = z(T)/\|z(T)\|$: every slice in the admissible window yields the *same* normalised state, so post-selection may accept the whole plateau window rather than a single outcome, summing the corresponding probabilities. The success probability of the slice at p^* is, up to p -discretisation error,

$$\Pr(p^*) = \frac{e^{-2p^*} \|z(T)\|^2}{N_0^2} \propto e^{-2p^*}, \quad (\text{B.9})$$

which at the threshold slice $p^* = \lambda^+ T$ is the $e^{-2\lambda^+ T}$ post-selection penalty of Section 3.4.3—the cost that the Lyapunov symmetrizer of Section 3.4.4 replaces by the T -independent factor $\kappa_2^2(S)$. On the statevector simulator used for the demonstrations of Section 3.5.3, the slice is read off rather than sampled: the final statevector is reshaped to an $N_p \times 2^{n_s}$ array, row j at p^* is extracted, and the result is rescaled by $e^{p_j} N_0$. This is deterministic and exact, so the reported errors measure the Trotter and discretisation error only; on hardware the same slice would additionally cost the factor e^{-2p^*} of Eq. (B.9) in repetitions.

APPENDIX C

VALIDATION PROTOCOL AND CONVERGENCE DATA

This appendix records the classical validation protocol behind every reflection number quoted in Section 3.4.2 and the raw convergence data behind the fitted orders of Section 3.5.2 [Figure 3.2(a-d)].

C.1 Gold-standard reflection protocol

Measuring the residual reflection of an absorbing layer against the absorbing dynamics itself is circular; the gold standard is the open-domain solution. The truncated domain has $N = 128$ points (seven grid qubits) with the CPML occupying n_{pml} points at each end. The initial data are a Gaussian velocity bump centred at $j = 64$ with width 3 cells and $w(0) = 0$, normalised to $\|z_0\|_2 = 1$. The reference embeds the *same* truncated data centred in a $4\times$ hard-wall domain ($N_{\text{ref}} = 512$, embedding offset $\Delta = (N_{\text{ref}} - N)/2 = 192$) and evolves it with the closed-domain unitary $e^{-iH_{\text{ref}}T}$. Both evolutions are computed by dense matrix exponentials, so the comparison contains no time-integration error. The reported error is the pointwise ℓ_2 difference on the interior window,

$$E(T)^2 = \sum_{f \in \{v, w\}} \sum_{j=n_{\text{pml}}}^{N-n_{\text{pml}}-1} |f_j(T) - f_{j+\Delta}^{\text{ref}}(T)|^2. \quad (\text{C.1})$$

Two checks make the protocol sound. (i) *Window alignment*: at $T = 0$ the windowed difference in Eq. (C.1) evaluates to exactly zero in floating point, pinning the embedding offset and the index conventions. (ii) *Causal validity*: with $c = 1$, $h = 1$ the pulse meets the layer interface at depth $j = n_{\text{pml}}$ and its reflection is back at the centre by $T \approx 2(64 - n_{\text{pml}}) \leq 120$ for $n_{\text{pml}} \geq 4$, whereas the far walls of the reference domain are first felt inside the comparison window only at $T \approx 380$. The plateau times $T \in \{100, 120, 140\}$ therefore capture the complete round-trip reflection of the truncated domain while the reference is still uncontaminated. The reported reflection is the maximum of $E(T)$ over the three plateau times, and the relative plateau spread $[(\max - \min)/\text{mean}]$ is the flatness diagnostic. A reduced instance of the same protocol ($N = 64$ versus 256, $T = 20$ and 60, asserted $< 10^{-3}$) is pinned in the unit-test suite (Section C.6).

C.2 Reflection versus layer width and design reflection

Table C.1 interprets as follows. The curve is the sum of two competing contributions. At the thinnest layer ($n_{\text{pml}} = 4$) the discrete transition error dominates: the measured reflection sits $9.4\times$ above the discrete wall-return prediction. For $n_{\text{pml}} \geq 8$ the transition error is negligible and the reflection is set by

Table C.1 Gold-standard true reflection (1D CPML, $N = 128$ versus the $N = 512$ hard-wall reference, $m = 2$): plateau error $\max_{T \in \{100, 120, 140\}} E(T)$ and relative plateau spread. Upper block: layer-width sweep at $R_0 = 10^{-3}$; lower block: design-reflection sweep at $n_{\text{pml}} = 12$.

n_{pml}	R_0	plateau error	plateau spread
4	10^{-3}	6.7×10^{-4}	1.0×10^{-9}
6	10^{-3}	2.6×10^{-4}	3.3×10^{-10}
8	10^{-3}	2.6×10^{-4}	2.2×10^{-10}
12	10^{-3}	3.4×10^{-4}	1.7×10^{-9}
16	10^{-3}	4.2×10^{-4}	3.3×10^{-5}
12	10^{-2}	4.5×10^{-3}	1.5×10^{-10}
12	10^{-3}	3.4×10^{-4}	1.7×10^{-9}
12	10^{-4}	4.3×10^{-5}	1.9×10^{-7}

the wall return, which *rises toward the design value from below* as the layer widens: node sampling of the polynomial profile overestimates $\int \sigma dx$ by a factor $1 + O(1/n_{\text{pml}})$ (exactly $\Sigma \sigma h / \int \sigma = 1.38, 1.25, 1.19, 1.13, 1.09$ for $n_{\text{pml}} = 4\text{--}16$), so the effective round-trip reflection is $R_{\text{eff}} = R_0^{\Sigma \sigma h / \int \sigma} = 7.1 \times 10^{-5}, 1.7 \times 10^{-4}, 2.7 \times 10^{-4}, 4.2 \times 10^{-4}, 5.2 \times 10^{-4}$ across the sweep; the measured values track this prediction within 5–20% for $n_{\text{pml}} \geq 8$. Thin layers therefore *over*-perform the nominal R_0 (at the price of larger transition error), and widening the layer converges the reflection up to the design value. We verified this directly in the SEISMIC.CPML reference implementation [7] (the 2D isotropic elastic code, unmodified kernels, same gold-standard protocol, $\alpha = 0$, $K = 1$): its P -wave reflection is likewise non-monotonic and rises from $n_{\text{pml}} = 12$ to 16 ($5.8 \times 10^{-4} \rightarrow 6.5 \times 10^{-4}$, against a predicted $6.5 \times 10^{-4} \rightarrow 7.2 \times 10^{-4}$), converging up toward the design R_0 and matching its own discrete-sum prediction within $\sim 10\%$ for $n_{\text{pml}} \geq 12$. Two scheme-dependent differences: the staggered elastic scheme samples the profile at both full and half points, so its quadrature ratio is $1 + 3/(4n_{\text{pml}})$ rather than our $\sim 1 + 3/(2n_{\text{pml}})$; and the minimum sits at $n_{\text{pml}} \approx 12$ there because S -wave transition reflection (whose design value $R_0^{c_p/c_s}$ is far smaller) and the coarser wavelength resolution keep the transition term dominant further out. The effect is invisible in normal use of that code, where the layer width is fixed at ~ 10 cells.

The miscalibration admits a one-line remedy, which we provide as an option (`calibration='discrete'`): rescale the staggered profile *pair* by a single common factor so that the characteristic damping sum $\Sigma(\sigma_v + \sigma_w)h/2$ equals the design integral exactly. The realised reflection is then flat at the design value, $0.81\text{--}1.08 R_0$ across the whole sweep $n_{\text{pml}} = 4\text{--}16$ [red squares in Figure 3.2(a)], i.e. the Collino–Tsogka formula becomes exact on the grid. The common factor is essential: rescaling each profile *separately* gives σ_v and σ_w different damping functions (the node and midpoint quadratures differ), an in-layer impedance mismatch that reflects at $O(1/n_{\text{pml}})$ —measured at 1.2×10^{-2} to 6.9×10^{-2} , i.e. ten

to seventy times *worse* than no calibration at all. This remedy applies verbatim to classical CPML codes. That the level is R_0 -controlled rather than an absolute discretisation floor is established by the lower block: at $n_{\text{pml}} = 12$ the measured reflection tracks the design value over two decades, with ratios $E/R_0 = 0.45, 0.34, 0.43$ —the Collino–Tsogka formula is predictive within a factor ~ 2.3 in this discretisation, and lowering R_0 lowers the true reflection commensurately, down to 4.3×10^{-5} at $R_0 = 10^{-4}$. This is the floor behaviour reported for classical CPML implementations by Komatitsch and Martin [7]. The plateau is flat to $\leq 2 \times 10^{-9}$ (relative) in six of the eight rows; even the two most weakly damped rows ($n_{\text{pml}} = 16$ and $R_0 = 10^{-4}$) vary by only 3.3×10^{-5} and 1.9×10^{-7} of their mean across $T \in \{100, 120, 140\}$, i.e. the round-trip transient has fully decayed and the quoted numbers are converged in T .

C.3 Recovery order in the p register

Table C.2 Schrödingerisation recovery error versus p -register size n_p for the kinked $e^{-|p|}$ profile and the C^1 cubic profile of Ref. [12] [1D CPML benchmark, $N = 32$, $n_{\text{pml}} = 8$, $\sigma_{\text{max}} = 1$, $T = 30$, $p \in [-18, 18]$, $\Delta p = 36/2^{n_p}$]. Errors are $\|z_{\text{rec}} - e^{AT} z_0\|_2$ with $\|z_0\|_2 = 1$, recovered at the library-default slice $p^* = 3\Delta p$ (valid at any $p^* > 0$ since $H_1 \leq 0$ here). The fitted orders are least-squares slopes of $\log_2 E$ versus n_p .

n_p	Δp	$e^{- p }$ profile	C^1 cubic profile
6	0.563	4.1×10^{-2}	1.3×10^{-2}
7	0.281	1.6×10^{-2}	2.6×10^{-3}
8	0.141	3.6×10^{-3}	1.7×10^{-4}
9	0.0703	1.5×10^{-3}	1.3×10^{-5}
10	0.0352	1.1×10^{-3}	5.9×10^{-6}
11	0.0176	6.3×10^{-4}	2.4×10^{-6}
order in Δp (all points)		1.23	2.64
order in Δp ($n_p = 6$ –9)		1.64	3.40

Table C.2 gives the recovery error of the full Schrödingerised pipeline (exact per-mode evolution, no Trotter) as a function of the p -register size, for the two warping profiles of Section 3.3.3. The errors are absolute per unit initial norm; by $T = 30$ the layer has absorbed all but $\|e^{AT} z_0\|_2 = 1.4 \times 10^{-3}$ of the state, so resolving the residual field demands the higher-order profile. Both profiles meet or exceed their nominal orders [$O(\Delta p)$ kinked, $O(\Delta p^2)$ cubic, Remark 4.9 of Ref. [12]]: the full-sweep fits give orders 1.23 and 2.64, and the pre-floor fits ($n_p = 6$ –9) give 1.64 and 3.40. Because the default slice $p^* = 3\Delta p$ moves with refinement, we re-evaluated both series at the fixed slice $p^* = 1$: the fitted orders become 1.64 and 3.25, so the profile ordering is not a slice artefact. At $n_p = 11$ the cubic profile is $\sim 260\times$ more accurate (2.4×10^{-6} versus 6.3×10^{-4}); the flattening of its last two points marks the approach to the recovery floor of this configuration.

Table C.3 Gate-level Trotter error versus step count [1D CPML, $N = 8$, $n_{\text{pml}} = 2$, $\sigma_{\text{max}} = 1$, $T = 4$, $n_p = 5$, $p_{\text{max}} = 8$; transpiled circuits on a statevector simulator; errors $\|z_{\text{circ}} - z_{\text{ref}}\|_2$ with $\|z_0\|_2 = 1$]. “vs exact S.” uses the exact Schrödingerised evolution (exact per-mode propagation at the same p discretisation) as reference, isolating the splitting error; “vs $e^{AT}z_0$ ” uses the full matrix-exponential reference and therefore contains the p -discretisation floor, measured independently as 1.29×10^{-2} . Order 2 symmetrises both Trotter levels (Strang outer split, palindromic term order inner).

steps	order 1		order 2	
	vs exact S.	vs $e^{AT}z_0$	vs exact S.	vs $e^{AT}z_0$
10	1.36×10^{-1}	1.32×10^{-1}	3.24×10^{-2}	3.40×10^{-2}
20	6.55×10^{-2}	6.27×10^{-2}	8.02×10^{-3}	1.46×10^{-2}
40	3.25×10^{-2}	3.11×10^{-2}	2.00×10^{-3}	1.29×10^{-2}
80	1.62×10^{-2}	1.75×10^{-2}	4.99×10^{-4}	1.29×10^{-2}
slope	−1.02	−0.98	−2.01	−0.44 ^a

^a Floor-limited: not a meaningful order, see text.

C.4 Trotter orders at gate level

Table C.3 separates the two error sources of the circuit pipeline of Section 3.5.2. Against the exact Schrödingerised evolution the splitting error is clean: fitted slope −1.02 at order 1 and −2.01 at order 2 with both Trotter levels symmetrised [Figure 3.2(d)]. Against the full $e^{AT}z_0$ reference the order-2 series saturates at 1.29×10^{-2} from 40 steps onward—exactly the independently measured p -discretisation floor of this deliberately coarse $n_p = 5$ register—so the nominal slope −0.44 in that column is floor-limited rather than a defect of the splitting. The order-1 series does not saturate within the sweep because its splitting error stays above the floor. Enlarging the p register lowers the floor at the rates of Table C.2.

C.5 Stagger ablation

The damping profiles are sampled at $x_j = (j + s)h$ with the stagger s matched to where each field lives (Section 3.3.2): for the forward-difference convention, v at integer nodes ($s = 0$) and w at the half-cells $x_{j-1/2}$ ($s = -1/2$). Rerunning the gold-standard protocol ($n_{\text{pml}} = 12$, $R_0 = 10^{-3}$) with the σ_w stagger ablated gives Table C.4: misplacing the profile by half a cell costs two orders of magnitude in true reflection. The stagger is load-bearing, not cosmetic.

Table C.4 Stagger ablation under the gold-standard protocol (Section C.1; 1D CPML, $n_{\text{pml}} = 12$, $R_0 = 10^{-3}$): plateau reflection versus the stagger offset s of the σ_w sampling, $x_j = (j + s)h$.

σ_w stagger s	plateau error	degradation
−1/2 (correct half-cell)	3.4×10^{-4}	1
0 (unstaggered)	4.1×10^{-2}	≈120
+1/2 (wrong sign)	8.1×10^{-2}	≈235

C.6 Test suite

The classical and circuit layers are pinned by 24 unit tests (`tests/test_absorbing.py`), grouped by the property classes they assert. (i) *Generator structure and spectra*: the Collino–Tsogka profile amplitude and support; $\max \operatorname{Re} \operatorname{eig}(A) \leq 0$ to round-off for the collapsed 1D, general CFS ($\kappa_{\max} = 2$, $\alpha_{\max} = 0.05$), and 2D memory-form generators; the Hermitian split $A = H_1 + iH_2$ with $H_1 = -\operatorname{diag}(\sigma) \preceq 0$ for the collapsed 1D form and the sponge. (ii) *Collapse*: the 1D memory form reproduces the collapsed diagonal damping to 10^{-10} . (iii) *Absorption and reflection bounds*: interior energy below 10^{-5} where the hard-wall reference retains more than 0.5; round-trip amplitude reflection below 5×10^{-3} ; 2D CPML and sponge energy decay; and the gold-standard true reflection below 10^{-3} on the reduced instance. (iv) *Recovery thresholds*: scalar decay recovered to 10^{-4} ; the cubic profile beating the kinked profile at equal resolution; 1D recovery below 2×10^{-3} with a plateau-flatness diagnostic; 2D sponge recovery below 10^{-3} with $\lambda^+ = 0$ asserted; and the 2D CPML threshold of Section 3.4.3 verified in both directions—accurate above $p^* = \lambda^+ T + 1$ and at least ten times worse below. (v) *Circuit-versus-matrix agreement*: the exact-Hamiltonian-gate path against the numpy per-mode pipeline to 10^{-8} (pinning the QFT/FFT conventions and qubit ordering); the Trotterised Bell circuit to 5×10^{-2} ; the separated 2D operator strings against the block Hamiltonian to 10^{-13} ; the per-step orders $O(\delta t^2)$ and $O(\delta t^3)$ of the projector-aware Bell circuit and its Strang variant; and the 17-qubit end-to-end gate-level run to 3×10^{-2} . (vi) *Guards and regression*: rejection of out-of-range recovery slices, degenerate layers, and inconsistent damping vectors; and a regression test pinning the untouched closed-domain library (the mixed Dirichlet/Neumann Laplacian stencil). Every bound above is asserted numerically, so any change that degrades a spectrum, a reflection, a recovery threshold, or a circuit convention fails the suite.

APPENDIX D

THE LYAPUNOV SYMMETRIZER: CONSTRUCTION, EXACT PROPERTIES, AND COMPRESSIBILITY DATA

This appendix collects the classical construction behind Section 3.4.4, the exact identity underlying Eq. (3.14) with its proof, the complete conditioning data of which Table 3.2 is an excerpt, the grid-size saturation measurements, and the compressibility study of the symmetrizer S .

D.1 Construction and classical cost

The absorbing generator A is marginally stable—we measure $\max \operatorname{Re} \operatorname{eig}(A) = 0$ to round-off (Section 3.4.2)—so the Lyapunov equation for A itself is singular. We therefore shift, $A_\epsilon = A - \epsilon I$, which is Hurwitz with margin ϵ , and solve

$$A_\epsilon^\dagger W + W A_\epsilon = -I, \quad W = \int_0^\infty e^{tA_\epsilon^\dagger} e^{tA_\epsilon} dt \succ 0, \quad (\text{D.1})$$

the integral being the unique Hermitian solution. The symmetrizer is the Hermitian square root $S = W^{1/2}$, with conditioning $\kappa_2(S) = \sqrt{\lambda_{\max}(W)/\lambda_{\min}(W)}$. The shift is undone classically by the scalar factor $e^{\epsilon T}$ [Eq. (3.13)].

Numerically, Eq. (D.1) is solved by the Bartels–Stewart algorithm (one complex Schur decomposition of $A_\epsilon^\dagger$ plus a triangular substitution; `scipy.linalg.solve_continuous_lyapunov`), and S , S^{-1} follow from one Hermitian eigendecomposition of W —three dense $O(n^3)$ kernels in total. Measured wall times at the largest completed size, $n = 4096$ (32×16 grid, $\sigma_{\max} = 0.5$): the Lyapunov solve takes 268 s at $\epsilon = 10^{-2}$ and 244 s at $\epsilon = 10^{-3}$, the eigendecomposition of W takes 31 s, the residual verification 2.5–2.6 s, and the sparse Lanczos evaluation of $\lambda^+ = \lambda_{\max}^+(H_1)$ is negligible (0.05 s). The $n = 8192$ point was gated off by cubic extrapolation of the measured $n = 4096$ time (projected 40.3 min per ϵ , over the 25-min budget) and by memory (a dense complex 8192^2 matrix occupies 1 GiB and the solve holds six to eight of them, against 5.1 GiB free). Across all 18 configurations of Table D.1 the relative Lyapunov residual $\|A_\epsilon^\dagger W + W A_\epsilon + I\|_F / \|W\|_F$ is at most 1.4×10^{-14} .

D.2 An exact identity for the transformed Hermitian part

The equality sign in Eq. (3.14) is not an estimate. With the unit right-hand side $-I$ in Eq. (D.1), the Hermitian part of the transformed generator is an explicit function of W :

Proposition 1 *Let $A \in \mathbb{C}^{n \times n}$ have spectral abscissa at most zero, fix $\epsilon > 0$, set $A_\epsilon = A - \epsilon I$, let $W \succ 0$ solve $A_\epsilon^\dagger W + W A_\epsilon = -I$, and let $S = W^{1/2}$ be the Hermitian positive-definite square root. Then, exactly,*

$$\frac{1}{2}(SA_\epsilon S^{-1} + S^{-1}A_\epsilon^\dagger S) = -\frac{1}{2}W^{-1} \prec 0, \quad (\text{D.2})$$

$$\frac{1}{2}(SAS^{-1} + S^{-1}A^\dagger S) = \epsilon I - \frac{1}{2}W^{-1}. \quad (\text{D.3})$$

In particular the ϵ -shifted transform is strictly dissipative with $\lambda_{\max}(\tilde{H}_1^{(\epsilon)}) = -1/(2\lambda_{\max}(W))$, and the transform of the original A has $\lambda_{\max}(\tilde{H}_1) = \epsilon - 1/(2\lambda_{\max}(W)) < \epsilon$.

Proof. Since A_ϵ is Hurwitz, the integral in Eq. (D.1) converges and is the unique Hermitian solution; it is positive definite, so $S = W^{1/2}$ exists, is Hermitian, and $S^2 = W$. Write $\tilde{A} = SA_\epsilon S^{-1}$ and take any $x \in \mathbb{C}^n$ with $x = Sy$. By congruence,

$$x^\dagger(\tilde{A} + \tilde{A}^\dagger)x = y^\dagger(WA_\epsilon + A_\epsilon^\dagger W)y = -\|y\|^2 = -x^\dagger W^{-1}x, \quad (\text{D.4})$$

using $S\tilde{A}S = WA_\epsilon$ and $S\tilde{A}^\dagger S = A_\epsilon^\dagger W$. Two Hermitian matrices with equal quadratic forms are equal, so $\tilde{A} + \tilde{A}^\dagger = -W^{-1}$, which is Eq. (D.2); its largest eigenvalue is $-\frac{1}{2}\lambda_{\min}(W^{-1}) = -1/(2\lambda_{\max}(W))$. For Eq. (D.3), $SAS^{-1} = SA_\epsilon S^{-1} + \epsilon I$ and ϵI is Hermitian, so the Hermitian part shifts by ϵI and the eigenvalues are $\epsilon - 1/(2\lambda_j(W))$. ■

The identity is verified in every measured configuration: in all 18 rows of Table D.1 the directly computed $\lambda_{\max}(\tilde{H}_1)$ (dense eigendecomposition of the Hermitian part of SAS^{-1}) agrees with the prediction $\epsilon - 1/(2\lambda_{\max}(W))$ to a relative deviation of at most 3.8×10^{-12} —far beyond the four-digit agreement quoted in the main text—and the shifted column equals the unshifted one minus ϵ to the same precision. A direct check of the matrix identity (D.2) on an $n = 50$ instance gives a Frobenius residual of 8.9×10^{-14} . Proposition 1 also explains the choice made in Section 3.4.4: the transform of the *original* A is not dissipative ($\lambda_{\max}(\tilde{H}_1) \approx \epsilon > 0$), whereas the transform of A_ϵ is, strictly; we therefore evolve the shifted system and undo $e^{\epsilon T}$ classically.

D.3 The full conditioning table

Table D.1 lists all 18 measured configurations (three grids, two damping strengths, three shifts). Three regularities are worth recording. (i) $\lambda_{\min}(W)$ is essentially independent of ϵ (0.66–0.68 at $\sigma_{\max} = 0.5$, 0.37 at $\sigma_{\max} = 1.0$), so the entire ϵ dependence of $\kappa_2(S)$ is carried by $\lambda_{\max}(W)$, consistent with Proposition 1, in which $\lambda_{\max}(W)$ is pinned to the inverse dissipativity margin of the shifted transform. (ii) Least-squares fits of $\ln \kappa_2(S)$ against $\ln \epsilon$ per (grid, $\sigma_{\max}$) configuration give slopes between -0.64 and -1.03 (the range quoted as $\epsilon^{-0.6 \text{ to } -1.0}$ in Section 3.4.4), steepening with grid size at fixed ϵ range. (iii) The crossover times $T^* = \ln \kappa^2/(2\lambda^+)$ span 6.7–16.0 and grow only logarithmically with $\kappa_2(S)$. For comparison, the eigenvector symmetrizer $W = (VV^\dagger)^{-1}$ yields $\kappa_2(S) = 2.9 \times 10^8$ – 6.9×10^8 where it exists and fails outright ($VV^\dagger$

Table D.1 Full Lyapunov-symmetrizer data set; Table 3.2 of the main text is the excerpt obtained by dropping the $\lambda_{\min}(W)$ and unshifted-transform columns. Grids are listed in points (n is the full state dimension including the eight blocks). $\lambda^+ = \lambda_{\max}^+(H_1)$ is the indefiniteness of the untransformed generator; $\kappa_2(S) = \sqrt{\text{cond}_2(W)}$; $\lambda_{\max}(\tilde{H}_1)$ is the largest Hermitian-part eigenvalue of the transform SAS^{-1} of the original generator and $\lambda_{\max}(\tilde{H}_1^{(\epsilon)})$ that of the ϵ -shifted transform $SA_\epsilon S^{-1}$; $T^* = \ln \kappa^2/(2\lambda^+)$ is the crossover time. Every row satisfies the identities of Proposition 1: $\lambda_{\max}(\tilde{H}_1)$ equals $\epsilon - 1/(2\lambda_{\max}(W))$ to a relative deviation $\leq 3.8 \times 10^{-12}$, and $\lambda_{\max}(\tilde{H}_1^{(\epsilon)}) = \lambda_{\max}(\tilde{H}_1) - \epsilon$. Relative Lyapunov residuals are $\leq 1.4 \times 10^{-14}$.

Grid	n	$\sigma_{\max}$	ϵ	$\lambda^+(H_1)$	$\lambda_{\min}(W)$	$\kappa_2(S)$	$\lambda_{\max}(\tilde{H}_1)$	$\lambda_{\max}(\tilde{H}_1^{(\epsilon)})$	T^*
8×8	512	0.5	10^{-2}	0.683	0.669	402.1	9.9954×10^{-3}	-4.63×10^{-6}	8.8
8×8	512	0.5	10^{-3}	0.683	0.681	3009	9.9992×10^{-4}	-8.10×10^{-8}	11.7
8×8	512	0.5	10^{-4}	0.683	0.683	1.105×10^4	9.9994×10^{-5}	-6.00×10^{-9}	13.6
8×8	512	1.0	10^{-2}	0.927	0.373	509.3	9.9948×10^{-3}	-5.17×10^{-6}	6.7
8×8	512	1.0	10^{-3}	0.927	0.376	2851	9.9984×10^{-4}	-1.64×10^{-7}	8.6
8×8	512	1.0	10^{-4}	0.927	0.377	9762	9.9986×10^{-5}	-1.39×10^{-8}	9.9
16×8	1024	0.5	10^{-2}	0.687	0.661	506.3	9.9970×10^{-3}	-2.95×10^{-6}	9.1
16×8	1024	0.5	10^{-3}	0.687	0.673	7009	9.9998×10^{-4}	-1.51×10^{-8}	12.9
16×8	1024	0.5	10^{-4}	0.687	0.674	3.052×10^4	9.9999×10^{-5}	-7.96×10^{-10}	15.0
16×8	1024	1.0	10^{-2}	0.956	0.369	749.6	9.9976×10^{-3}	-2.41×10^{-6}	6.9
16×8	1024	1.0	10^{-3}	0.956	0.372	6981	9.9997×10^{-4}	-2.76×10^{-8}	9.3
16×8	1024	1.0	10^{-4}	0.956	0.373	2.732×10^4	9.9998×10^{-5}	-1.80×10^{-9}	10.7
16×16	2048	0.5	10^{-2}	0.694	0.655	566.9	9.9976×10^{-3}	-2.37×10^{-6}	9.1
16×16	2048	0.5	10^{-3}	0.694	0.667	9856	9.9999×10^{-4}	-7.71×10^{-9}	13.3
16×16	2048	0.5	10^{-4}	0.694	0.669	6.612×10^4	1.0000×10^{-4}	-1.71×10^{-10}	16.0
16×16	2048	1.0	10^{-2}	0.972	0.367	857.7	9.9981×10^{-3}	-1.85×10^{-6}	6.9
16×16	2048	1.0	10^{-3}	0.972	0.370	1.249×10^4	9.9999×10^{-4}	-8.67×10^{-9}	9.7
16×16	2048	1.0	10^{-4}	0.972	0.370	6.347×10^4	1.0000×10^{-4}	-3.35×10^{-10}	11.4

numerically singular) on both 16×8 configurations; the Lyapunov construction is four to six orders of magnitude better conditioned and never fails.

D.4 Saturation of $\kappa_2(S)$ with grid size

Table D.2 extends the $\sigma_{\max} = 0.5$ series to $n = 4096$. At $\epsilon = 10^{-2}$ the conditioning is flat: $\kappa_2(S)$ decreases from 566.9 to 564.5 between $n = 2048$ and $n = 4096$ (local log-log slope -0.006). At $\epsilon = 10^{-3}$ it still rises but visibly flattens. Power-law fits confirm that a power law is the wrong model: at $\epsilon = 10^{-2}$ the fitted exponent drops from 0.25 ($r^2 = 0.963$, $n = 512$ – 2048) to 0.16 ($r^2 = 0.823$, $n = 512$ – 4096), and at $\epsilon = 10^{-3}$ from 0.86 ($r^2 = 0.943$) to 0.62 ($r^2 = 0.875$)—falling exponents with degrading fit quality, i.e. κ is saturating, not following a power law. The drivers saturate individually: λ^+ grows from 0.6829 to 0.6940 over the eightfold range in n and is then identical to ten digits between $n = 2048$ and $n = 4096$, and $\lambda_{\min}(W)$ stays at 0.655–0.681 throughout. The conditioning is therefore ϵ -controlled, not grid-controlled: mesh refinement does not inflate the symmetrizer cost.

Table D.2 $\kappa_2(S)$ and $\lambda_{\min}(W)$ versus grid size at $\sigma_{\max} = 0.5$ (grids in points; n is the state dimension). The $n = 8192$ point was gated off by the projected runtime and memory (Section D.1).

Grid	n	$\lambda^+(H_1)$	$\epsilon = 10^{-2}$		$\epsilon = 10^{-3}$	
			$\kappa_2(S)$	$\lambda_{\min}(W)$	$\kappa_2(S)$	$\lambda_{\min}(W)$
8×8	512	0.6829	402.1	0.669	3009	0.681
16×8	1024	0.6875	506.3	0.661	7009	0.673
16×16	2048	0.6940	566.9	0.655	9856	0.667
32×16	4096	0.6940	564.5	0.655	11241	0.668

D.5 Compressibility of S

Protocol. At $\epsilon = 10^{-2}$, $\sigma_{\max} = 0.5$, we measured the structure of S on the $n = 512$ grid and confirmed on the $n = 1024$ grid, and scored 25 candidate approximate symmetrizers $S_b \approx S$ drawn from six families (Manhattan-distance-banded, magnitude-thresholded, block-pattern, identity-plus-low-rank, low-rank-plus-band hybrid, and banded- S^{-1}). Each candidate is scored by its residual indefiniteness $\lambda_b = \lambda_{\max}(\text{Herm}(S_b A_\epsilon S_b^{-1}))$; for the exact S , $\lambda_b = -1/(2\lambda_{\max}(W)) < 0$ by Proposition 1. A candidate *passes* if $\lambda_b \leq 2 \times 10^{-3}$ (the ϵ -mode recovery at the default slice remains valid); a candidate with $2 \times 10^{-3} < \lambda_b \leq 0.5$ is *rescue-eligible*: the classical shift is enlarged from ϵ to $s = \epsilon + \lambda_b$, so the evolved generator $S_b(A - sI)S_b^{-1}$ is again dissipative, the undo factor becomes e^{sT} , and the post-selection penalty relative to the exact S has exponent $2\lambda_b T$. End-to-end recoveries (errors relative to $e^{AT} z_0$ at $T \in \{10, 40\}$, $n_p = 10$) were run for the exact S and one representative per surviving family. Table D.3 lists all candidates on the $n = 1024$ confirmation grid.

Structure of S . Three exact or near-exact structural facts. (i) *Dead-index identity.* On the 328 of 1024 indices (32%; 176 of 512, i.e. 34%, at $n = 512$) that carry no dynamics—interior memory components and the zero padding block— A acts as zero, so Eq. (D.1) reduces to $2\epsilon w = 1$ there and S is *exactly* $(2\epsilon)^{-1/2}I$ on this subspace: measured deviation 2.0×10^{-14} on the diagonal, off-diagonal entries below 7.8×10^{-15} , and coupling to the active subspace below 1.1×10^{-12} . This factor is free structure for any implementation. (ii) *The active support is about half the matrix:* the exact S has 48.1% ($n = 1024$) and 51.1% ($n = 512$) nonzero entries (at threshold 10^{-14} of the largest). (iii) *The correction is full rank but strip-localised:* with the optimal scalar $\alpha = 16.7$, the correction $E = S - \alpha I$ carries 94.5% of the Frobenius norm of S , has 999 of 1024 singular values above 10^{-2} of the largest, and needs rank 995 to capture 99.99% of its energy—numerically full rank—while 99.6% of that energy sits on index pairs both lying in the absorbing-layer strips (72 of the 1024 indices are strip-free interior).

Findings. The full-rank measurement rules out identity-plus-low-rank outright: for ranks 8–64 the residual $\lambda_b = 0.674$ – 0.677 equals the *untransformed* $\lambda^+ = 0.687$ to within two percent, i.e. a short LCU correction has no symmetrizing effect at all. Narrow bands, aggressive thresholds, block patterns, and banded inverses are either indefinite or carry $\lambda_b \gtrsim 1$. Moderate compression, however, works under the rescue accounting: magnitude thresholding at 35% density recovers to 8.7×10^{-6} ($T = 10$) and 5.4×10^{-5} ($T = 40$); the Manhattan band $d = 12$ at 38.6% reaches 8.7×10^{-6} and 3.1×10^{-4} ; the rank-32 plus band-8 hybrid at 26.5% reaches 1.3×10^{-5} and 6.6×10^{-4} . The rescue cost is mild at the measured residuals: the largest rescued $\lambda_b = 0.066$ gives an undo factor $e^{\lambda_b T} = e^{2.6} \approx 14$ and a post-selection penalty $e^{2\lambda_b T} = e^{5.3} \approx 2 \times 10^2$ at $T = 40$, against the T -independent baseline $\kappa_2^2(S) \approx 2.6 \times 10^5$. The rescue frame is also the robust one: the dissipativity margin of the exact S is only $|\lambda_b| = 3.0 \times 10^{-6}$, and a perturbation of S of relative spectral norm 6.4×10^{-7} suffices to push λ_b past the 2×10^{-3} gate, so *strict* dissipativity of any approximation is fragile, whereas absorbing the measured λ_b into the classical shift is exact bookkeeping. Taken together—half-dense active support, full-rank strip-localised correction, free dead-index identity factor—these measurements support the implementation verdict of Section 3.4.4: block-encoding of a $\sim 35\%$ -dense operator rather than a local circuit or a short LCU, with a structured-ansatz Lyapunov solve as the open problem. The compression study is limited to $n \leq 1024$.

D.6 Choosing the shift ϵ

The shift enters the cost twice and the accuracy nowhere: the recovered state is exact up to the classically applied factor $e^{\epsilon T}$, while $\kappa_2(S)$ grows as $\epsilon^{-0.64}$ – $\epsilon^{-1.03}$ (Section D.3) and the evolved norm of the shifted system costs $e^{2\epsilon T}$ in post-selection. At the recommended $\epsilon = 10^{-2}$ these are $\kappa_2(S) = 4.0 \times 10^2$ – 8.6×10^2 across all configurations and $e^{\epsilon T} \leq e^{0.4}$ ($e^{2\epsilon T} \leq e^{0.8} \approx 2.2$) over the horizons

Table D.3 All 25 candidate approximate symmetrizers on the $n = 1024$ (16×8) confirmation grid at $\epsilon = 10^{-2}$, $\sigma_{\max} = 0.5$. Density is the fraction of nonzero entries of the sparse object; for the $\alpha I + \text{rank-}r$ family it is $(r+1)/n$ (the number of LCU terms relative to n), and for the banded- S^{-1} family it is the density of the banded S^{-1} (state preparation by sparse linear solve). $\lambda_b = \lambda_{\max}(\text{Herm}(S_b A_\epsilon S_b^{-1}))$ is the residual indefiniteness. Status: “pass” = $\lambda_b \leq 2 \times 10^{-3}$, valid in ϵ mode at the default slice; “rescue” = run end to end with the enlarged shift $\epsilon + \lambda_b$; “elig.” = rescue-eligible ($\lambda_b \leq 0.5$) but not run end to end; “fail” = S_b indefinite or $\lambda_b > 0.5$. The top-50% threshold and the support-of- A^2 patterns reproduce the exact S to machine precision (relative Frobenius deviation $\leq 6 \times 10^{-16}$); the end-to-end errors of every usable candidate are obtained at conditioning $\text{cond}(S_b) = 5.06 \times 10^2$ – 5.11×10^2 , equal to the exact value 5.06×10^2 .

Candidate S_b	density (%)	λ_b	status	err. ($T=10$)	err. ($T=40$)
exact S (active support)	48.1	-3.0×10^{-6}	pass	7.7×10^{-6}	4.9×10^{-5}
Manhattan band, $d = 1$	1.8	2.36	fail	–	–
Manhattan band, $d = 2$	4.2	1.32	fail	–	–
Manhattan band, $d = 4$	10.8	3.09	fail	–	–
Manhattan band, $d = 8$	26.5	0.062	elig.	–	–
Manhattan band, $d = 12$	38.6	0.056	rescue	8.7×10^{-6}	3.1×10^{-4}
magnitude threshold, top 1%	1.0	104	fail	–	–
magnitude threshold, top 5%	5.0	3.03	fail	–	–
magnitude threshold, top 10%	10.0	0.734	fail	–	–
magnitude threshold, top 25%	25.0	0.054	elig.	–	–
magnitude threshold, top 35%	35.0	4.2×10^{-3}	rescue	8.7×10^{-6}	5.4×10^{-5}
magnitude threshold, top 50%	48.1	-3.0×10^{-6}	pass	–	–
block diagonal (per-field)	7.4	6.83	fail	–	–
support of A	29.9	8.58	fail	–	–
support of A^2	48.1	-3.0×10^{-6}	pass	7.7×10^{-6}	4.9×10^{-5}
$\alpha I + \text{rank } 8$	0.9	0.677	fail	–	–
$\alpha I + \text{rank } 16$	1.7	0.677	fail	–	–
$\alpha I + \text{rank } 32$	3.2	0.674	fail	–	–
$\alpha I + \text{rank } 64$	6.3	0.674	fail	–	–
$\alpha I + \text{rank } 32 + \text{band } 2$	4.2	0.374	elig.	–	–
$\alpha I + \text{rank } 32 + \text{band } 8$	26.5	0.066	rescue	1.3×10^{-5}	6.6×10^{-4}
$\alpha I + \text{rank } 64 + \text{band } 2$	4.2	0.349	elig.	–	–
$\alpha I + \text{rank } 64 + \text{band } 8$	26.5	0.033	elig.	–	–
banded S^{-1} , $d = 2$	4.0	5.02	fail	–	–
banded S^{-1} , $d = 8$	25.3	2.23	fail	–	–

tested ($T \leq 40$)—the shift is negligible against κ_2^2 , and decreasing ϵ by a decade buys only that trivial factor while inflating κ_2^2 by one and a half to two decades. The shifted transform is what makes the large- ϵ choice safe: with the *unshifted* transform of A , Proposition 1 gives $\lambda_{\max}(\tilde{H}_1) = \epsilon - 1/(2\lambda_{\max}(W)) \approx \epsilon > 0$, so the recovery threshold [Eq. (3.7)] would demand $p^* \gtrsim \epsilon T$ —at the default slice $p^* = 0.117$ ($n_p = 10$, $p_{\max} = 20$) and $T = 40$ this would force $\epsilon \lesssim 2.9 \times 10^{-3}$, excluding the best-conditioned setting, and would couple ϵ to the horizon and the p -register extent. The shifted transform has $\tilde{H}_1^{(\epsilon)} \preceq -1/(2\lambda_{\max}(W))I \prec 0$, so any $p^* > 0$ is valid at every T and ϵ acts purely as a conditioning knob.